GRANDE FINALE V4: EXACT THRESHOLD COMPILERS, PRIMITIVE Q, AND ORDER-32 REED–SOLOMON MCA

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Abstract. This paper is a complete companion to CAP25 v13.2 [Cho26]. It imports by citation, and does not reprove, the support-wise MCA conventions, universal cap, four deployed unsafe endpoints, identity-prefix and simple-pole constructions, aperiodic and subfield floors, prefix rigidity and exact second moment, quotient and extension reductions, balanced-core reduction, and normalized finite ledger established there.

The first half develops the full safe-side compiler. It proves exact support sparsification, syndrome–secant and fixed-union compilers, recursive affine-span and generalized-weight bounds, common-zero and directional Johnson estimates, codimension-one and arbitrary-codimension support-flag recursions, a minimum-distance sharpening using the actual direction-code support, fixed-mismatch puncturing, a quadratic mean-overlap staircase, exact quotient–remainder prefix normal forms, effective-image Fourier normalization, primitive Q after an explicit Sidon payment, saturation, a moving-root theorem for one-parameter split pencils, and the implication that a row-sharp Q theorem pays the primitive shift-pair ledger.

The second half develops a different MCA completion mechanism. A rank-two reduced intersection matrix gives an exact kernel dictionary and shows that the deployed critical order is 32. Exact dyadic descent identifies the quotient branch. Maximal-support interpolation excludes all primitive slope degrees below 18, while support-collapsed interpolation extracts every low-complexity explanation into a scalar-locator rational atom. Rational atoms satisfy collision rigidity and local-to-global coherence. A finite critical example shows that primitive spread cores can exist, so the remaining theorem is abundance rather than nonexistence. Canonical owner localization pays every small-owner atom, component-free owner pencils contribute at most seven rich points, core-compatible slopes contribute at most 58, a complete correction ray contributes at most $2(n-m+1)+963$, and proper or clone-tolerant correction spaces are paid in the explicit deployed ranges.

The manuscript does not assert the adjacent safe rows unconditionally. For MCA, the remaining inputs are the classification and payment of high-dimensional positive-dimensional correction components, the row-sharp image bound for one coherent large-owner atom—most sharply the Mersenne-31 factor-9.57219783037 prefix target—and complete routing of denominator-root, extension, clone, imperfect-quotient, and near-sunflower exceptions. For the list rows, the remaining inputs are the row-sharp high-nullity prefix and arbitrary-word interior bounds, extension payment, and one exact summed integer certificate.

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Date: July 2026.

2020 *Mathematics Subject Classification.* 94B35, 11T71, 05B40.

Key words and phrases. Reed–Solomon codes, mutual correlated agreement, primitive Q, Sidon moments, generalized Hamming weights, order-32 rank, rational atoms, spread cores.

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1. Scope and relation to CAP25 v13.2

Throughout, CAP25 v13.2 [Cho26] is the foundational source for the deployed unsafe frontier and the normalized first-match ledger. Its theorems are invoked by citation rather than renumbered here. This paper contains all of the additional threshold, compiler, rank, atom, and spread-incidence results needed for the current completion programme.

Imported from CAP25 v13.2	Proved in this paper
MCA conventions, universal cap, active unsafe endpoints	Exact sparsification, syndrome–secant, fixed-union, affine-span, generalized-weight, support-flag, and minimum-distance compilers
Identity-prefix, pole-line, aperiodic, subfield, quotient, and extension packages	Quadratic threshold staircases, exact profile normal forms, effective Fourier normalization, and primitive Q after the stated Sidon payment
Prefix rigidity, exact second moment, balanced-core reduction, normalized ledger	Saturation, moving-root BC, Q $\Rightarrow$ SP, order-32 rank, rational-atom extraction and coherence, owner localization, and spread-core correction incidence

The paper uses the closed integer-ball convention of CAP25 v13.2. An adjacent certificate consists of the imported unsafe inequality at agreement a_0 and a proved upper inequality at $a_0 + 1$. Structural reductions, asymptotic estimates, and route-cut boundaries are not called adjacent certificates unless they fit the literal row budget.

2. Main theorem package

Theorem 2.1 (Integrated theorem package)**.** *The following statements hold.*

(i) *The exact support, syndrome–secant, fixed-union, affine-span, generalized-weight, common-zero, codimension-one, support-flag, minimum-distance, puncturing, directional-Johnson, paving, mean-overlap, packing, and explicit prime-row results of Part I are unconditional.*

(ii) *The quotient–remainder normal form, partial-occupancy disintegration, effective-image Fourier identities, and profile add-back of Part II are unconditional. Primitive Q follows from the Sidon-moment payment of Part III, and the shallow Fourier theorem proves that payment in its displayed range.*

(iii) *Saturation, the one-parameter moving-root theorem, and Q $\Rightarrow$ SP are unconditional. The finite compilers pay the displayed low-nullity, low-affine-span, common-zero, support-flag, and direction-defect ranges.*

(iv) *The rank-two kernel dictionary, critical order* 32*, global affine-block bound, dyadic pair descent, rational-profile converse, slope-degree barrier, low-complexity atom extraction, collision rigidity, atom coherence, owner localization, owner-pencil incidence bound, core-compatible abundance bound, complete correction-ray bound, and the stated proper and clone-tolerant correction-space bounds are unconditional.*

(v) *The two deployed MCA adjacent rows follow from the three explicit inputs in Theorem 46.2; the two list adjacent rows follow from the exact summed certificate in Theorem 54.2. Those remaining inputs are not asserted in this paper.*

Part 1. Exact threshold refinements beyond CAP25 v13.2

3. Exact decomposition of the mutual layer

Let $C \subseteq \mathbb{F}^D$ be linear, let a be an agreement threshold, and put $t = |D| - a$. We use the foundation's definitions of support-wise MCA-badness and of the challenge-restricted numerator $B^{\mathrm{MCA}}_{C,\Gamma}(a)$ [Cho26, Secs. 1–2].

A received pair is *column-far at agreement* a when no support of size at least a simultaneously explains its two coordinates. Let $B^{\mathrm{CA}}_{C,\Gamma}(a)$ be the maximum number of challenge slopes whose affine combination is explained on some support of size at least a, maximized over column-far pairs. Define the sparse mutual numerator

$$\sigma_{C,\Gamma}(a) = \max \#\{\gamma \in \Gamma : \gamma \text{ is MCA-bad for } (e_0, e_1)\},$$

where the maximum is over pairs with $|\operatorname{supp}(e_0) \cup \operatorname{supp}(e_1)| \le t$.

Theorem 3.1 (Exact sparsification of the mutual layer)**.** *For every linear code, every nonempty challenge set Γ, and every a,*

$$B^{\mathrm{MCA}}_{C,\Gamma}(a) = \max\{B^{\mathrm{CA}}_{C,\Gamma}(a), \sigma_{C,\Gamma}(a)\}. \tag{3.1}$$

Proof. Both terms on the right maximize over subclasses of received pairs, so the left side dominates them. Conversely fix a pair. If it is column-far, its MCA-bad and CA-bad challenge slopes coincide. Otherwise translate it by a codeword pair that explains it on one common support of size at least a. The translated pair is supported on at most t coordinates, and translation preserves, slope by slope and support by support, both affine explanation and simultaneous explanation. Its bad set is therefore bounded by $\sigma_{C,\Gamma}(a)$. □

The foundation supplies the exact-agreement reduction, support uniqueness, tangent floor, and deep-regime theorem used below [Cho26, Secs. 13–14]. We invoke those interfaces only by citation.

4. Syndrome–secant geometry and finite ray compilers

The *syndrome* of a word is the result of applying a parity-check matrix; it vanishes exactly on codewords. Thus syndrome space forgets the part of a received word already lying in the code and retains only its error class. This description separates possible error supports from distinct line parameters. Put $R = n - k$, the redundancy (and syndrome-space dimension), and choose the generalized Vandermonde parity-check matrix H with columns

$$h_x = \lambda_x (1, x, \ldots, x^{R-1})^{\mathsf{T}}, \qquad \lambda_x = \left(\prod_{y \in D \setminus \{x\}} (x - y) \right)^{-1}. \tag{4.1}$$

Lagrange interpolation gives $\sum_{x \in D} \lambda_x x^j = 0$ for $0 \le j \le n-2$, so this matrix annihilates every degree-less-than-k evaluation. Every set of at most R columns is independent. For $E \subseteq D$, write $V_E = \operatorname{Span}\{h_x : x \in E\}$; it is exactly the set of syndromes of words supported on E. We write $s(u) = Hu^{\mathsf{T}}$ for the syndrome of u.

Proposition 4.1 (Syndrome-line normal form)**.** *Let $S = D \setminus E$. A word u is explained on S if and only if $s(u) \in V_E$. Consequently γ is MCA-bad on S for a pair (u_0, u_1) if and only if*

$$s(u_0) + \gamma s(u_1) \in V_E, \qquad \{s(u_0), s(u_1)\} \not\subseteq V_E. \tag{4.2}$$

For fixed E, there is at most one such finite slope.

Proof. If $u = c + e$, where $c \in C$ and e is supported on E, then $s(u) = s(e) \in V_E$. Conversely, if $s(u) = \sum_{x \in E} e_x h_x$, subtract the word supported on E with values e_x; the result has zero syndrome and belongs to C. Apply this equivalence to the line point and to both coordinates. Passing (4.2) to $\mathbb{F}^R/V_E$ shows that two distinct solutions would put both projected syndromes at zero, contradicting badness. $\square$

Since $\lambda_x \neq 0$, one has $[h_x] = [1 : x : \cdots : x^{R-1}]$; these are the points of the rational normal curve in projective syndrome space, while the weights λ_x merely choose vector representatives. A *secant span* is the linear span V_E of a selected set of those points. We call the meeting of a syndrome line with V_E *transverse* when the entire line is not contained in V_E, equivalently when its two generators are not both in V_E. The next result is a *compiler* in the sense that it translates the bad-slope problem, exactly and in both directions, into this line–secant incidence problem. A *ray* means a one-dimensional direction in syndrome or coefficient space; different raw witnesses may represent the same ray and hence the same slope.

Theorem 4.2 (Exact syndrome–secant compiler)**.** *Put $t = n - a$, and for a syndrome line $\ell_{y_0,y_1} = \{y_0 + \gamma y_1 : \gamma \in \mathbb{F}\}$ define*

$$\Theta_t(y_0, y_1) = |\{\gamma \in \mathbb{F} : \exists E \subseteq D,\ |E| \leq t, y_0 + \gamma y_1 \in V_E,\ \{y_0, y_1\} \not\subseteq V_E\}|. \tag{4.3}$$

Then

$$B_C^{\mathrm{MCA}}(a) = \max_{y_0, y_1 \in \mathbb{F}^{n-k}} \Theta_t(y_0, y_1). \tag{4.4}$$

For each fixed E, the affine line has at most one transverse intersection parameter with V_E. For a Reed–Solomon code the spaces V_E are the spans of at most t points of the weighted rational normal curve in (4.1)*. Thus the higher-dimensional ray problem is exactly a transverse line–secant incidence problem, with repeated intersection parameters deduplicated.*

Proof. Apply Proposition 4.1 with $S = D \setminus E$, and take the union over $|E| \leq t$. The syndrome map is surjective, so maximizing over received pairs is the same as maximizing over (y_0, y_1). In the quotient $\mathbb{F}^{n-k}/V_E$, a transverse intersection solves $\overline{y}_0 + \gamma \overline{y}_1 = 0$ with the two projected vectors not both zero, and therefore has at most one finite solution. The last assertion is the Vandermonde description of the parity columns. $\square$

A *parity-column circuit* is a minimally linearly dependent set of columns of H. The next lemma says that many distinct transverse rays cannot all be supported inside an independent union of columns.

Lemma 4.3 (Independent-union ray bound)**.** *Let every set of at most R parity columns be independent, put $t < R$, and fix a syndrome line. For each slope in a transverse set Z, choose an error set E_γ, $|E_\gamma| \leq t$, witnessing* (4.3)*. If $U = \bigcup_{\gamma \in Z} E_\gamma$ has size at most R, then $|Z| \leq t + 1$. Consequently, more than $t+1$ transverse slopes force the witness union to contain a parity-column circuit.*

Proof. If $|Z| \leq 1$, there is nothing to prove; assume that Z contains two distinct slopes. Two distinct line points lie in V_U, hence $y_0, y_1 \in V_U$. Column independence gives unique coefficient lifts $e_0, e_1 \in \mathbb{F}^U$, and every chosen lift is $e_\gamma = e_0 + \gamma e_1$. Let U' be the set of s coordinates where the pair (e_0, e_1) is nonzero. If $s \leq t$ and E_γ contained all of U', then $y_0, y_1 \in V_{U'} \subseteq V_{E_\gamma}$, contradicting the transversality of this particular witness. Hence some $x \in U' \setminus E_\gamma$ exists, and uniqueness of the lift gives $e_0(x) + \gamma e_1(x) = 0$. Each nonzero affine coordinate function has at most one root, so $|Z| \leq s \leq t$. If $s > t$, every chosen lift of weight at most t has at least $s - t$ zero coordinates. Double counting the pairs (γ, x) with $e_0(x) + \gamma e_1(x) = 0$ gives $|Z|(s - t) \leq s$, whence $|Z| \leq \lfloor s/(s - t) \rfloor \leq t + 1$. If $|U| \leq R$, independence holds; the contrapositive proves the circuit assertion. $\square$

Theorem 4.4 (Bounded residual-kernel ray compiler)**.** *Let H be a parity-check matrix over $\mathbb{F}$, let U be a set of columns, and put*

$$\kappa(U) = \dim_{\mathbb{F}} \ker(H_U : \mathbb{F}^U \longrightarrow \mathbb{F}^{n-k}).$$

Fix a syndrome line $y_0+\gamma y_1$, an integer $t\ge 0$, and a set Z of finite transverse slopes. Suppose that for every $\gamma\in Z$ there is $E_\gamma\subseteq U$, $|E_\gamma|\le t$, such that

$$y_0+\gamma y_1\in V_{E_\gamma},\qquad \{y_0,y_1\}\not\subseteq V_{E_\gamma}.$$

Then

$$|Z|\le (t+1)|\mathbb{F}|^{\kappa(U)}. \tag{RC$_{\mathrm{ker}}$}$$

For a Reed–Solomon parity check of redundancy $R=n-k$, $\kappa(U)=\max\{0,|U|-R\}$. Hence a certified residual chart with $(|U|-R)_+\log|\mathbb{F}|=o(n)$ has a direct subexponential ray payment with the displayed exact finite loss.

Proof. The assertion is immediate when $|Z|\le 1$. Otherwise two line points belong to V_U, so $y_0,y_1\in V_U$. Choose lifts $b_0,b_1\in\mathbb{F}^U$ with $H_Ub_i=y_i$. For every γ, choose a lift c_γ supported in E_γ and put $z_\gamma=c_\gamma-b_0-\gamma b_1\in\ker H_U$.

Fix $z\in\ker H_U$, and consider the slopes for which $z_\gamma=z$. Put

$$U_z=\{x\in U:(b_0(x)+z(x),b_1(x))\ne(0,0)\},\qquad s=|U_z|.$$

Every active coordinate is a nonzero affine function of γ, and therefore vanishes for at most one slope. Hence the total number of active zero incidences in this kernel class is at most s. If $s>t$, the weight bound on c_γ forces at least $s-t$ active zeros, so

$$|Z_z|(s-t)\le s,\qquad |Z_z|\le\lfloor s/(s-t)\rfloor\le t+1.$$

If $s\le t$, transversality forces at least one active zero: without one, $U_z=\operatorname{supp}c_\gamma\subseteq E_\gamma$, while both b_0+z and b_1 are supported in U_z, putting y_0,y_1 in V_{E_γ}. Thus this kernel class contains at most $s\le t$ slopes. Summing over the $|\mathbb{F}|^{\kappa(U)}$ kernel elements proves the bound. The MDS column property gives the Reed–Solomon formula for $\kappa(U)$. $\square$

Theorem 4.5 (Multiplicity-amplified fixed-union ray compiler)**.** *Let H be a Reed–Solomon parity check of redundancy R, let $U\subseteq D$ have size $N=R+\nu$ with $\nu\ge1$, and put $C_U=\ker H_U$. Fix $t<R$, a syndrome line $y_0+\gamma y_1$, and a set Z of distinct finite transverse slopes. Suppose that for every $\gamma\in Z$ there is $e_\gamma\in\mathbb{F}^U$ such that*

$$H_Ue_\gamma=y_0+\gamma y_1,\qquad \mathrm{wt}(e_\gamma)\le t,\qquad \{y_0,y_1\}\not\subseteq V_{\operatorname{supp}e_\gamma}.$$

Then

$$|Z|\le\left\lfloor\frac{\binom{R+\nu}{\nu+1}}{R-t}\right\rfloor. \tag{4.5}$$

The estimate is independent of $|\mathbb{F}|$; it replaces the factor $|\mathbb{F}|^\nu$ in Theorem 4.4 by an exact incidence multiplicity.

Proof. There is nothing to prove when $|Z|\le1$. Otherwise two line points lie in V_U, hence $y_0,y_1\in V_U$. Choose lifts $b_0,b_1\in\mathbb{F}^U$, and choose a basis of the ν-dimensional kernel C_U. Write its generator matrix as K, with row K_x at $x\in U$. Every chosen error has a unique expression

$$e_\gamma=b_0+\gamma b_1+Kc_\gamma,\qquad c_\gamma\in\mathbb{F}^\nu.$$

Put $p_\gamma=(\gamma,c_\gamma)\in\mathbb{F}^{\nu+1}$. A zero coordinate of e_γ is the affine hyperplane incidence

$$b_0(x)+\gamma b_1(x)+K_xc_\gamma=0,\qquad x\in U, \tag{4.6}$$

whose normal is $n_x=(b_1(x),K_x)\in\mathbb{F}^{\nu+1}$. Since $\mathrm{wt}(e_\gamma)\le t$, the point p_γ lies on at least $h=N-t$ of these hyperplanes.

We claim that the normals incident with p_γ span $\mathbb{F}^{\nu+1}$. Otherwise a nonzero pair (δ,v) is orthogonal to all of them. Then

$$q=\delta b_1+Kv$$

vanishes at every zero coordinate of e_γ, so $\operatorname{supp}q\subseteq\operatorname{supp}e_\gamma$ and $\mathrm{wt}(q)\le t$. If $\delta=0$, then $q=Kv\ne0$ is a kernel word of weight at most $t<R+1$, contradicting the MDS distance of $C_U=[R+\nu,\nu,R+1]$. If $\delta\ne0$, divide by δ. The word q/δ represents y_1, and $e_\gamma-\gamma q/\delta$ represents y_0; both are supported inside $\operatorname{supp}e_\gamma$, contradicting transversality.

Every normal n_x is nonzero. Indeed, if the coordinate functional of C_U vanished at x, then puncturing at x would leave kernel dimension ν, whereas the MDS rank of $H_{U\setminus\{x\}}$ makes that dimension $\nu-1$. Thus the incident normals form a loopless rank-$(\nu+1)$ matroid on at least h elements. Fix one basis. Each of the other incident elements enters a basis by a fundamental-circuit exchange, producing a distinct basis. Hence p_γ is incident with at least

$$h-(\nu+1)+1 = R-t$$

independent $(\nu+1)$-subsets of hyperplanes. An independent subset has at most one affine intersection point. Counting pairs consisting of a slope point and one of its incident bases gives

$$|Z|(R-t) \le \binom{R+\nu}{\nu+1},$$

which is (4.5). □

Corollary 4.6 (Single MDS-circuit ray compiler, multiplicity form)**.** *Under the hypotheses of the former single-circuit chart, namely* $|U| = R+1$*,*

$$|Z| \le \left\lfloor \frac{R(R+1)}{2(R-t)} \right\rfloor. \tag{4.7}$$

Thus the pair-charge $\binom{R+1}{2}$ *is divided by the exact number* $R-t$ *of independent zero-coordinate pairs forced by every transverse slope.*

Proof. Apply Theorem 4.5 with $\nu = 1$. □

Theorem 4.7 (Fixed-syndrome GRS–Johnson compiler)**.** *Let* H *be a Reed–Solomon parity check of redundancy* R*, let* $U \subseteq D$ *have size* $N = R+\nu$*, and fix a syndrome* y*. Put*

$$\mathcal{L}_U(y,t) = \{e \in \mathbb{F}^U : H_U e = y,\ \operatorname{wt}(e) \le t\}, \qquad h = N-t,$$

where $t < R$*. If* $\nu = 0$*, then* $|\mathcal{L}_U(y,t)| \le 1$*. If* $\nu \ge 1$ *and*

$$h^2 > N(\nu-1), \tag{4.8}$$

then

$$|\mathcal{L}_U(y,t)| \le \left\lfloor \frac{N(h-\nu+1)}{h^2 - N(\nu-1)} \right\rfloor. \tag{4.9}$$

Equivalently, the same bound holds for the number of codewords in one Reed–Solomon list whose error supports are all contained in the fixed union U*.*

Proof. For $\nu = 0$, the columns in U are independent, so a syndrome has at most one lift. Assume $\nu \ge 1$, choose one lift b of y, and put $C_U = \ker H_U$. This is an $[N, \nu, R+1]$ generalized Reed–Solomon code. The set $\mathcal{L}_U(y,t)$ is the low-weight part of the affine coset $b + C_U$. Thus its members are in bijection with codewords of C_U that agree with the fixed word $-b$ on at least $h = N-t$ coordinates.

For each such codeword choose an h-subset A_i of its agreement set. Distinct codewords of the $[N,\nu]$ GRS code agree on at most $\nu-1$ coordinates, so $|A_i \cap A_j| \le \nu-1$. If there are L sets, and d_x is the number containing x, then

$$\sum_x d_x = Lh, \qquad \sum_x \binom{d_x}{2} \le \binom{L}{2}(\nu-1).$$

Cauchy–Schwarz gives

$$\sum_x \binom{d_x}{2} \ge \frac{1}{2}\left(\frac{L^2h^2}{N} - Lh\right).$$

Combining the two inequalities and using (4.8) yields

$$L(h^2 - N(\nu-1)) \le N(h-\nu+1),$$

which proves (4.9). □

Theorem 4.8 (Recursive affine-span Reed–Solomon list compiler)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, K]$ have length n, let $u \in \mathbb{F}^D$, and put $m = K + w \le n$. Let $\mathcal{A} = c_0 + C' \subseteq C$ be an affine subspace of codewords whose direction space has dimension $s \ge 0$. Then*

$$|\mathcal{L}_{\mathcal{A}}(u, m)| \le \left\lfloor \frac{(n - K + s)^{\underline{s}}}{(w + 1)^{\overline{s}}} \right\rfloor = \left\lfloor \frac{\binom{n-K+s}{s}}{\binom{w+s}{s}} \right\rfloor. \tag{4.10}$$

The empty products at $s = 0$ are one. In particular, a one-dimensional codeword pencil contains at most $\lfloor (n - K + 1)/(w + 1) \rfloor$ members of the list.

Proof. We induct on s. The case $s = 0$ is immediate. Assume $s \ge 1$, and let

$$G = \{x \in D : c(x) = 0 \text{ for every } c \in C'\}, \qquad z = |G|.$$

The polynomial space representing C' has dimension s. Division by the locator of G embeds it into the polynomials of degree less than $K - z$, so

$$z \le K - s. \tag{4.11}$$

Let $g \le z$ be the number of coordinates in G at which the common value of the affine flat agrees with u. Every member of $\mathcal{L}_{\mathcal{A}}(u, m)$ therefore has at least $m - g$ agreements outside G.

For $x \in D \setminus G$, the evaluation functional on C' is nonzero. Hence

$$\mathcal{A}_x = \{c \in \mathcal{A} : c(x) = u(x)\}$$

is either empty or an affine subspace of dimension $s - 1$. By the induction hypothesis, the number of listed codewords in $\mathcal{A}_x$ is at most

$$B_{s-1} := \frac{\binom{n-K+s-1}{s-1}}{\binom{w+s-1}{s-1}}.$$

Double-counting pairs (c, x) with $c \in \mathcal{L}_{\mathcal{A}}(u, m)$, $x \notin G$, and $c(x) = u(x)$ gives

$$|\mathcal{L}_{\mathcal{A}}(u, m)|(m - g) \le (n - z) B_{s-1}. \tag{4.12}$$

Put $b = z - g$ and $q = K - s - z$. Both are nonnegative by (4.11), and the exact identity

$$\begin{aligned} &(m - g)(n - K + s) - (n - z)(w + s) \\ &\quad = b(n - K + s) + q(n - m) \ge 0 \end{aligned} \tag{4.13}$$

shows that

$$\frac{n - z}{m - g} \le \frac{n - K + s}{w + s}.$$

Combining this with (4.12) yields

$$|\mathcal{L}_{\mathcal{A}}(u, m)| \le B_{s-1} \frac{n - K + s}{w + s} = \frac{\binom{n-K+s}{s}}{\binom{w+s}{s}}.$$

Taking the integer floor proves (4.10). □

Theorem 4.9 (Generalized-weight rank-flat list compiler)**.** *Retain the hypotheses of Theorem 4.8, assume $s \ge 1$, and let*

$$d_j = d_j(C') := \min_{\substack{V \le C' \\ \dim V = j}} |\operatorname{supp}(V)|, \qquad 1 \le j \le s,$$

be the generalized Hamming weights of the direction code. Put

$$R = n - K, \qquad t = n - m,$$

let G be the common-zero set of C', let $z = |G| = n - d_s$, let g be the number of common agreements with u on G, and put $b = z - g$. Then

$$|\mathcal{L}_{\mathcal{A}}(u, m)| \le \left\lfloor \frac{d_s^{\underline{s}}}{\prod_{j=1}^{s} (d_j - t + b)} \right\rfloor. \tag{4.14}$$

Every factor in the denominator is positive. More precisely,

$$d_j \ge R + j, \qquad d_j - t + b \ge w + j + b. \tag{4.15}$$

Proof. Choose a basis of C' and identify the affine flat with $\mathbb{F}^s$. For each active coordinate $x \in D \setminus G$, agreement with u is an affine hyperplane whose nonzero normal is the evaluation functional $\ell_x \in (C')^*$. Every listed point is incident with at least

$$h = m - g = d_s - t + b$$

of these hyperplanes.

Suppose that r independent incident normals have already been chosen and span $W \le (C')^*$. The coordinates whose normals lie in W are common zeros of the $(s-r)$-dimensional subcode $W^\perp$. This subcode has support at least d_{s-r}; after the common-zero set G is removed, at most

$$(n - d_{s-r}) - z = d_s - d_{s-r}$$

active normals lie in W. Hence there are at least

$$h - (d_s - d_{s-r}) = d_{s-r} - t + b$$

choices for the next independent incident normal. Each listed point is therefore incident with at least

$$\prod_{j=1}^{s} (d_j - t + b)$$

ordered bases of coordinate hyperplanes. There are at most $d_s^{\underline{s}}$ ordered coordinate bases globally, and an independent affine-hyperplane basis has at most one common point. Double counting proves (4.14).

Finally, if $V \le C'$ has dimension j and common-zero set of size z_V, division by its locator embeds the representing polynomial space into the polynomials of degree less than $K - z_V$. Thus $j \le K - z_V$, so $|\operatorname{supp}(V)| = n - z_V \ge n - K + j = R + j$. This proves (4.15) and positivity. $\square$

Corollary 4.10 (Over-budget rank-flat concentration). *In Theorem 4.9, let $s \ge 2$ and let B be a nonnegative integer. Define*

$$\Theta_{s,B}(d_s, b) = \max\left\{ q \in \mathbb{Z}_{\ge 0} : q^{s-1} \le \left\lfloor \frac{d_s^{\underline{s}}}{(B+1)(d_s - t + b)} \right\rfloor \right\}. \tag{4.16}$$

If $|\mathcal{L}_{\mathcal{A}}(u, m)| > B$, then some proper direction subcode has dimension $1 \le j < s$ and

$$d_j \le t - b + \Theta_{s,B}(d_s, b). \tag{4.17}$$

A row-independent coarse version is obtained by replacing $d_s^{\underline{s}}$ by $n^{\underline{s}}$ and $d_s - t + b$ by $w + s$.

Proof. If the list contains at least $B+1$ points, (4.14) implies

$$\prod_{j=1}^{s-1} (d_j - t + b) \le \left\lfloor \frac{d_s^{\underline{s}}}{(B+1)(d_s - t + b)} \right\rfloor.$$

At least one of the $s-1$ integer factors is at most the integer root in (4.16), proving (4.17). The coarse form follows from $d_s \le n$ and (4.15). $\square$

Theorem 4.11 (Common-zero Johnson list compiler). *In the notation of Theorem 4.9, put*

$$N = d_s, \qquad h = m - g = d_s - t + b, \qquad \alpha = d_s - d_1.$$

If

$$h^2 > N\alpha, \tag{4.18}$$

then

$$|\mathcal{L}_{\mathcal{A}}(u, m)| \le \left\lfloor \frac{N(h - \alpha)}{h^2 - N\alpha} \right\rfloor. \tag{4.19}$$

Equivalently, write $z = K - \nu$. Then $N = R + \nu$, $h = w + \nu + b$, and the generalized Singleton bound gives the universal specialization

$$|\mathcal{L}_{\mathcal{A}}(u, m)| \le \left\lfloor \frac{(R+\nu)(w+b+1)}{(w+\nu+b)^2 - (R+\nu)(\nu-1)} \right\rfloor \tag{4.20}$$

whenever the displayed denominator is positive.

Proof. All codewords of the affine flat have the same values on G. Delete those coordinates and, for each listed codeword, choose an h-subset of its active agreement set. If two distinct codewords are chosen, their difference is a nonzero word of C', has weight at least d_1, and therefore vanishes on at most $N - d_1 = \alpha$ active coordinates. The standard Johnson incidence count now gives

$$L(h^2 - N\alpha) \le N(h - \alpha),$$

which proves (4.19).

If $z = K - \nu$, then $N = n - z = R + \nu$ and $h = m - (z - b) = w + \nu + b$. By (4.15), $d_1 \ge R + 1$, so $\alpha \le \nu - 1$. The Johnson fraction is nondecreasing in α while its denominator is positive. Replacing α by $\nu - 1$ gives (4.20). □

5. Coset-free codimension-one rank-flat recursion

The rank-flat concentration theorem produces a low-support direction subcode, but a naive partition into its affine cosets introduces the field size. The next theorem removes that loss when the support closure has codimension one. It counts the whole affine chart at once by two disjoint resources: bases using an agreement in the new support layer, and bases whose final coordinate lies in the old support.

Theorem 5.1 (Two-resource codimension-one recursion)**.** *Let*

$$C = \mathrm{RS}[\mathbb{F}, D, K], \qquad n = |D|, \qquad m = K + w, \qquad t = n - m,$$

and let $\mathcal{A} = c_0 + C' \subseteq C$ be an affine codeword flat with $\dim C' = j + 1$, where $j \ge 1$. Let $V < C'$ be a hyperplane, put

$$U = \mathrm{supp}(V), \qquad S = \mathrm{supp}(C'), \qquad T = S \setminus U, \qquad H = D \setminus S,$$

and assume $e := |T| \ge 1$. Write $d = |U|$. Then

$$V = \{v \in C' : \mathrm{supp}(v) \subseteq U\}; \tag{5.1}$$

in particular every word of $C' \setminus V$ is nonzero at every coordinate of T.

Fix a received word $u \in \mathbb{F}^D$. Let b_0 be the number of coordinates in H on which the common value of $\mathcal{A}$ disagrees with u. For $0 \le b \le e$, let L_b count the members $c \in \mathcal{A}$ satisfying

$$\mathrm{agr}(u, c) \ge m, \qquad |\{x \in T : c(x) \ne u(x)\}| = b.$$

Let $f_i = d_i(V)$, $1 \le i \le j$, be the generalized Hamming weights of V, and put

$$P_b = \prod_{i=1}^{j} (f_i - t + b_0 + b), \qquad Q = w + 1 + b_0. \tag{5.2}$$

Then every factor in P_b is positive and

$$\sum_{b=0}^{e} L_b (e - b) P_b \le e\, d^{\underline{j}}, \tag{5.3}$$

$$\sum_{b=0}^{e} L_b (Q - e + b)_+ P_b \le d^{\underline{j+1}}. \tag{5.4}$$

Moreover, with $L = \sum_b L_b$,

$$L \le \left\lfloor \frac{(n - j) d^{\underline{j}}}{Q P_0} \right\rfloor. \tag{5.5}$$

More generally, if $x, y \ge 0$ satisfy

$$x(e - b)P_b + y(Q - e + b)_+ P_b \ge 1 \qquad (0 \le b \le e), \tag{5.6}$$

then

$$L \le \left\lfloor x e\, d^{\underline{j}} + y\, d^{\underline{j+1}} \right\rfloor. \tag{5.7}$$

Thus the affine cosets of V are never summed separately.

Proof. For $x \in T$, every word of V vanishes at x, whereas some word of C' is nonzero there. Evaluation at x therefore induces a nonzero functional on the one-dimensional quotient C'/V. It is nonzero on every nonzero quotient class, proving (5.1).

Parameterize $\mathcal{A}$ by the direction space C'. Agreement at a coordinate x is an affine hyperplane with normal $\ell_x \in (C')^*$. A listed point counted by L_b has at least

$$m - (|H| - b_0) - (e - b) = d - t + b_0 + b$$

agreements in U. Restrict the corresponding normals to V. After r independent restricted normals have been selected, their span annihilates a $(j-r)$-dimensional subcode of V, whose support has size at least f_{j-r}. Hence at most $d - f_{j-r}$ normals of U lie in the old span, and there are at least

$$f_{j-r} - t + b_0 + b$$

choices for the next independent normal. Every listed point counted by L_b is therefore incident with at least P_b ordered V-normal bases. The generalized Singleton inequalities $f_i \geq n - K + i$ show at the same time that all factors in P_b are at least $w + i + b_0 + b$.

The span in $(C')^*$ of a lifted V-normal basis meets the one-dimensional annihilator $V^\perp$ trivially. Every agreeing coordinate of T has a nonzero normal in $V^\perp$, and therefore extends the basis. A point in class b supplies at least $(e-b)P_b$ ordered bases of this type. Globally there are at most $e\, d^{\underline{j}}$ ordered choices, and an independent affine hyperplane basis cuts out at most one point. This proves (5.3).

Fix now one lifted V-normal basis and let $h \in C' \setminus V$ generate its one-dimensional annihilator. A coordinate normal extends the basis exactly when h is nonzero at that coordinate. The nonzero Reed–Solomon word h has at most $K-1$ zeros. All coordinates of H are zeros of h, and exactly $|H| - b_0$ of them are agreements. Thus among the at least m agreement coordinates of the listed point, at most $K - 1 - b_0$ fail to extend the basis. There are consequently at least

$$m - (K - 1 - b_0) = Q$$

extending agreement coordinates. All $e - b$ agreeing coordinates of T already extend, so at least $(Q - e + b)_+$ extending agreements lie in U. Double-counting the resulting ordered $(j+1)$-tuples inside U proves (5.4).

The same argument without separating U from T gives at least $QP_b \geq QP_0$ full bases through every listed point. There are at most $(n-j)d^{\underline{j}}$ ordered choices of a V-basis and one additional coordinate, proving (5.5). Finally multiply (5.3) by x, multiply (5.4) by y, and use (5.6); this gives (5.7). □

Lemma 5.2 (Profile interpolation)**.** *Let*

$$P(z) = \prod_{i=1}^{j} (a_i + z), \qquad a_i > 0,$$

and let $e, Q > 0$. For every integer $0 \leq b \leq e$,

$$\frac{(e-b)P(b)}{eP(0)} + \frac{(Q-e+b)_+ P(b)}{QP(e)} \geq 1 \tag{5.8}$$

under either of the following conditions:

(a) *$e \leq Q$;*
(b) *$e > Q$ and*

$$\left(\frac{P(e)}{P(0)}\right)^{1-Q/e} \geq \frac{e}{Q}. \tag{5.9}$$

If $j \geq 2$, $A = \max_i a_i$, and $(j-1)Q \geq A$, then (5.9) holds for every $e \geq Q$.

Proof. Put $r = P(e)/P(0) \geq 1$. Since $\log P$ is concave, with $\theta = b/e$ one has

$$\frac{P(b)}{P(0)} \geq r^{\theta}, \qquad \frac{P(b)}{P(e)} \geq r^{-(1-\theta)}.$$

If $e \le Q$, then $(Q-e+b)/Q \ge \theta$, and weighted AM–GM gives

$$(1-\theta)r^{\theta} + \theta r^{-(1-\theta)} \ge 1.$$

This proves (5.8) in case (a).

Assume $e > Q$, put $c = Q/e$, and first take $0 \le \theta \le 1-c$. The second summand in (5.8) vanishes, while concavity gives the lower bound $(1-\theta)r^{\theta}$. The logarithm of this expression is concave in θ; it is zero at 0, and at $1-c$ it is nonnegative exactly when (5.9) holds.

For $1-c \le \theta \le 1$, write $\alpha = 1-\theta \in [0,c]$. After multiplication by r^{α}, the lower bound supplied by log-concavity is

$$\alpha r + 1 - \frac{\alpha}{c}.$$

Convexity of r^{α} on $[0,c]$ gives

$$r^{\alpha} \le 1 + \frac{\alpha}{c}(r^{c}-1).$$

The former expression dominates the latter precisely when $cr \ge r^{c}$, which is again (5.9). This proves case (b).

Finally, if $A = \max_i a_i$, then

$$r \ge (1+e/A)^{j}.$$

For $e \ge Q$, set

$$\phi(e) = j(1-Q/e)\log(1+e/A) - \log(e/Q).$$

One has $\phi(Q) = 0$, and the elementary inequality $\log(1+x) \ge x/(1+x)$ gives

$$\phi'(e) \ge \frac{j}{A+e} - \frac{1}{e} = \frac{(j-1)e-A}{e(A+e)} \ge 0.$$

Thus $\phi(e) \ge 0$, which is (5.9). □

Corollary 5.3 (MDS-soft codimension-one compiler)**.** *Retain the hypotheses of Theorem 5.1, and put*

$$\Pi_b = (d-t+b_0+b)\prod_{i=1}^{j-1}(w+i+b_0+b). \tag{5.10}$$

Suppose the profile interpolation inequality (5.8) holds for $P = \Pi$, with $Q = w+1+b_0$. Then

$$L \le \left\lfloor \frac{d^{j}}{\Pi_0} + \frac{d^{j+1}}{Q\Pi_e} \right\rfloor. \tag{5.11}$$

Independently of profile interpolation,

$$L \le \left\lfloor \frac{(n-j)d^{j}}{Q\Pi_0} \right\rfloor. \tag{5.12}$$

In particular, (5.11) is a bound for the whole affine chart, not a per-coset estimate.

Proof. The generalized Singleton inequalities give $f_i \ge n-K+i$ for $i<j$, while $f_j = d$. Hence $P_b \ge \Pi_b$. The two resource inequalities of Theorem 5.1 remain true after replacing P_b by Π_b. Under profile interpolation, the choices

$$x = \frac{1}{e\Pi_0}, \qquad y = \frac{1}{Q\Pi_e}$$

are feasible in (5.6), and (5.7) gives (5.11). The all-extender estimate follows directly from (5.5). □

6. Higher-codimension support-flag recursion

The codimension-one theorem removes the field-size loss for a single support-saturated hyperplane. The same principle extends to arbitrary codimension, but one must keep all possible choices of "new-layer" and "old-support" extenders. The resulting object is a finite exact linear program with one resource inequality for each binary flag pattern. In particular, no affine coset is ever counted separately.

Definition 6.1 (Support saturation). *If $V \le W \le C$, the* support closure of V in W *is*

$$\overline{V}^W = \{c \in W : \operatorname{supp}(c) \subseteq \operatorname{supp}(V)\}.$$

We call V support-saturated in W *when $\overline{V}^W = V$.*

Lemma 6.2 (Generalized-weight minimizers are support-saturated). *Let W be a finite-length linear code, and let $V \le W$ have dimension j and support size $d_j(W)$. Then V is support-saturated in W. More generally, the generalized Hamming weights satisfy*

$$d_1(W) < d_2(W) < \cdots < d_{\dim W}(W).$$

Proof. Let $Y \le W$ have dimension $i+1$ and support size $d_{i+1}(W)$. Choose a coordinate on which some word of Y is nonzero. The kernel of evaluation at that coordinate is an i-dimensional subcode whose support omits that coordinate, so

$$d_i(W) \le d_{i+1}(W) - 1.$$

This proves strict increase. If the support closure of a j-dimensional minimizer V had dimension $k > j$, then

$$d_k(W) \le |\operatorname{supp}(V)| = d_j(W),$$

contradicting strict increase. □

Lemma 6.3 (Existence of a support-saturated flag). *Let V be support-saturated in W and put $r = \dim W - \dim V$. There is a flag*

$$V = V_0 < V_1 < \cdots < V_r = W, \qquad \dim V_\ell = \dim V + \ell,$$

such that $V_{\ell-1}$ is support-saturated in V_ℓ for every $1 \le \ell \le r$. Consequently, with

$$U_\ell = \operatorname{supp}(V_\ell), \qquad T_\ell = U_\ell \setminus U_{\ell-1},$$

every layer T_ℓ is nonempty.

Proof. Construct the flag downward. Suppose $V < X \le W$. Since V is support-saturated in W, some word of $X \setminus V$ is nonzero at a coordinate $x \notin \operatorname{supp}(V)$. Put

$$X' = \{c \in X : c(x) = 0\}.$$

Then $V \le X'$, X' is a hyperplane of X, and $x \notin \operatorname{supp}(X')$. If $c \in X$ has support contained in $\operatorname{supp}(X')$, then $c(x) = 0$, hence $c \in X'$. Thus X' is support-saturated in X. Repeating until the dimension reaches $\dim V$ gives the claimed flag; the terminal subspace contains V and has the same dimension, so it is V. A proper support-saturated inclusion cannot have equal supports, proving $T_\ell \ne \varnothing$. □

Theorem 6.4 (Exact support-flag resource recursion). *Let*

$$C = \mathrm{RS}[\mathbb{F}, D, K], \qquad n = |D|, \qquad m = K + w, \qquad t = n - m,$$

and let $\mathcal{A} = c_0 + C' \subseteq C$ be an affine codeword flat. Assume that

$$V_0 < V_1 < \cdots < V_r = C'$$

is a support-saturated flag with

$$\dim V_0 = j \ge 1, \qquad \dim V_\ell = j + \ell.$$

Put

$$U_\ell = \operatorname{supp}(V_\ell), \quad d_\ell = |U_\ell|, \quad T_\ell = U_\ell \setminus U_{\ell-1}, \quad e_\ell = |T_\ell|,$$

and $H = D \setminus U_r$*. Fix a received word* $u \in \mathbb{F}^D$*, and let* b_0 *be the number of coordinates of* H *on which the common value of* $\mathcal{A}$ *disagrees with* u*.*

For a profile $\mathbf{b} = (b_1, \ldots, b_r)$ *with* $0 \le b_\ell \le e_\ell$*, let* $L_{\mathbf{b}}$ *count the words* $c \in \mathcal{A}$ *satisfying*

$$\operatorname{agr}(u,c) \ge m, \qquad |\{x \in T_\ell : c(x) \ne u(x)\}| = b_\ell \quad (1 \le \ell \le r).$$

Let $f_i = d_i(V_0)$*,* $1 \le i \le j$*, and define*

$$P_{\mathbf{b}} = \prod_{i=1}^{j} \left(f_i - t + b_0 + \sum_{q=1}^{r} b_q \right), \tag{6.1}$$

$$Q_\ell(\mathbf{b}) = w + 1 + b_0 + \sum_{q>\ell} b_q, \tag{6.2}$$

$$A_\ell(\mathbf{b}) = e_\ell - b_\ell, \qquad B_\ell(\mathbf{b}) = \big(Q_\ell(\mathbf{b}) - e_\ell + b_\ell\big)_+. \tag{6.3}$$

For a binary pattern $\sigma \in \{T, U\}^r$*, put*

$$C_\sigma(\mathbf{b}) = P_{\mathbf{b}} \prod_{\ell:\sigma_\ell = T} A_\ell(\mathbf{b}) \prod_{\ell:\sigma_\ell = U} B_\ell(\mathbf{b}), \tag{6.4}$$

$$G_\sigma = d_0^j \prod_{\ell:\sigma_\ell = T} e_\ell \prod_{\ell:\sigma_\ell = U} (d_{\ell-1} - (j + \ell - 1)). \tag{6.5}$$

Then every factor in $P_{\mathbf{b}}$ *is positive and, for every* $\sigma \in \{T, U\}^r$*,*

$$\boxed{\sum_{\mathbf{b}} L_{\mathbf{b}} C_\sigma(\mathbf{b}) \le G_\sigma.} \tag{6.6}$$

Consequently, if nonnegative weights x_σ *satisfy*

$$\sum_{\sigma \in \{T,U\}^r} x_\sigma C_\sigma(\mathbf{b}) \ge 1 \qquad \text{for every profile } \mathbf{b}, \tag{6.7}$$

then, with $L = \sum_{\mathbf{b}} L_{\mathbf{b}}$*,*

$$\boxed{L \le \left\lfloor \sum_{\sigma \in \{T,U\}^r} x_\sigma G_\sigma \right\rfloor.} \tag{6.8}$$

The bound is for the whole affine chart and contains no factor depending on $|\mathbb{F}|$ *or on the number of affine cosets of* V_0*.*

Proof. Parameterize $\mathcal{A}$ by C'. Agreement at a coordinate x is an affine hyperplane with normal $\ell_x \in (C')^*$. A point of profile $\mathbf{b}$ has at least

$$d_0 - t + b_0 + \sum_{q=1}^{r} b_q$$

agreements in U_0. Applying the ordered-basis argument of Theorem 4.9 inside V_0 shows that it is incident with at least $P_{\mathbf{b}}$ ordered coordinate bases whose restrictions form a basis of V_0^*.

Suppose an adapted basis has already been chosen through level $V_{\ell-1}$. Its annihilator in V_ℓ is one-dimensional; let $h_\ell \in V_\ell \setminus V_{\ell-1}$ generate it. At a coordinate of T_ℓ, evaluation vanishes on $V_{\ell-1}$ but not on V_ℓ; on the one-dimensional quotient $V_\ell / V_{\ell-1}$ it is therefore a nonzero functional. Hence h_ℓ is nonzero at every coordinate of T_ℓ. Moreover, h_ℓ vanishes outside U_ℓ. The number of mismatches outside U_ℓ is

$$b_0 + \sum_{q>\ell} b_q.$$

Since a nonzero degree-$< K$ word has at most $K-1$ zeros, at most $K - 1 - b_0 - \sum_{q>\ell} b_q$ agreement coordinates fail to extend the current basis. Hence there are at least $Q_\ell(\mathbf{b})$ extending agreement coordinates in total. The $A_\ell(\mathbf{b})$ agreeing coordinates of T_ℓ all extend, so at least $B_\ell(\mathbf{b})$ extending agreement coordinates lie in $U_{\ell-1}$.

Fix a pattern σ. Multiplying the preceding lower bounds over the flag levels gives at least $C_\sigma(\mathbf{b})$ adapted ordered bases of pattern σ through every point of profile $\mathbf{b}$. Globally, the initial ordered basis has at most d_0^j choices. At a T-level there are at most e_ℓ coordinate choices. At a U-level, all $j+\ell-1$ previously selected coordinates lie in $U_{\ell-1}$, and an extending normal cannot repeat one of them; there are therefore at most $d_{\ell-1}-(j+\ell-1)$ choices. An independent affine-hyperplane basis cuts out at most one parameter point. Double counting proves (6.6). Multiplying those inequalities by x_σ, summing over σ, and using (6.7) proves (6.8). Positivity follows from $f_i \geq n-K+i$. $\square$

Corollary 6.5 (All-extender flag bound)**.** *In the notation of Theorem 6.4, put*

$$P_0 = \prod_{i=1}^{j}(f_i - t + b_0), \qquad Q = w + 1 + b_0.$$

Then

$$L \leq \boxed{\left\lfloor \frac{d_0^j \prod_{\ell=1}^{r}(d_\ell - (j+\ell-1))}{P_0 Q^r} \right\rfloor}. \tag{6.9}$$

In particular, the numerator may be replaced by $d_0^j (n-j)^{\underline{r}}$.

Proof. Every point has at least P_0 initial bases. At level ℓ it has at least $Q_\ell(\mathbf{b}) \geq Q$ extending agreement coordinates, without separating the two resources. The global number of adapted extensions at that level is at most $d_\ell - (j+\ell-1)$. Multiplication and double counting give the first bound. Since $d_\ell \leq n$, each factor is at most $n-j-\ell+1$, giving the second statement. $\square$

Lemma 6.6 (One-level interpolation defect)**.** *For $e, Q > 0$ and $\rho \geq 1$, define*

$$\eta(e,Q,\rho) = \begin{cases} 1, & e \leq Q, \\ \min\left\{1, \dfrac{Q}{e}\rho^{1-Q/e}\right\}, & e > Q. \end{cases} \tag{6.10}$$

Then, for every real $0 \leq b \leq e$,

$$\rho^{b/e}\left(\frac{e-b}{e} + \frac{(Q-e+b)_+}{Q\rho}\right) \geq \eta(e,Q,\rho). \tag{6.11}$$

Thus the right side is 1 precisely in the profile-interpolation range used in Lemma 5.2.

Proof. If $e \leq Q$, the proof is the weighted AM–GM argument of Lemma 5.2. Suppose $e > Q$, put $c = Q/e$ and $\theta = b/e$. On $0 \leq \theta \leq 1-c$, the logarithm of the left side is $\log(1-\theta) + \theta \log \rho$, a concave function, so its minimum is at an endpoint and equals at least $\min\{1, c\rho^{1-c}\}$. On $1-c \leq \theta \leq 1$, put $\alpha = 1-\theta \in [0,c]$. The left side becomes

$$\rho^{-\alpha}\left(\alpha\rho + 1 - \frac{\alpha}{c}\right).$$

Its logarithm is a linear function plus the logarithm of a positive affine function, hence is concave. Its endpoint values are again 1 and $c\rho^{1-c}$. This proves the claim. $\square$

Corollary 6.7 (Tensorized support-flag certificate)**.** *Retain the notation of Theorem 6.4. Put*

$$E = \sum_{\ell=1}^{r} e_\ell, \qquad P(z) = \prod_{i=1}^{j}(f_i - t + b_0 + z), \qquad \rho = \frac{P(E)}{P(0)},$$

and define

$$\rho_\ell = \rho^{e_\ell/E}, \qquad \eta_\ell = \eta(e_\ell, Q, \rho_\ell), \qquad Q = w + 1 + b_0.$$

Then

$$L \leq \boxed{\left\lfloor \frac{d_0^j}{P(0)\prod_{\ell=1}^{r}\eta_\ell} \prod_{\ell=1}^{r}\left(1 + \frac{d_{\ell-1} - (j+\ell-1)}{Q\rho_\ell}\right) \right\rfloor}. \tag{6.12}$$

If every $\eta_\ell = 1$, this is a closed higher-codimension analogue of (5.11). For $r = 1$ it specializes to the codimension-one two-resource certificate, with the explicit defect factor when profile interpolation fails.

Proof. Let $B = \sum b_\ell$. Concavity of $\log P$ on $[0, E]$ gives

$$\frac{P(B)}{P(0)} \geq \rho^{B/E} = \prod_{\ell=1}^{r} \rho_\ell^{b_\ell/e_\ell}.$$

Also $Q_\ell(\mathbf{b}) \geq Q$, so $B_\ell(\mathbf{b}) \geq (Q - e_\ell + b_\ell)_+$. By Lemma 6.6,

$$\frac{P_{\mathbf{b}}}{P(0)} \prod_{\ell=1}^{r} \left(\frac{A_\ell(\mathbf{b})}{e_\ell} + \frac{B_\ell(\mathbf{b})}{Q\rho_\ell} \right) \geq \prod_{\ell=1}^{r} \eta_\ell.$$

Use the tensor-product dual weights

$$x_\sigma = \frac{1}{P(0) \prod_\ell \eta_\ell} \prod_{\ell:\sigma_\ell = T} \frac{1}{e_\ell} \prod_{\ell:\sigma_\ell = U} \frac{1}{Q\rho_\ell}.$$

They satisfy (6.7); substituting them into (6.8) and factoring the sum over binary patterns gives (6.12). □

Remark 6.8 (What the flag theorem closes). The uncontrolled field-size coset sum is now removed in every codimension. For a concrete finite chart, (6.6) is an exact rational linear-program certificate, while (6.12) is a closed analytic certificate. What can still fail is numerical payment: a low-mismatch flag with several short layers may have a certified bound above the row budget. Such a flag is a specific finite residual atom, not an unproved multiplicity principle.

Proposition 6.9 (Counterexample to the affine-span transverse MCA compiler)**.** *The affine-span MCA bound formerly printed here is false even under the direction-separation hypothesis. There is a received line for*

$$C = \mathrm{RS}[\mathbb{F}_{1009}, \{0, \ldots, 99\}, 1], \qquad (n, K, m, w, s) = (100, 1, 21, 20, 1),$$

with 31 *distinct selected slopes and explanations in the one-dimensional constant code, while*

$$\max_{c \in C} \mathrm{agr}(r_1, c) = 20 < 21 = m$$

and the claimed bound was

$$\left\lfloor \max \left\{ \frac{100^2}{21 \cdot 20}, \frac{100^2}{20^2} \right\} \right\rfloor = 23 < 31.$$

Every selected maximal agreement support has size exactly 21 *and is same-support pair-noncontained.*

Proof. Partition the domain into sets C_0, E, T, W of sizes $20, 30, 21, 29$. Put $(r_0, r_1) = (0, 0)$ on C_0. Index E by $1 \leq i \leq 30$ and put $(r_0, r_1) = (-i^2, i)$ at its i-th point. For slope i, the zero codeword then agrees exactly on C_0 and that point of E.

On T, put $r_0 = 1$ and choose 21 distinct fresh nonzero direction values avoiding $1 + ir_1 = 0$ for every $1 \leq i \leq 30$. At slope zero, the constant-one codeword agrees exactly on T. On W, choose fresh direction values and choose each base value outside $\{0, 1\} \cup \{-ir_1 : 1 \leq i \leq 30\}$. All choices exist in $\mathbb{F}_{1009}$, and they remove every unintended selected agreement.

The direction is zero on C_0 and takes distinct nonzero values elsewhere, so its maximum agreement with a constant is exactly 20. The selected explanations are zero and one, hence have affine span one. On a zero-explanation support, the core forces a simultaneous constant pair to be $(0, 0)$, but the remaining direction value is nonzero. On T, the direction values are distinct. Thus every support is pair-noncontained.

The proof failure is the asserted proper-subspace occupancy bound for the incident normals. On each zero-explanation support, 20 normals lie on one line and one is transverse. There are only 40 ordered bases, not the claimed lower count $mw = 420$. The complete deterministic reconstruction is in `experimental/verify_mca_affine_span_incidence_counterexample_v1.py`. □

Remark 6.10 (Scope of the route cut). The counterexample retracts only the MCA incidence compiler and statements that use its ordered-basis denominator. It does not refute the ordinary affine-span list theorem, common-core cancellation, the directional Johnson compiler, codeword-direction gauge equivalence, or the selector-free all-LineRay error-affine-core set-pair theorem, whose proof uses zero-mask restriction and the Bollobas set-pair inequality instead.

Theorem 6.11 (Pair-noncontained proper-subspace MCA compiler). *Let $C = \mathrm{RS}[\mathbb{F}, D, K]$ have length n, let $r_\gamma = r_0 + \gamma r_1$, put $w = m - K > 0$, and let $\mathcal{A} = c_0 + C' \subseteq C$ have $\dim C' = s$, where $1 \le s \le K$. Let $Z \subseteq \mathbb{F}$ be a set of distinct slopes such that for every $\gamma \in Z$, some $c_\gamma \in \mathcal{A}$ has agreement at least m, and its maximal agreement support is same-support pair-noncontained. Define*

$$e = \min_{b \in C} \mathrm{wt}(r_1 - b), \qquad L = \max\{1, e - (n - m)\}.$$

Then

$$|Z| \le \left\lfloor \frac{1}{L} \max\left\{ \frac{n^{\underline{s+1}}}{m(w+1)^{\overline{s-1}}}, \frac{(n-K+s)^{\underline{s+1}}}{(w+1)^{\overline{s}}} \right\} \right\rfloor. \tag{6.13}$$

For $s = 1$, the empty rising product in the first denominator is one.

Proof. Choose a basis $c_1, \ldots, c_s$ of C'. As in the rejected proof, the agreement hyperplane at coordinate x has normal

$$v_x = (r_1(x), -c_1(x), \ldots, -c_s(x)) \in \mathbb{F}^{s+1}.$$

The normals incident with every selected parameter point span $\mathbb{F}^{s+1}$. Indeed, an annihilator with zero slope coordinate would give a nonzero codeword with at least $m > K - 1$ roots. An annihilator with nonzero slope coordinate would give simultaneous codeword explanations for r_0 and r_1 on the same maximal support, contrary to pair noncontainment.

Let $W < \mathbb{F}^{s+1}$ have dimension $j < s$, and put $X_W = \{x : v_x \in W\}$. If the slope coordinate vanishes on $W^\perp$, then $s + 1 - j$ independent codewords vanish on X_W. Otherwise the kernel of that coordinate supplies $s - j$ independent codewords. The MDS generalized-weight formula gives in either case

$$|X_W| \le K - s + j.$$

Consequently, after $j = 1, \ldots, s-1$ independent incident normals have been selected, there are at least $w + s - j$ choices outside their span. These factors multiply to $(w+1)^{\overline{s-1}}$.

After s independent normals have been selected, local full rank leaves at least one final choice. If their hyperplane has an annihilator with nonzero slope coordinate, it contains at most $n - e$ coordinate normals and leaves at least $e - (n - m)$ incident choices. If that slope coordinate is zero, the ordinary root bound leaves at least $w + 1$. Interpolation on K coordinates gives $e \le n - K$, so every case leaves at least L final choices.

Let z be the number of zero normals and let $g \le z$ be those whose agreement hyperplane contains every parameter point. Common-zero MDS gives $z \le K - s$. Every selected point is therefore incident with at least

$$(m - g)(w+1)^{\overline{s-1}} L$$

ordered normal bases, while there are at most $(n - z)^{\underline{s+1}}$ ordered independent coordinate tuples. For fixed z, the resulting ratio is largest at $g = z$. Its successive ratio in z has the sign of $n - (s+1)m + sz$, so its maximum on $0 \le z \le K - s$ is at an endpoint. The endpoints $z = 0$ and $z = K - s$ are precisely the two terms in (6.13). ☐

Remark 6.12 (Exact first shortened-row calibration). For a shortened row $(n, K, m) = (R + K, K, d + K)$, the last factor is $L = \max\{1, e - (R - d)\} \le d$. At the first residual dimensions from the common-core staircase, exact integer evaluation gives

row	K	paid for every e	additional paid walls
KoalaBear	14	$s \le 9$	$s = 10, 11, 12, 13: \ e \ge 981108, 981153, 981861, 992852$
Mersenne-31	6	$s \le 1$	$s = 2, 3, 4, 5: \ e \ge 981144, 981363, 984779, 1037876$

At top rank, the required factors are $182530 > d = 67472$ and $882143 > d = 67448$, respectively, so this compiler does not pay those cells. The exact replay is in `experimental/verify_mca_proper_subspace_occupancy_compiler_v1.py`.

Theorem 6.13 (Sparse-direction punctured Johnson MCA profile)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, K]$ have*

$$N = R + K, \qquad m = d + K,$$

and let $r_\gamma = r_0 + \gamma r_1$. Suppose

$$r_1 = b + q, \qquad b \in C, \qquad E = \mathrm{supp}(q), \qquad |E| = e < d.$$

For every selected slope, assume an exact size-m, same-support pair-noncontained agreement witness with explanation c_γ, and put $a_\gamma = c_\gamma - \gamma b$. For each distinct selected transformed explanation define

$$h_a = m - |\{x \in D \setminus E : a(x) = r_0(x)\}|.$$

Then $1 \le h_a \le e$, and one deficit-h explanation owns at most $\lfloor e/h \rfloor$ selected slopes.

Assume

$$\Delta_e = (m-e)^2 - (N-e)(K-1) > 0. \tag{6.14}$$

Put $J_0 = 0$ and, for $1 \le h \le e$,

$$J_h = \left\lfloor \frac{(N-e)(m-h-K+1)}{(m-h)^2 - (N-e)(K-1)} \right\rfloor. \tag{6.15}$$

The selected slope set Z satisfies

$$|Z| \le \sum_{h=1}^{e} (J_h - J_{h-1}) \left\lfloor \frac{e}{h} \right\rfloor \le (e-1) J_{\lfloor e/2 \rfloor} + J_e. \tag{6.16}$$

For $e = 1$, the term indexed by zero in the last expression is zero.

Proof. Outside E, the direction equals b. If a agreed with r_0 on at least m outside coordinates, an m-subset there would be simultaneously explained by (a, b), contrary to same-support pair noncontainment. Hence $h_a \ge 1$. A size-m witness can use at most e coordinates of E, so $h_a \le e$.

For $x \in E$, agreement at slope γ is equivalent to

$$\frac{a(x) - r_0(x)}{q(x)} = \gamma.$$

These ratio fibers are disjoint. A deficit-h explanation needs at least h inside agreements, and therefore owns at most $\lfloor e/h \rfloor$ slopes.

Puncture E and fix h. Each explanation of deficit at most h has an agreement set of size at least $A = m - h$ in length $n' = N - e$. Agreement sets of two distinct degree-$< K$ polynomials intersect in at most $K - 1$ coordinates. If there are L explanations and s_x is the incidence multiplicity at coordinate x, then

$$\sum_x s_x \ge LA, \qquad \sum_x \binom{s_x}{2} \le (K-1)\binom{L}{2}.$$

Cauchy–Schwarz gives

$$\sum_x \binom{s_x}{2} \ge \frac{1}{2}\left(\frac{L^2 A^2}{n'} - LA\right).$$

Since $A^2 > n'(K-1)$, comparison yields $L \le J_h$. Condition (6.14) is the weakest, $h = e$, denominator and hence makes every J_h legal. The Johnson fractions are nondecreasing in h. Applying their cumulative caps to the nonincreasing owner weights proves the first inequality in (6.16).

Finally put $u = \lfloor e/2 \rfloor$. Bound all weights with $h \le u$ by e; every weight with $h > u$ equals one. The two telescoping blocks give

$$eJ_u + (J_e - J_u) = (e-1)J_u + J_e.$$

∎

Remark 6.14 (Exact sparse-direction support walls). At the first shortened KoalaBear and Mersenne-31 rows, exact integer evaluation gives

row	K	last paid e	Δ_e	$(e-1)J_{\lfloor e/2 \rfloor} + J_e$
KoalaBear	14	63908	1218	4607583
Mersenne-31	6	65236	2794	2605443

against budgets 274980728111395087 and 16777215. At the adjacent supports the denominators are -5924 and -1636, so this theorem makes no claim there. The exact scan is in `experimental/verify_mca_sparse_direction_punctured_johnson_profile_v1.py`.

Theorem 6.15 (Near-Johnson centered-Gram list compiler)**.** *Let $S_1, \ldots, S_L$ be distinct A-subsets of an n-set such that $|S_i \cap S_j| \le c$ for $i \ne j$, where $A > c$. Put*

$$g = nc - A^2 \ge 0, \qquad G = (A-c)^2 - cg.$$

If $G > 0$, then

$$L \le \left\lfloor \frac{nA(A-c)}{G} \right\rfloor. \tag{6.17}$$

Proof. Let B be the L-by-n incidence matrix and put

$$H = BB^\intercal - c\mathbf{1}\mathbf{1}^\intercal.$$

Every row of B has sum A, so $\mathbf{1} = A^{-1}B\mathbf{1}_n$. The images of both summands defining H therefore lie in $\operatorname{col}(B)$, and

$$\operatorname{rank}(H) \le \operatorname{rank}(B) \le n. \tag{6.18}$$

Write $\delta_{ij} = c - |S_i \cap S_j|$. Then $0 \le \delta_{ij} \le c$ is integral and

$$\operatorname{tr}(H) = L(A-c), \qquad \operatorname{tr}(H^2) = L(A-c)^2 + 2\sum_{i<j} \delta_{ij}^2.$$

If s_x is the incidence multiplicity at coordinate x, then Cauchy–Schwarz gives

$$2\sum_{i<j} |S_i \cap S_j| = \sum_x s_x^2 - LA \ge \frac{L^2A^2}{n} - LA.$$

Thus, for $\Delta = \sum_{i<j} \delta_{ij}$,

$$2\Delta \le L\left(\frac{Lg}{n} + A - c\right).$$

Since $\delta_{ij}^2 \le c\delta_{ij}$,

$$\operatorname{tr}(H^2) \le L\left((A-c)^2 + c(A-c) + \frac{cLg}{n}\right).$$

The symmetric trace-rank inequality $\operatorname{rank}(H) \ge \operatorname{tr}(H)^2/\operatorname{tr}(H^2)$, together with (6.18), yields

$$L((A-c)^2 - cg) \le nA(A-c),$$

which is (6.17). ∎

Corollary 6.16 (Sparse-direction near-Johnson MCA payment)**.** *Retain the hypotheses of Theorem 6.13, put*

$$u = \lfloor e/2 \rfloor, \qquad n = N - e, \qquad c = K - 1, \qquad A = m - e,$$

and assume

$$(m-u)^2 - nc > 0, qquad g = nc - A^2 \ge 0, qquad G = (A-c)^2 - cg > 0.$$

Define

$$J_u = \left\lfloor \frac{n(m-u-K+1)}{(m-u)^2 - nc} \right\rfloor, \qquad Q_e = \left\lfloor \frac{nA(A-c)}{G} \right\rfloor.$$

Then

$$|Z| \le (e-1)J_u + Q_e. \tag{6.19}$$

Proof. Let N_u be the number of transformed explanations with outside deficit at most u, and let N_e be their total number. The deficit-owner bound gives

$$|Z| \le eN_u + (N_e - N_u) = (e-1)N_u + N_e.$$

The ordinary Johnson argument on the punctured row gives $N_u \le J_u$. For each explanation counted by N_e, choose exactly $A = m - e$ outside agreement coordinates. Distinct degree-$< K$ explanations give distinct blocks with intersections at most $c = K - 1$, so Theorem 6.15 gives $N_e \le Q_e$. □

Remark 6.17 (Exact centered-Gram support walls). Together with the preceding Johnson prefix, exact integer evaluation of (6.19) expands the low-support walls to

row	K	last paid e	endpoint bound
KoalaBear	14	64037	198047217
Mersenne-31	6	65418	16759641

against budgets 274980728111395087 and 16777215. At KoalaBear $e = 64038$, the centered-Gram denominator is -36911. At Mersenne $e = 65419$, it remains positive, but the valid bound 18212004 exceeds budget. The exact replay is in `experimental/verify_mca_sparse_direction_near_johnson_gram_rank_v1.py`.

Theorem 6.18 (Mean-centered Gram list compiler)**.** *Let $S_1, \dots, S_L$ be distinct A-subsets of an n-set with $|S_i \cap S_j| \le c$ for $i \ne j$, where $A > c$. Put*

$$g = nc - A^2, \qquad T = (n-A)^2 - (n-1)g.$$

If

$$g \ge 0, \qquad 2A^2 \ge nc, \qquad T > 0,$$

then

$$L \le \left\lfloor \frac{(n-1)n^2(A-c)}{AT} \right\rfloor. \tag{6.20}$$

Proof. Let B be the incidence matrix and put

$$P = I_n - \frac{1}{n}J_n, \qquad H = BPB^{\mathsf{T}}.$$

Then H is positive semidefinite and has rank at most $n-1$. Set $p = A^2/n$. The diagonal entries are $A - p$, while an off-diagonal entry is

$$x_{ij} = |S_i \cap S_j| - p \in [-p, c-p].$$

Convexity gives the endpoint-chord inequality

$$x_{ij}^2 \le (c-2p)x_{ij} + p(c-p).$$

The assumption $2A^2 \ge nc$ makes the chord slope nonpositive. Since H is positive semidefinite,

$$2\sum_{i<j} x_{ij} \ge -L(A-p).$$

With $C = p(c-p) = A^2 g/n^2$, the preceding two displays imply

$$\operatorname{tr}(H^2) \le CL^2 + A(A-c)L.$$

Now $\operatorname{tr}(H) = L(A-p)$. The PSD trace-rank inequality gives

$$L((A-p)^2 - (n-1)C) \le (n-1)A(A-c).$$

Finally,

$$(A-p)^2 - (n-1)C = \frac{A^2T}{n^2},$$

which proves (6.20). □

Corollary 6.19 (Exact mean-centered sparse-direction profile)**.** *Retain the sparse-direction notation of Theorem 6.13. For each deficit threshold $1 \le h \le e$, put $A_h = m - h$ and let C_h be the ordinary Johnson cap when $A_h^2 > (N-e)(K-1)$, and otherwise the cap in Theorem 6.18 whenever its three hypotheses hold. Assume every C_h is defined, and put*

$$B_0 = 0, \qquad B_h = \min_{h \le v \le e} C_v.$$

Then

$$|Z| \le \sum_{h=1}^{e} (B_h - B_{h-1}) \left\lfloor \frac{e}{h} \right\rfloor . \tag{6.21}$$

Proof. Let N_h count distinct transformed explanations with outside deficit at most h. Puncturing the direction support and applying the appropriate ordinary-list theorem gives $N_h \le C_h$. For every $v \ge h$, also $N_h \le N_v \le C_v$, hence $N_h \le B_h$. The suffix minima are nondecreasing. Apply the deficit owner weight $\lfloor e/h \rfloor$ and sum by parts. □

Remark 6.20 (Exact mean-centered support walls)*.* The complete exact profile in (6.21) expands the walls to

row	K	last paid e	endpoint profile
KoalaBear	14	64047	181731868
Mersenne-31	6	65454	16101127

At KoalaBear $e = 64048$, the endpoint denominator is $T = -1499457466$. At Mersenne $e = 65455$, every cap remains defined, but the exact profile is $17120123 > 16777215$. The replay is in `experimental/verify_mca_sparse_direction_mean_centered_gram_profile_v1.py`.

Theorem 6.21 (Terminal-deficit affine-line cap)**.** *Retain the sparse-direction notation of Theorem 6.13, assume $e \ge K$, and put*

$$n = N - e, \qquad A = m - e, \qquad c = K - 1.$$

If $A > c$, then the number L_e of transformed explanations that own a selected slope and have exact outside deficit e satisfies

$$L_e \le \left\lfloor \frac{n-c}{A-c} \right\rfloor . \tag{6.22}$$

Proof. An exact-deficit-e explanation has exactly $A = m - e$ outside agreements. To own a selected slope it must therefore agree at all e coordinates of the direction support E. If two such explanations have slopes $\gamma \ne \delta$, their normalized difference is a nonzero degree-$< K$ codeword p whose restriction to E is the gauged direction $q|_E$. Every further terminal explanation gives the same normalized difference: the two codewords agree on E, and evaluation on E is injective because $e \ge K$. Thus all terminal explanations lie on one affine codeword line.

Outside E, let G be the common agreement core on the zeros of p, and write $g = |G| \le K - 1 = c$. Away from G, a coordinate can be an agreement for at most one affine-line parameter. Since each terminal explanation has $A - g$ agreements away from the core,

$$L_e(A-g) \le n - g.$$

The ratio $(n-g)/(A-g)$ increases with g, so $g \le c$ gives (6.22). The case $L_e \le 1$ is immediate. □

Corollary 6.22 (Prefix-plus-terminal sparse-direction profile)**.** *For every $1 \le h < e$, let C_h be a proved cumulative cap on the number N_h of transformed explanations with outside deficit at most h, and put*

$$B_0 = 0, \qquad B_h = \min_{h \le v < e} C_v.$$

Then

$$|Z| \le \sum_{h=1}^{e-1} (B_h - B_{h-1}) \left\lfloor \frac{e}{h} \right\rfloor + \left\lfloor \frac{N-e-(K-1)}{m-e-(K-1)} \right\rfloor . \tag{6.23}$$

Proof. For $h < e$, monotonicity gives $N_h \le \min_{h \le v < e} C_v = B_h$. Sum the nonincreasing owner weights $\lfloor e/h \rfloor$ by parts over the prefix layers, then apply Theorem 6.21 to the exact terminal layer, whose owner weight is one. □

Remark 6.23 (Exact terminal-line support walls). Using the Johnson/mean-centered cumulative caps for $h < e$, (6.23) pays one further support in each deployed row:

row	K	last paid e	terminal cap	total profile
KoalaBear	14	64048	287	181326343
Mersenne-31	6	65455	493	16100647

At KoalaBear $e = 64049$, the cumulative cap at $h = e - 1$ is unavailable. At Mersenne $e = 65456$, the valid profile is $17119507 > 16777215$. The remaining full-lift intervals are therefore $64049 \le e \le 1044238$ and $65456 \le e \le 1044241$. This is a support-wall refinement, not a proof of either deployed row. The exact replay is in `experimental/verify_mca_sparse_direction_terminal_deficit_line_payment_v1.py`.

Theorem 6.24 (Full-explanation lifted-rank gauge dichotomy)**.** *Let $C \le \mathbb{F}^D$ have dimension K, let $r_\gamma = r_0 + \gamma r_1$, and let $Z \subseteq \mathbb{F}$ contain distinct slopes. For every $\gamma \in Z$, choose $c_\gamma \in C$, and assume that the c_γ have full affine rank K. Fix $\gamma_0 \in Z$ and put*

$$V = \operatorname{span}\{(\gamma - \gamma_0, c_\gamma - c_{\gamma_0}) : \gamma \in Z\} \le \mathbb{F} \oplus C.$$

Then $\dim V \in \{K, K+1\}$, and exactly one of the following holds.

1. *If $\dim V = K$, there is a unique nonzero linear functional $\ell : C \to \mathbb{F}$ such that $V = \{(\ell(u), u) : u \in C\}$. The gauge*
$$r_1 \longmapsto r_1 - b, \qquad c_\gamma \longmapsto c_\gamma - \gamma b$$
drops the explanation affine rank to exactly $K - 1$ precisely for $b \in C$ satisfying $\ell(b) = 1$. Thus the rank-dropping gauges form an affine hyperplane in C.
2. *If $\dim V = K + 1$, then $V = \mathbb{F} \oplus C$, and every $b \in C$ leaves the explanation affine rank equal to K.*

If, in addition, every chosen maximal agreement support is same-support pair-noncontained, then $r_1 \notin C$, and the selected error vectors $r_\gamma - c_\gamma$ have affine rank exactly $\dim V$.

Proof. Projection of V to C is onto because the explanations have full affine rank. Hence $K \le \dim V \le K + 1$. If $\dim V = K$, projection is an isomorphism and V is the graph of a unique functional ℓ. It is nonzero because the slopes are not all equal. On anchored differences, the gauge is the map

$$T_b : V \longrightarrow C, \qquad T_b(a, u) = u - ab.$$

For $V = \{(\ell(u), u) : u \in C\}$, this map is $u \mapsto u - \ell(u)b$. It has rank $K - 1$ exactly when $\ell(b) = 1$, in which case its image is $\ker \ell$; otherwise it is invertible. If $\dim V = K + 1$, then $V = \mathbb{F} \oplus C$, and T_b is surjective for every b.

Suppose now that $r_1 \in C$. On any selected agreement support, r_1 explains the direction and $c_\gamma - \gamma r_1 \in C$ explains the base, contradicting pair noncontainment. Thus $r_1 \notin C$. The linear map

$$(a, u) \longmapsto a r_1 - u$$

is consequently injective on $\mathbb{F} \oplus C$, and it sends the anchored lifted differences to the anchored error-vector differences. Their affine rank is therefore $\dim V$. □

Theorem 6.25 (Full-lift near-MDS extension reduction)**.** *Retain the hypotheses of Theorem 6.24, assume $\dim V = K + 1$, write $N = |D|$, and put*

$$W = C + \langle r_1 \rangle, \qquad e = \min_{b \in C} \operatorname{wt}(r_1 - b).$$

Then $\dim W = K + 1$, and the generalized Hamming weights of W are

$$d_1(W) = e, \qquad d_j(W) = N - K + j - 1 \quad (2 \le j \le K + 1). \tag{NM}$$

The selected errors $r_\gamma - c_\gamma$ lie in one affine coset of W, have full affine rank $K+1$, and determine their slopes uniquely. Moreover, if every selected maximal agreement support has size greater than K, restriction of W to each such support is injective.

Proof. Pair noncontainment gives $r_1 \notin C$, so W has dimension $K+1$. Interpolation on K coordinates gives $e \le N-K$. Every nonzero word of W is either a nonzero RS word, of weight at least $N-K+1$, or a nonzero scalar multiple of $r_1 - b$, of weight at least e; a nearest b attains e. Thus $d_1(W) = e$.

Let $A \le W$ have dimension $j \ge 2$. If $A \le C$, the RS MDS hierarchy gives $|\operatorname{supp} A| \ge N-K+j$. Otherwise $\dim(A \cap C) = j-1$, and that RS subcode has at most $K-j+1$ common zero coordinates. Hence

$$|\operatorname{supp} A| \ge N-K+j-1.$$

The generalized Singleton bound for the $[N, K+1]$ code W is the reverse inequality, proving (NM).

The map $(a,u) \mapsto ar_1 - u$ is an isomorphism $\mathbb{F} \oplus C \to W$, so the full lifted rank is exactly the error affine rank and equal errors have equal slopes. Finally, a nonzero $ar_1 - u \in W$ vanishing on a selected support either contradicts the RS root bound when $a=0$, or supplies simultaneous direction and base explanations when $a \ne 0$. Pair noncontainment excludes both cases. ☐

Remark 6.26 (Exact top-rank split at the first shortened rows)*.* The gauge preserves agreement supports, pair noncontainment, and $e = \min_{b\in C} \operatorname{wt}(r_1 - b)$. Therefore the $\dim V = K$ branch may use Theorem 6.11 at explanation rank $K-1$. Combining this with the already separate low-support and recursive-direction gates gives the following exact routing table; q is explanation affine rank and $h = \dim V$, equivalently the error affine rank.

row	q	h	currently paid support range
KoalaBear	14	14	$e \le 64047$ or $e \ge 992852$
KoalaBear	14	15	$e \le 64047$ or $e \ge 1044239$
Mersenne-31	6	6	$e \le 65454$ or $e \ge 1037876$
Mersenne-31	6	7	$e \le 65454$ or $e \ge 1044242$

The improvement is only in the $h = K$ rows. The full-lift $h = K+1$ middle intervals remain unpaid and are the wider top-rank residuals. After Corollary 6.19, those full-lift intervals are exactly $64048 \le e \le 1044238$ for KoalaBear and $65455 \le e \le 1044241$ for Mersenne-31. Theorem 6.25 shows that every generalized weight from rank two onward is already MDS-sharp. Even at the best endpoint $e = N-K$, the generic corrected compiler gives

$$743896698428332665 > 274980728111395087 \quad \text{and} \quad 219426634 > 16777215$$

on KoalaBear and Mersenne-31, respectively. Closing these cells therefore requires structure of the codimension-one RS extension beyond its weight hierarchy, or a sharper row-specific affine-list theorem. A finite-field audit and the exact adjacent arithmetic are in `experimental/verify_mca_full_explanation_lifted_rank_gauge_dichotomy_v1.py`.

Theorem 6.27 (Directional Johnson ray compiler)**.** *Retain the hypotheses and notation of Theorem 4.5, but allow $\nu \ge 0$. Choose a lift $b_1 \in \mathbb{F}^U$ of y_1, put $C_U = \ker H_U$, and define the lift-invariant directional agreement*

$$\alpha_U(y_1) = \max_{c\in C_U} |\{x \in U : b_1(x) + c(x) = 0\}|. \tag{6.24}$$

If $h = N - t$ and

$$h^2 > N\alpha_U(y_1), \tag{6.25}$$

then

$$|Z| \le \left\lfloor \frac{N(h-\alpha_U(y_1))}{h^2 - N\alpha_U(y_1)} \right\rfloor. \tag{6.26}$$

Moreover, with

$$d_U(y_1) = \min\{\operatorname{wt}(e) : e \in \mathbb{F}^U,\ H_U e = y_1\},$$

one has

$$\alpha_U(y_1) = N - d_U(y_1) \geq \nu. \tag{6.27}$$

Equality holds exactly when the direction syndrome has no representation on fewer than R columns of U.

Proof. Changing the lift b_1 by a kernel word merely permutes the maximization in (6.24), so $\alpha_U(y_1)$ is intrinsic. Choose a lift b_0 of y_0. For every slope write

$$e_\gamma = b_0 + \gamma b_1 + k_\gamma, \qquad k_\gamma \in C_U,$$

and choose an h-set A_γ of zero coordinates of e_γ. On A_γ, the codeword $-k_\gamma$ agrees with $b_0 + \gamma b_1$. If $\gamma \neq \delta$, then on $A_\gamma \cap A_\delta$,

$$b_1 = \frac{-k_\gamma + k_\delta}{\gamma - \delta},$$

which is a codeword of C_U. Hence $|A_\gamma \cap A_\delta| \leq \alpha_U(y_1)$. The same Johnson incidence count as in Theorem 4.7, with $\alpha_U(y_1)$ in place of $\nu - 1$, gives (6.26).

Finally, the words $b_1 + c$, $c \in C_U$, are exactly all lifts of y_1 supported in U. Maximizing their number of zero coordinates is therefore equivalent to minimizing their weight, proving the equality in (6.27). Any R columns of H_U span syndrome space, so $d_U(y_1) \leq R$ and $\alpha_U(y_1) \geq N - R = \nu$. The equality criterion is immediate. □

Corollary 6.28 (Rank-regular fixed-union ray bound). *In Theorem 6.27, suppose the direction syndrome is rank-regular on U, meaning $d_U(y_1) = R$. If $(R+\nu-t)^2 > (R+\nu)\nu$, then*

$$|Z| \leq \left\lfloor \frac{(R+\nu)(R-t)}{(R+\nu-t)^2 - (R+\nu)\nu} \right\rfloor. \tag{6.28}$$

Thus every failure of this stronger low-nullity payment is either a high-nullity chart or an explicit direction-rank-defect cell $d_U(y_1) < R$.

Proof. By (6.27), rank regularity is exactly $\alpha_U(y_1) = \nu$. Substitute this value into (6.26). □

Corollary 6.29 (Quantitative direction-rank-defect payment). *Retain the notation of Theorem 6.27, and write*

$$d = R - t, \qquad j = R - d_U(y_1), \qquad N = R + \nu, \qquad D_0 = (d+\nu)^2 - N\nu.$$

Thus $j \geq 0$ measures the direction-rank defect and $\alpha_U(y_1) = \nu + j$. Whenever $D_0 - Nj > 0$,

$$|Z| \leq \left\lfloor \frac{N(d-j)}{D_0 - Nj} \right\rfloor. \tag{6.29}$$

For an integer budget $B > 1$, define

$$J_B(\nu) = \min\left\{ d-1, \left\lfloor \frac{D_0 - 1}{N} \right\rfloor, \left\lfloor \frac{BD_0 - Nd}{N(B-1)} \right\rfloor \right\}. \tag{6.30}$$

If $0 \leq j \leq J_B(\nu)$, then the whole fixed-union ray chart costs at most B slopes. Hence the residual direction cell at a fixed nullity can be stated quantitatively as $j > J_B(\nu)$, rather than merely as $j > 0$.

Proof. By (6.27), $\alpha_U(y_1) = N - d_U(y_1) = \nu + j$. Substitution into (6.26) gives (6.29). The second term in (6.30) makes its denominator positive, and the first makes its numerator positive. The final term is exactly the rearrangement

$$N(d-j) \leq B(D_0 - Nj).$$

Thus the real fraction in (6.29) is at most B, and so is its floor. □

Theorem 6.30 (Paving-basis direction-defect ray compiler). *Retain the fixed-union MCA notation of Theorems 4.5 and 6.27. Write*

$$N = R + \nu, \qquad h = N - t, \qquad d = R - t, \qquad j = R - d_U(y_1).$$

Assume $0 \le j < d$, equivalently that the direction word is outside the agreement ball on this chart. Then

$$|Z| \le \left\lfloor \frac{(\nu+1)\binom{R+\nu}{\nu+1}}{(R-t-j)\binom{R+\nu-t}{\nu}} \right\rfloor. \tag{6.31}$$

This estimate is independent of the Johnson denominator and is strongest for small nullity, including large direction defect.

Proof. Choose a minimum-weight lift $q \in \mathbb{F}^U$ of the direction syndrome, so $\mathrm{wt}(q) = R - j$, and choose a basis $k_1, \dots, k_\nu$ of the MDS kernel $C_U = \ker H_U$. In the parameter space with coordinates $(\gamma, \lambda_1, \dots, \lambda_\nu)$, agreement at coordinate x has normal

$$v_x = (q_x, k_1(x), \dots, k_\nu(x)).$$

Every set of at most ν such normals is independent, because its projection to the last ν coordinates is a set of columns of a generator matrix of the $[R+\nu, \nu, R+1]$ MDS code C_U.

Every rank-ν flat of these normals contains at most $\nu + j$ coordinates. Indeed, such a flat is annihilated by a nonzero functional (δ, a). If $\delta \ne 0$, the vanishing coordinates are zeros of a lift $\delta q + c$ of δy_1, which has weight at least $R - j$. If $\delta = 0$, they are zeros of a nonzero MDS kernel word and number at most $\nu - 1$.

Fix a slope point and choose h incident coordinate hyperplanes. Put $r = \nu + 1$ and $\alpha = \nu + j$. Every $(r-1)$-subset of their normals is independent. A dependent r-subset is counted by each of its r subsets of size $r - 1$, while the closure of any such subset contains at most α coordinate normals. Hence the number of independent r-subsets among the h incident normals is at least

$$\binom{h}{r} - \frac{\alpha - r + 1}{r}\binom{h}{r-1} = \frac{h-\alpha}{r}\binom{h}{r-1} = \frac{d-j}{\nu+1}\binom{R+\nu-t}{\nu}.$$

Globally there are at most $\binom{R+\nu}{\nu+1}$ independent coordinate bases, and each basis cuts out at most one parameter point. Double counting proves (6.31). □

Corollary 6.31 (Exact paving defect threshold)**.** *For an integer budget $B \ge 0$, define*

$$J_B^{\mathrm{pav}}(\nu) = \max\left\{-1, d - 1 - \left\lfloor \frac{(\nu+1)\binom{R+\nu}{\nu+1}}{(B+1)\binom{R+\nu-t}{\nu}} \right\rfloor \right\}. \tag{6.32}$$

Every separated fixed-union MCA chart with $0 \le j \le J_B^{\mathrm{pav}}(\nu)$ costs at most B slopes. Combining this with Corollary 6.29, the exact certified defect range is

$$0 \le j \le \max\{J_B(\nu), J_B^{\mathrm{pav}}(\nu)\}.$$

Proof. The floor in (6.31) is at most B exactly when its numerator is strictly smaller than $B + 1$ times its denominator. Solving this integer inequality for j gives (6.32). □

7. Minimum-distance sharpening and the exact primitive wedge

Retain the notation of Theorem 6.4. Let

$$\mu = d_1(C') = \min_{0 \ne c \in C'} |\mathrm{supp}(c)|.$$

The generalized Singleton bound gives

$$\mu \ge n - K + 1 = R + 1, \qquad R = n - K.$$

The general flag theorem uses only the ambient MDS lower bound for each flag-annihilator direction. Every such direction actually belongs to C', so the true minimum support gives a sharper resource count.

Theorem 7.1 (Minimum-distance enhanced flag resources). *Retain the hypotheses and notation of Theorem 6.4. For a profile $\mathbf{b} = (b_1, \dots, b_r)$ define*

$$P^\sharp_{\mathbf{b}} = \prod_{i=1}^{j} \left(f_i - t + b_0 + \sum_{q=1}^{r} b_q \right), \tag{7.1}$$

$$Q^\sharp_\ell(\mathbf{b}) = \mu - t + b_0 + \sum_{q>\ell} b_q, \tag{7.2}$$

$$A_\ell(\mathbf{b}) = e_\ell - b_\ell, \qquad B^\sharp_\ell(\mathbf{b}) = \left(Q^\sharp_\ell(\mathbf{b}) - e_\ell + b_\ell\right)_+. \tag{7.3}$$

For $\sigma \in \{T, U\}^r$, put

$$C^\sharp_\sigma(\mathbf{b}) = P^\sharp_{\mathbf{b}} \prod_{\ell:\sigma_\ell = T} A_\ell(\mathbf{b}) \prod_{\ell:\sigma_\ell = U} B^\sharp_\ell(\mathbf{b}), \tag{7.4}$$

$$G_\sigma = d_0^j \prod_{\ell:\sigma_\ell = T} e_\ell \prod_{\ell:\sigma_\ell = U} (d_{\ell-1} - (j + \ell - 1)). \tag{7.5}$$

Then

$$\boxed{\sum_{\mathbf{b}} L_{\mathbf{b}} C^\sharp_\sigma(\mathbf{b}) \le G_\sigma} \tag{7.6}$$

for every binary resource pattern. Consequently every nonnegative family x_σ satisfying

$$\sum_\sigma x_\sigma C^\sharp_\sigma(\mathbf{b}) \ge 1 \quad \textit{for every profile } \mathbf{b}$$

gives the whole-chart bound

$$\boxed{L \le \left\lfloor \sum_\sigma x_\sigma G_\sigma \right\rfloor.} \tag{7.7}$$

The tensor and all-extender corollaries remain valid after replacing $w + 1 + b_0$ by $\mu - t + b_0$ and replacing every old-support factor by (7.3).

Proof. Fix a flag level ℓ and an adapted coordinate basis through $V_{\ell-1}$. Its annihilator in V_ℓ is generated by a nonzero word

$$h_\ell \in V_\ell \setminus V_{\ell-1} \subseteq C'.$$

Therefore $\mathrm{wt}(h_\ell) \ge \mu$. The word vanishes outside U_ℓ. The number of mismatches outside U_ℓ is

$$b_{\mathrm{out}} = b_0 + \sum_{q>\ell} b_q,$$

and all of those coordinates are zeros of h_ℓ. Among the zero coordinates of h_ℓ, at most

$$(n - \mu) - b_{\mathrm{out}}$$

are agreements. Since every listed point has at least m agreements, at least

$$m - ((n - \mu) - b_{\mathrm{out}}) = \mu - t + b_{\mathrm{out}} = Q^\sharp_\ell(\mathbf{b})$$

agreement coordinates are nonzeros of h_ℓ and hence extend the adapted basis. The $A_\ell = e_\ell - b_\ell$ agreeing coordinates in the new layer all extend, so at least $B^\sharp_\ell$ extending agreements lie in the old support. The global basis capacities are unchanged. The double count and finite dual are therefore identical to the proof of Theorem 6.4 with the sharpened factors. □

Corollary 7.2 (Minimum-word all-extender bound). *Choose $V_0 = \langle h \rangle$, where h is a minimum-weight word of C'. If $s = \dim C'$, then*

$$\boxed{L \le \left\lfloor \frac{\mu \prod_{\ell=1}^{s-1} (d_\ell - \ell)}{(\mu - t + b_0)^s} \right\rfloor \le \left\lfloor \frac{\mu (n - s + 1)^{s-1}}{(\mu - t + b_0)^s} \right\rfloor.} \tag{7.8}$$

Writing $\mu = R + a$, $a \geq 1$, the soft form is

$$L \leq \left\lfloor \frac{(R+a)(n-s+1)^{s-1}}{(w+a+b_0)^s} \right\rfloor. \tag{7.9}$$

Proof. A generalized-weight minimizer is support-saturated by Lemma 6.2, so the minimum-weight line admits a support-saturated flag. Apply the all-extender form of Theorem 7.1. The initial basis count is μ, and $\mu - t + b_0$ is a lower bound at every extension level. Since every remaining support layer is nonempty and $d_{s-1} \leq n$, one has $d_\ell - \ell \leq n - s + 1$. ☐

Proposition 7.3 (Exact active minimum-support transitions)**.** *At $b_0 = 0$, (7.9) pays every first-unpaid affine chart at and above the following support excesses:*

row	s	a_{paid}	*bound at* a_{paid}	*bound at* $a_{\mathrm{paid}} - 1$
KoalaBear MCA	15	70832	274974324324469113	275003903412638889
KoalaBear list	15	70832	274974569967317804	275004149082130935
Mersenne-31 MCA	7	111529	16776653	16777295
Mersenne-31 list	7	111529	16776668	16777309

Thus a surviving zero-mismatch chart has $a \leq 70831$ on KoalaBear or $a \leq 111528$ on Mersenne-31. Positive b_0 only shrinks this residual.

Proposition 7.4 (Exact mismatch curve)**.** *For selected support excesses, the least outside mismatch paid by (7.9) is*

	$a = 1$	$a = 1000$	$a = 10000$	$a = 50000$	$a = 70000$
KoalaBear MCA/list	70230	69240	60318	20659	825
Mersenne-31 MCA/list	108962	107987	99202	60141	40599

The curve is strictly decreasing in the actual direction-code support excess.

Remark 7.5 (Combination with the actual first generalized weight). Let the full support have size $d_s = R + \nu$. Replacing the ambient lower bound in the common-zero Johnson compiler by $d_1(C') = R + a$ gives

$$L \leq \left\lfloor \frac{(R+\nu)(w+a+b_0)}{(w+\nu+b_0)^2 - (R+\nu)(\nu-a)} \right\rfloor \tag{7.10}$$

when the denominator is positive. Combining (7.10) with the nonuniform form of (7.8) gives the exact residual wedge in (a, ν, b_0).

Theorem 7.6 (Fixed-mismatch puncturing compiler)**.** *Let $\mathcal{L} \subseteq \mathcal{A}$ be a family of codewords, each agreeing with a received word u on at least m coordinates. Suppose a fixed set $Z \subseteq D$, $|Z| = z$, is a mismatch set for every member of $\mathcal{L}$. Then $z \leq t$, and if the affine span of $\mathcal{L}$ has direction dimension at most s, then*

$$\boxed{|\mathcal{L}| \leq \left\lfloor \frac{\binom{R-z+s}{s}}{\binom{w+s}{s}} \right\rfloor}. \tag{7.11}$$

Proof. Every listed word has at most t mismatches, so $z \leq t$. Puncture the coordinates in Z. Since $z < t < R = n - K$, evaluation of degree-$< K$ polynomials remains injective on $D \setminus Z$, no agreement is deleted, and the affine direction dimension does not increase. Apply the recursive affine-span compiler to the punctured code of length $n - z$. ☐

Corollary 7.7 (Active puncture thresholds)**.** *At the first unpaid affine dimensions, a subfamily with a common fixed mismatch set is paid at the following first integers:*

row	z_{paid}	*bound at* z_{paid}	*bound at* $z_{\text{paid}} - 1$
KoalaBear MCA	67313	274978201175970494	274982404608707323
KoalaBear list	67314	274978201175970494	274982404608707323
Mersenne-31 MCA	322321	16777133	16777295
Mersenne-31 list	322322	16777133	16777295

Proposition 7.8 (Exchange defect dominates support excess)**.** *In one boundary prefix fiber, let the associated codeword direction space have minimum support $R + a$. For distinct supports M, M' of size m, write*

$$e = |M \setminus M'| = |M' \setminus M| = w + r.$$

Then

$$\boxed{r \geq a.} \tag{7.12}$$

Proof. The prefix-rigidity identity of CAP25 v13.2 gives

$$\Lambda_M - \Lambda_{M'} = G(A - B), \qquad \deg(A - B) \leq r - 1,$$

where G is the locator of $M \cap M'$ and has degree $m - e$. The corresponding nonzero codeword difference vanishes on the $m - e$ common support coordinates, so its weight is at most

$$n - (m - e) = t + e = R + r.$$

By the definition of a, its weight is at least $R + a$. Hence $r \geq a$. □

Corollary 7.9 (Top-seam localization)**.** *The top seam $e = w + 1$ can occur only in the exact-minimum-distance branch $a = 1$. Every chart with $a \geq 2$ has first possible exchange defect at least a.*

Theorem 7.10 (Exact primitive residual after minimum-distance payment)**.** *After all unconditional first-match payments in Parts I–IV, an over-budget first-unpaid Q or list chart must lie in the following simultaneous wedge:*

- (R1) *its support excess a lies below the exact curve (7.9);*
- (R2) *its full-support nullity lies beyond both (7.10) and the minimum-word flag bound;*
- (R3) *it contains no fixed mismatch block meeting Theorem 7.6;*
- (R4) *every boundary neighbour has exchange defect at least a, and the top seam is restricted to $a = 1$;*
- (R5) *it is not in an earlier quotient, occupancy, extension, common-zero, saturation, or rank-defect cell.*

8. The quadratic mean-overlap staircase

Lemma 8.1 (Large-overlap collapse)**.** *Fix a received pair and an exact agreement $a = n-r \geq k+1$. Choose one noncommon exact witness support S_z and explaining polynomial p_z for every slope z in its bad set Z. If two distinct chosen supports satisfy*

$$|S_z \cap S_w| \geq k + r, \tag{MO1}$$

then $|Z| \leq r + 1$.

Proof. Put

$$c_1 = \frac{p_z - p_w}{z - w}, \qquad c_0 = p_z - zc_1.$$

On $S_z \cap S_w$, the two line equations imply that the received pair equals (c_0, c_1). Let C_0 be the full common support of this codeword pair and write $c = |C_0| \geq k + r$. For every $u \in Z$, $|S_u \cap C_0| \geq a + c - n \geq k$, so the explaining polynomial p_u equals $c_0 + uc_1$.

Subtract this codeword pair from the received pair. At every coordinate outside C_0 at least one of the two residual values is nonzero, and a coordinate can cancel at at most one finite slope. A noncommon witness for u must contain at least $\max\{1, a-c\}$ such cancellations: the support-size condition gives $a-c$, and one is still necessary when $a \le c$. Double counting therefore gives

$$|Z| \le \left\lfloor \frac{n-c}{\max\{1, a-c\}} \right\rfloor \le r+1. \tag{MO2}$$

Indeed, for $c \ge a$ the numerator is at most r; for $c = a-1$ it is $r+1$; and, with $x = a-c \ge 2$, the quotient is $(r+x)/x \le r+1$. ☐

Theorem 8.2 (Quadratic mean-overlap theorem)**.** *Let $C = \mathrm{RS}_{\mathbb{F}}(D,k)$, put $R = n-k$, let $0 \le r \le R-1$, and let $\varnothing \ne \Gamma \subseteq \mathbb{F}$. If*

$$(n-r)^2 \ge n(k+r), \qquad \textit{equivalently} \qquad r^2 \ge n(3r-R), \tag{MO3}$$

then

$$B^{\mathrm{MCA}}_{C,\Gamma}(n-r) = \min\{|\Gamma|, r+1\}. \tag{MO4}$$

The same upper bound holds for every linear $[n,k]$ MDS code.

Proof. The universal tangent construction imported from [Cho26, Sec. 13] gives the lower bound. For the upper bound, fix a received pair and its challenge-restricted bad set Z. If $|Z| \le r+1$ there is nothing to prove. Otherwise choose $r+2$ slopes and distinct exact witness supports $S_1, \dots, S_{r+2}$, each of size $a = n-r$. Suppose every pair had intersection at most $k+r-1$, and put $d_x = \#\{i : x \in S_i\}$. Then

$$\sum_x \binom{d_x}{2} \le \binom{r+2}{2}(k+r-1), \tag{MO5}$$

whereas Cauchy–Schwarz and $a^2/n \ge k+r$ give

$$\sum_x \binom{d_x}{2} \ge \frac{(r+2)^2 a^2/n - (r+2)a}{2} \ge \frac{(r+2)^2(k+r) - (r+2)a}{2}. \tag{MO6}$$

The last lower bound exceeds the upper bound in (MO5) by

$$\frac{r+2}{2}(k+3r-n+1).$$

When $3r > R$ this is positive. When $3r \le R$, the exact deep theorem already gives the desired upper bound. Thus in the remaining case some pair satisfies (MO1), and Lemma 8.1 contradicts $|Z| \ge r+2$. The exact-support reduction and large-overlap lemma use only the MDS information-set property, proving the stated extension. ☐

Corollary 8.3 (Exact quadratic staircase)**.** *Put*

$$r_{\mathrm{quad}}(n,k) = \left\lfloor \frac{3n - \sqrt{n(5n+4k)}}{2} \right\rfloor. \tag{MO7}$$

For every $0 \le r \le r_{\mathrm{quad}}(n,k)$ and every nonempty challenge set Γ,

$$B^{\mathrm{MCA}}_{C,\Gamma}(n-r) = \min\{|\Gamma|, r+1\}.$$

If $k/n \to \rho$, then $r_{\mathrm{quad}}(n,k)/n \to \alpha_\rho = (3 - \sqrt{5+4\rho})/2$.

Proof. The smaller root of $r^2 - 3nr + n(n-k) = 0$ is the real number inside the floor in (MO7). Apply Theorem 8.2; the limiting formula follows by division by n. ☐

Proposition 8.4 (Limit of pairwise-overlap information)**.** *Let $k/n \to \rho \in (0,1)$, $r/n \to \alpha \in (0,1)$, and suppose*

$$\alpha^2 < 3\alpha - (1-\rho). \tag{MO8}$$

For all sufficiently large n, there are $r+2$ distinct r-subsets T_i such that, whenever $j = 3r-(n-k) \le r$, $|T_i \cap T_\ell| \le j-1$ for every $i \ne \ell$. If $j > r$, that intersection condition is automatic. Hence pairwise support overlap alone cannot extend the exact theorem past the *quadratic boundary; further progress must use the syndrome, determinant, or fiber incidence. This is a limitation of the method, not a construction of an MCA-bad line.*

Proof. When $j \le r$, choose $r+2$ independent uniform r-subsets. The intersection of a fixed pair is hypergeometric with mean $r^2/n = (\alpha^2 + o(1))n$, whereas $j = (3\alpha - (1-\rho) + o(1))n$. More explicitly,

$$\Pr(|T_1 \cap T_2| = s) = \frac{\binom{r}{s}\binom{n-r}{r-s}}{\binom{n}{r}}.$$

After writing $s = xn$, Stirling's formula gives the exponent

$$\alpha h(x/\alpha) + (1-\alpha)h((\alpha - x)/(1-\alpha)) - h(\alpha),$$

which is strictly concave and has its unique maximum zero at $x = \alpha^2$. The strict gap in (MO8), followed by summing at most $n+1$ terms, therefore gives probability $e^{-\Omega(n)}$ that one fixed intersection reaches j. A union bound over $O(n^2)$ pairs is smaller than one. A successful family is automatically distinct, because equal members have intersection $r \ge j$. When $j > r$, choose any $r+2$ distinct r-sets. $\square$

The following complementary theorem can be stronger when the overlap defect $j = 3r - R$ is small. Its only new input is a packing inequality; in particular it uses no smoothness or Fourier estimate.

Theorem 8.5 (Exact overlap–packing criterion)**.** *Let $C = \mathrm{RS}_{\mathbb{F}}(D,k)$, put $R = n-k$, and fix an integer $0 \le r \le R-1$. Set $a = n-r$ and $j = 3r - R$. If either*

$$j \le 0, \tag{OP0}$$

or

$$1 \le j \le r, \qquad \binom{n}{j} \le (r+1)\binom{r}{j}, \tag{OP}$$

then

$$B_C^{\mathrm{MCA}}(n-r) = r+1. \tag{OP$'$}$$

The same upper bound holds for every linear $[n,k]$ MDS code; the matching lower bound holds whenever the field contains at least $r+1$ distinct slope parameters.

Proof. Condition (OP0) is the imported deep-regime case [Cho26, Sec. 13], so assume (OP). Fix an arbitrary received pair (f,g), let Z be its bad-slope set, and use the exact-agreement reduction imported from [Cho26, Sec. 13] to choose, for every $z \in Z$, an exact witness support S_z, $|S_z| = a$, and an explaining polynomial $p_z \in \mathbb{F}[X]_{<k}$. A single noncommon support cannot carry two distinct slopes, so the supports chosen for distinct elements of Z are distinct.

For the stated MDS extension, the same exact-support reduction is available: interpolate the unexplained component on any k-subset of the original witness support; because every k-subset is an information set, nonexplanation is detected at one further coordinate, and that $(k+1)$-set can be enlarged to size a.

First suppose that two of them have a large intersection:

$$|S_z \cap S_w| \ge n + k - a = k + r \qquad (z \ne w). \tag{OP1}$$

Then Lemma 8.1 gives $|Z| \le r+1$.

It remains to consider the complementary case, in which every pair of chosen supports satisfies $|S_z \cap S_w| \le n + k - a - 1$. Put $T_z = D \setminus S_z$, so $|T_z| = r$. For distinct slopes,

$$|T_z \cap T_w| = n - 2a + |S_z \cap S_w| \le 3r - R - 1 = j - 1. \tag{OP3}$$

No j-subset of D can therefore lie in two different T_z's. Counting pairs (z,J) with $J \in \binom{T_z}{j}$ yields

$$|Z|\binom{r}{j} \le \binom{n}{j} \le (r+1)\binom{r}{j},$$

and hence $|Z| \le r+1$. The universal tangent construction gives the reverse inequality. The proof used only linearity, the fact that agreement of two codewords on k coordinates forces equality, and the exact-support reduction; these are MDS properties. The coordinate tangent construction gives the stated MDS lower bound. ☐

Proposition 8.6 (General overlap–packing upper bound). *For $1 \le j = 3r-(n-k) \le r$, let $A(n,r,j)$ be the largest size of a family of r-subsets of an n-set whose distinct members intersect in at most $j-1$ points. Then*

$$B^{\mathrm{MCA}}_{C,\Gamma}(n-r) \le \min\{|\Gamma|, \max(r+1, A(n,r,j))\} \tag{OP5}$$

and

$$A(n,r,j) \le \left\lfloor \frac{\binom{n}{j}}{\binom{r}{j}} \right\rfloor. \tag{OP6}$$

Proof. For a fixed received pair, either two chosen exact witness supports have intersection at least $k+r$, in which case Lemma 8.1 gives at most $r+1$ slopes, or every pair of complementary r-sets intersects in at most $j-1$ points. The latter family has size at most $A(n,r,j)$. Finally, no j-subset can lie in two members of such a packing, so counting contained j-subsets proves (OP6). ☐

9. Exact thresholds for explicit prize rows

Table 1. Explicit maximal-dimension prize-admissible rows. In each row $k = 2^{40}$, $B = \lfloor p/2^{128} \rfloor = r_\rho(n)+1$, and (s,a_0) is part of the Proth certificate in Proposition 9.1.

rate	length	B	bits of p	(s,a_0)
1/2	2^{41}	389500552609	167	$(92,3)$
1/4	2^{42}	1210584858040	169	$(93,13)$
1/8	2^{43}	2879806199253	170	$(95,5)$
1/16	2^{44}	6233898019554	171	$(97,5)$

For completeness, the four field orders are

$$\begin{aligned} p_{1/2} &= 132540169958804033333249306710494641010898987122689, \\ p_{1/4} &= 411940680852499481698306614369841346700408394874881, \\ p_{1/8} &= 979947269755402568812854322316630667196565607677953, \\ p_{1/16} &= 2121285573237585848299875619011192262679065433997313. \end{aligned}$$

Their odd Proth coefficients u, in the same order, are

$$u_{1/2} = 26766274163673319604503, \qquad u_{1/4} = 41595378994516821279015,$$
$$u_{1/8} = 24737346889219389259839, \qquad u_{1/16} = 13387194060291799253121.$$

For $F_{n,k}(r) = r^2 - n(3r-(n-k))$, the exact boundary checks are

(PC0)

ρ	$F_{n,k}(B-1)$	$F_{n,k}(B)$
1/2	5154112775168	-663955886271
1/4	7590647904465	-3182321912768
1/8	13908181940112	-6720484728007
1/16	19335616403905	-20973145690236.

Proposition 9.1 (Exact arithmetic and primality certificates). *For the four rows in Table 1, the displayed integer p satisfies*

$$p = u\,2^s + 1, \qquad u \text{ odd}, \qquad u < 2^s, \qquad a_0^{(p-1)/2} \equiv -1 \pmod{p}. \tag{PC1}$$

Consequently every p is prime. Moreover

$$n \mid p-1, \qquad p < 2^{256}, \qquad B\,2^{128} \le p < (B+1)2^{128}, \tag{PC2}$$

where n and B are the corresponding entries of the table. They also satisfy $F_{n,k}(B-1) \ge 0 > F_{n,k}(B)$. Since $F'_{n,k}(r) = 2r - 3n < 0$ on $[0, n]$, these two signs locate its smaller root and give $B - 1 = r_\rho(n) = r_{\mathrm{quad}}(n, k)$. Thus $\mathbb{F}_p^*$ contains a subgroup of order n, and each displayed code exists with the asserted slope budget.

Proof. All identities and inequalities in (PC0)–(PC2) are direct integer calculations; the values of (s, a_0), u, p, n, and B have all been printed above so that the calculation is independently reproducible. We recall why the certificate proves primality. If a prime ℓ divides p, then $a_0^{u2^{s-1}} \equiv -1 \pmod{\ell}$, whereas $a_0^{u2^s} \equiv 1 \pmod{\ell}$. The multiplicative order of a_0 modulo ℓ therefore has exact 2-part 2^s, so $2^s \mid \ell - 1$. Since $u < 2^s$, one has $2^s > \sqrt{p}$; hence a composite p would have a prime divisor both at most $\sqrt{p}$ and at least $2^s + 1$, a contradiction. This is Proth's theorem in the special form used here. Finally $s \ge \log_2 n$ gives $n \mid p - 1$, and cyclicity of $\mathbb{F}_p^*$ gives the required subgroup. □

Theorem 9.2 (Exact first adjacent row over a separating field)**.** *Let $C = \mathrm{RS}_{\mathbb{F}}(D, k)$, put $R = n - k$ and $M = \binom{n}{R-1} = \binom{n}{k+1}$, and define*

$$Q_{\mathrm{sep}}(M) = \max\left\{M, \binom{M}{2}\right\}.$$

If $|\mathbb{F}| > Q_{\mathrm{sep}}(M)$, then for the full-field challenge

$$B_C^{\mathrm{MCA}}(k+1) = M. \tag{AD1}$$

If $R \ge 2$ and $b = \lfloor \varepsilon |\mathbb{F}| \rfloor$, the exact target threshold therefore satisfies

$$\begin{aligned} b \ge M \qquad &\Longrightarrow \quad a^* = k+1, \\ \binom{n}{R-2} \le b < M \quad &\Longrightarrow \quad a^* = k+2. \end{aligned} \tag{AD2}$$

Thus the first finite adjacent comparison has literal binomial constants; no $e^{o(n)}$ or unspecified polynomial factor occurs.

Proof. The upper bound in (AD1) is the exact support-atlas bound imported from [Cho26, Sec. 13]. For the reverse inequality use the Reed–Solomon parity check of redundancy R. For every $(R-1)$-set $E \subseteq D$, its column span V_E is a hyperplane in $\mathbb{F}^R$, and distinct sets give distinct hyperplanes: otherwise the union of two distinct $(R-1)$-sets would contain at least R columns in one $(R-1)$-space, contradicting the MDS column property.

Every proper linear hyperplane in $\mathbb{F}^R$ has $|\mathbb{F}|^{R-1}$ points. Since $M < |\mathbb{F}|$, the union of the M spaces V_E has fewer than $|\mathbb{F}|^R$ points; choose y_1 outside it. For each E, choose a nonzero linear functional ℓ_E with kernel V_E. The line $y_0 + \gamma y_1$ meets V_E at

$$\gamma_E = -\ell_E(y_0)/\ell_E(y_1).$$

For $E \ne E'$, equality $\gamma_E = \gamma_{E'}$ cuts out a proper hyperplane in the variable y_0, because the normalized functionals $\ell_E/\ell_E(y_1)$ and $\ell_{E'}/\ell_{E'}(y_1)$ are distinct. There are at most $\binom{M}{2} < |\mathbb{F}|$ collision hyperplanes; their union also has fewer than $|\mathbb{F}|^R$ points, so choose y_0 outside it. Since $y_1 \notin V_E$, the line is not contained in V_E, and every intersection is transverse. The M parameters γ_E are then finite and pairwise distinct. Syndrome surjectivity and Theorem 4.2 realize them as M bad slopes at agreement $k+1$, proving (AD1).

If $b \ge M$, the support upper bound makes the first allowed agreement safe. If $\binom{n}{R-2} \le b < M$, equality (AD1) makes $k+1$ unsafe, whereas the exact support-atlas bound imported from [Cho26, Sec. 13] makes $k+2$ safe. Monotonicity proves (AD2). □

10. Asymptotic consequences of the imported prefix floor

Theorem 10.1 (Unconditional shallow-prefix RS–MCA exponent)**.** *Let $C_n = \mathrm{RS}_{\mathbb{F}_n}(D_n, k_n)$, where $D_n \subseteq \mathbb{B}_n \subseteq \mathbb{F}_n$, $|D_n| = n$, and $k_n/n \to \rho \in (0,1)$. Let $a_n \ge k_n + 1$, put $w_n = a_n - k_n - 1$, and assume*

$$\frac{a_n}{n} \to \alpha \in (0,1), \qquad w_n \log|\mathbb{B}_n| = o(n), \tag{AS1}$$

$|\mathbb{F}_n| > n$, and, for some finite $\phi \ge 0$,

$$\frac{1}{n}\log|\mathbb{F}_n| \to \phi, \qquad \frac{1}{n}\log(|\mathbb{F}_n| - n) \to \phi. \tag{AS2}$$

For the full-field challenge,

$$\frac{1}{n}\log B^{\mathrm{MCA}}_{C_n}(a_n) = \min\{h(\alpha), \phi\} + o(1), \tag{AS3}$$

and therefore

$$\frac{1}{n}\log e_{\mathrm{MCA}}\left(C_n, 1 - \frac{a_n}{n}\right) = -(\phi - h(\alpha))_+ + o(1). \tag{AS4}$$

Because $|\mathbb{B}_n| \ge n$, condition (AS1) forces $\alpha = \rho$. No smoothness, Fourier estimate, or auxiliary ray hypothesis is assumed.

Proof. Put $M_n = \binom{n}{a_n}$ and $L_n = \lceil M_n |\mathbb{B}_n|^{-w_n} \rceil$. Exact-agreement reduction and fixed-support slope uniqueness give the unconditional upper bound

$$B^{\mathrm{MCA}}_{C_n}(a_n) \le \min\{|\mathbb{F}_n|, M_n\}. \tag{AS5}$$

The exact locator-prefix list and collision-aware pole compiler imported from [Cho26, Secs. 14–18] give

$$B^{\mathrm{MCA}}_{C_n}(a_n) \ge \left\lceil \frac{L_n(|\mathbb{F}_n| - n)}{|\mathbb{F}_n| - n + k_n(L_n - 1)} \right\rceil. \tag{AS6}$$

By Stirling's formula and (AS1), $\log L_n = h(\alpha)n + o(n)$. For positive x, y, $xy/(x + k_n y) \ge \frac{1}{2}\min\{y, x/k_n\}$; hence (AS2), $\log k_n = o(n)$, and the ceiling show that the right side of (AS6) has logarithm $n\min\{h(\alpha), \phi\} + o(n)$. The upper bound (AS5) has the same exponent, proving (AS3); subtracting $\log|\mathbb{F}_n|$ proves (AS4). Finally $w_n \log n = o(n)$, so $(a_n - k_n)/n \to 0$ and $\alpha = \rho$. □

Corollary 10.2 (Unconditional large-challenge RS–MCA threshold)**.** *Under the row hypotheses of Theorem 10.1, let the full-field target satisfy $\varepsilon_n = \exp(-\lambda n + o(n))$ for a finite $\lambda \ge 0$. If*

$$\lambda < \phi - h(\rho), \tag{AS7}$$

then, for every sufficiently large n,

$$a^*_n = k_n + 1, \qquad \delta^*_n = 1 - \frac{k_n + 1}{n} = 1 - \rho + o(1). \tag{AS8}$$

Conversely, if $\lambda > (\phi - h(\rho))_+$, every agreement sequence satisfying $(a_n - k_n - 1)\log|\mathbb{B}_n| = o(n)$ is unsafe. The critical boundary $\lambda = (\phi - h(\rho))_+$ is governed by the exact finite constants and is not decided by exponent asymptotics. This is a large scalar-extension regime. If $|\mathbb{F}_n|$ is polynomial in n, then $\phi = 0$ and (AS7) cannot hold; the corollary does not settle that Proximity-Prize parameter regime.

Proof. At $a = k_n + 1$, (AS5) is $B^{\mathrm{MCA}}_{C_n}(a) \le \binom{n}{k_n+1} = \exp(h(\rho)n + o(n))$. Under (AS7), $\varepsilon_n|\mathbb{F}_n| = \exp((\phi - \lambda)n + o(n))$ has strictly larger exponent, and taking the integer floor does not change the comparison. Thus the first permitted agreement is safe and (AS8) follows. The converse follows from (AS4) and the strict reverse exponential inequality. □

Part 2. Exact profile refinements and the profile envelope

11. Profile envelope and the sequel compiler

A realized first-match profile λ has a full support slice Ω^0_λ, a coefficient field $\mathbb{B}_\lambda$, a proved boundary map $\Phi_\lambda : \Omega^0_\lambda \to \mathbb{B}^{R_\lambda}_\lambda$, realized image size $L_\lambda = |\Phi_\lambda(\Omega^0_\lambda)|$, and image-normalized average

$$\bar{N}_\lambda = \frac{|\Omega^0_\lambda|}{L_\lambda}.$$

For a received line and agreement a, let $\Lambda(a)$ be the realized first-match profile set and define

$$\mathfrak{E}_n(a) = 1 + (n - a + 1) + \sup_{\text{received lines}} \sum_{\lambda \in \Lambda(a)} (1 + \bar{N}_\lambda). \tag{11.1}$$

The formal codomain size $|\mathbb{B}_\lambda|^{R_\lambda}$ may replace L_λ only after a full-image certificate $L_\lambda \geq e^{-o(n)}|\mathbb{B}_\lambda|^{R_\lambda}$ has been proved. Otherwise the leaf stays at realized-image scale or is routed to an earlier effective-image-collapse cell.

Definition 11.1 (Admissible sequel compiler)**.** *A row sequence is* admissible *when: (i) every bad witness belongs to an ordered first-match atlas with $e^{o(n)}$ realized profiles; (ii) all quotient, planted, field-drop, tangent, rank, extension, and directly counted ray cells have proved distinct-slope payments; (iii) every remaining primitive profile has a fixed-density Boolean support slice, an exact boundary map, and image normalization; (iv) its nontrivial effective Fourier characters satisfy the certified minor/major payment of Definitions 13.2, 13.6 and 13.9, or an equivalent direct Sidon payment; and (v) every residual ray chart has either a direct slope bound or the ray compiler of Hypothesis 22.13. All profile fields and quotient scales are included in* (11.1)*.*

Theorem 11.2 (Conditional profile-envelope compiler)**.** *Every admissible row sequence satisfies*

$$B^{\mathrm{MCA}}_{C_n}(a_n) \leq e^{o(n)}\mathfrak{E}_n(a_n). \tag{11.2}$$

For a finite row the same proof is literal after every $e^{o(n)}$ term is replaced by its exact integer constant and all first-match cells are summed once.

Proof. The first-match atlas pays the algebraic cells. The effective Fourier/Sidon step and Theorem 21.2 pay primitive Q, while the ray compiler and Proposition 22.14 convert the support bounds to distinct slopes. Summation over the realized profile set gives (11.2). The finite statement is the same disjoint first-match count with integer constants. □

Definition 11.3 (Complete-fiber folding map)**.** *Let $D \subseteq \mathbb{B}$ be finite. A polynomial map $\phi : D \to Q = \phi(D)$ is a c-fold complete-fiber folding map when every fiber has exactly c points. A support is complete at this scale when it is a union of full fibers. Multiplicative power maps and the Chebyshev twin-coset foldings of the foundation* [Cho26, Secs. 7 and 11] *are the motivating examples.*

12. Quotient–remainder prefix normal form

The quotient-prefix floors and descent mechanisms of CAP25 v13.2 are imported from [Cho26, Secs. 3, 8, and 14]. The normal forms and obstruction theorems below refine that package for the sequel's Q compiler.

The quotient and remainder variables can be separated exactly at the locator-prefix level. This removes a spurious factor coming from summing over marked remainder sets, but it also exposes a genuine scaled quotient- coefficient-field term which prevents an unrestricted comparison with the identity profile.

Definition 12.1 (Quotient/remainder profile and field drop)**.** *Let $\phi : D \to Q$ be a c-fold complete-fiber folding map in the sense of Definition 11.3. A* complete-fiber-plus-remainder support *is*

$$S = \phi^{-1}(E) \sqcup R,$$

where $E \subseteq Q$ and $R \subseteq D \setminus \phi^{-1}(E)$. The set R is the support remainder*: it records the selected points in fibers that are not declared complete. A* quotient/remainder profile *fixes ϕ, its degree c, the size $r = |R|$, the relevant coefficient field, and any planted remainder data, while E and the unfixed part of R range subject to the displayed disjointness. A fixed-R profile is the subprofile in which the remainder support itself is part of the label. The adjective* Euclidean *used below refers to separating the locator factor of degree $r < c$ from the composed quotient locator; it does not mean that an arbitrary support remainder has size less than c.*

We call $\mathbb{B}_\phi \subseteq \mathbb{B}$ a scaled quotient coefficient field *when, for some $\eta \in \mathbb{B}^\times$,*

$$Q = \phi(D) \subseteq \eta\mathbb{B}_\phi.$$

When this field is recorded in a profile, it is taken minimal among the subfields with this property. Then the j-th elementary quotient coefficient lies in $\eta^j\mathbb{B}_\phi$. A field drop *is the use of a proper* *such subfield. It is a* positive-rate field drop *along a sequence when*

$$1 - \frac{\log|\mathbb{B}_\phi|}{\log|\mathbb{B}|}$$

stays bounded away from zero and a positive-density number on the entropy scale of quotient-prefix slots is live. The payment saved in each live slot is the factor $|\mathbb{B}|/|\mathbb{B}_\phi|$; this is the term quantified in Proposition 12.12.

Definition 12.2 (Balanced quotient core)**.** *A* split locator *is a monic squarefree polynomial all of whose roots lie in the evaluation domain D. After common locator factors, fixed quotient data, and planted remainder points have been removed, let A and B be two monic split locators of the same degree e. They form a* depth-w balanced core *if they have the same first w nonleading coefficients, equivalently*

$$\deg(A - B) \le e - w - 1.$$

It is a balanced quotient core *when the remaining locator factors are pullbacks through the same folding map ϕ. The word* balanced *records equality of degree and leading normalization; the* core *consists of the roots still allowed to move after common and planted factors are removed. If the resulting projective locator family is one-dimensional it is a split pencil $Q_0 + \lambda Q_1$; in higher dimension this definition supplies no distinct-ray bound, and the separate ray compiler is required.*

For a monic polynomial

$$A(X) = X^d + a_1 X^{d-1} + \cdots + a_d$$

and an integer $0 \le u \le d$, write $\mathrm{pref}_u(A) = (a_1, \ldots, a_u)$. Equivalently, if $A^\vee(Z) := Z^d A(Z^{-1})$, then $\mathrm{pref}_u(A)$ is the coefficient vector of $A^\vee(Z) \bmod Z^{u+1}$, with its constant coefficient omitted.

Theorem 12.3 (Exact quotient–remainder prefix normal form)**.** *Let $\mathbb{B} \subseteq \mathbb{F}$ be finite fields, let $D \subseteq \mathbb{B}$ have n distinct elements, and let $\phi \in \mathbb{B}[X]$ be monic of degree $c \ge 2$. Assume that the restriction*

$$\phi : D \longrightarrow Q := \phi(D)$$

has complete fibers of size c; in particular, $N := |Q| = n/c$. Fix integers m, r with $0 \le r < c$ and $cm + r \le n$. For

$$E \in \binom{Q}{m}, \qquad R \in \binom{D \setminus \phi^{-1}(E)}{r},$$

put

$$\begin{aligned} V_E(Y) &:= \prod_{y \in E} (Y - y) = Y^m + \sum_{j=1}^{m} v_j(E) Y^{m-j}, \\ P_R(X) &:= \prod_{x \in R} (X - x) = X^r + \sum_{i=1}^{r} p_i(R) X^{r-i}, \qquad \text{(QR1)} \\ L_{E,R}(X) &:= P_R(X) V_E(\phi(X)). \end{aligned}$$

Then $L_{E,R}$ is the monic locator of the support $\phi^{-1}(E) \sqcup R$, of degree $A = cm + r$.

Let $0 \le w \le A$ and put $d = \lfloor w/c \rfloor$.

(i) *If $w \ge r$, every nonempty fiber of $(E, R) \mapsto \mathrm{pref}_w(L_{E,R})$ determines R uniquely and, after R is recovered, is exactly one quotient-prefix fiber. More precisely, for every prefix value z there are unique data R_z and $\eta_z \in \mathbb{B}^d$, whenever the fiber is nonempty, such that*

$$\begin{aligned} &\{(E, R) : \mathrm{pref}_w(L_{E,R}) = z\} \\ &\quad = \{(E, R_z) : E \in \tbinom{Q \setminus \phi(R_z)}{m},\ \mathrm{pref}_d(V_E) = \eta_z\}. \qquad \text{(QR2)} \end{aligned}$$

Thus no factor counting possible remainder sets occurs in a fixed global prefix fiber.

(ii) *If $w < r$, then necessarily $w < c$, and the quotient set E is invisible to the first w coefficients. There is a fixed unitriangular bijection $T_{\phi,m,w} : \mathbb{B}^w \to \mathbb{B}^w$ such that*

$$\operatorname{pref}_w(L_{E,R}) = T_{\phi,m,w}(\operatorname{pref}_w(P_R)) \tag{QR3}$$

for every admissible (E, R). Consequently, with the convention $\binom{u}{v} = 0$ for $v > u$,

$$\begin{aligned} &\#\{(E, R) : \operatorname{pref}_w(L_{E,R}) = z\} \\ &= \sum_{\substack{R\in\binom{D}{r} \\ \operatorname{pref}_w(P_R)=T^{-1}_{\phi,m,w}(z)}} \binom{N - |\phi(R)|}{m}. \end{aligned} \tag{QR4}$$

Thus this case is a remainder-prefix problem, multiplied by the number of quotient sets still available after R is fixed.

Proof. For a monic degree-e polynomial A, set $A^\vee(Z) = Z^e A(Z^{-1})$. Also set

$$\Phi(Z) := Z^c\phi(Z^{-1}) = 1 + \phi_1 Z + \cdots + \phi_c Z^c.$$

Taking reciprocal polynomials in (QR1) gives the exact formal-power-series identity

$$L^\vee_{E,R}(Z) = P^\vee_R(Z)\left(\Phi(Z)^m + \sum_{j=1}^{m} v_j(E)Z^{cj}\Phi(Z)^{m-j}\right). \tag{QR5}$$

All factors on the right have constant coefficient one except for the displayed scalar coefficients $v_j(E)$.

Suppose first that $w \geq r$. Since $r < c$, no term containing a v_j occurs below degree c. For $1 \leq i \leq r$, the coefficient of Z^i on the right of (QR5) is $p_i(R)$ plus an expression depending only on $p_1(R), \ldots, p_{i-1}(R)$, on m, and on Φ. The first r coefficients therefore recover $p_1(R), \ldots, p_r(R)$ successively. They recover the monic polynomial P_R, hence its root set R, uniquely.

We may now divide $L^\vee_{E,R}$ by the unit $P^\vee_R$ modulo Z^{w+1}. In the remaining factor of (QR5), the coefficient of Z^{cj} is $v_j(E)$ plus an expression depending only on $v_1(E), \ldots, v_{j-1}(E)$, on m, and on Φ. Hence the coefficients at degrees $c, 2c, \ldots, dc$ recover $v_1(E), \ldots, v_d(E)$ successively. Conversely, P_R and these d coefficients determine the right side of (QR5) modulo Z^{w+1}, because every term containing v_j with $j > d$ starts in degree $cj > w$. This proves (QR2); admissibility of E is exactly the condition $E \cap \phi(R) = \varnothing$.

If $w < r < c$, every term involving a v_j starts in degree at least $c > w$, so modulo Z^{w+1} the bracket in (QR5) is the fixed unit $\Phi(Z)^m$. Multiplication by this unit sends the first w coefficients of $P^\vee_R$ to the first w coefficients of $L^\vee_{E,R}$ by a unitriangular transformation. It is independent of E and invertible. For a fixed R, the admissible E are precisely the m-subsets of $Q \setminus \phi(R)$, of which there are $\binom{N-|\phi(R)|}{m}$. Summing over the indicated remainder-prefix fiber proves (QR4). □

Proposition 12.4 (Complete support factorization and its limitation)**.** *Under the complete-fiber hypotheses of Theorem 12.3, every support $S \subseteq D$ has the unique decomposition*

$$E(S) = \{y \in Q : \phi^{-1}(y) \subseteq S\}, \qquad R(S) = S \setminus \phi^{-1}(E(S)), \tag{QR5a}$$

and $Q_S(X) = P_{R(S)}(X)V_{E(S)}(\phi(X))$. The remainder meets every nonfull ϕ-fiber in at most $c-1$ points, but its total size may be $\Theta(n)$. The reciprocal identity (QR5) remains valid for this arbitrary remainder. When $|R(S)| \geq c$, however, the quotient coefficients and the remainder coefficients begin to interlace at degree c, so the triangular recovery theorem above does not give a comparison without an additional partial-occupancy atlas.

Proof. A fiber is put into $E(S)$ exactly when all of its points lie in S; all other selected points form $R(S)$. This proves existence, disjointness, the occupancy bound, and uniqueness. Multiplying the corresponding linear factors gives the locator identity. The reciprocal calculation used for (QR5) is purely multiplicative and does not require $\deg P_R < c$. If $\deg P_R \geq c$, both $p_c(R)$ and $v_1(E)$ occur in degree c, which proves the stated limitation of the triangular separation. □

Theorem 12.5 (Exact partial-occupancy disintegration)**.** *Let $D = D_0 \sqcup X$, where $|X| = b$, and let $\phi : D_0 \twoheadrightarrow Q$ have exactly $N = |Q|$ fibers, each of cardinality c. For $S \subseteq D$, put*

$$X_S = S \cap X, \qquad E(S) = \{y \in Q : \phi^{-1}(y) \subseteq S\}, \qquad R(S) = (S \cap D_0) \setminus \phi^{-1}(E(S)).$$

Write

$$t = |X_S|, \qquad m = |E(S)|, \qquad p = |\phi(R(S))|, \qquad r = |R(S)|.$$

Then this datum is unique, every fiber meeting $R(S)$ contains between one and $c-1$ points of $R(S)$, and $|S| = t + cm + r$. If $\Omega_{t,m,p,r}$ denotes the family of supports with these four parameters, then

$$|\Omega_{t,m,p,r}| = \binom{b}{t}\binom{N}{p}\binom{N-p}{m}[x^r]((1+x)^c - 1 - x^c)^p. \tag{PO1}$$

Consequently the nonempty families $\Omega_{t,m,p,r}$ partition $\binom{D}{a}$, and their exact add-back is

$$\sum_{\substack{t,m,p,r\\ t+cm+r=a}} |\Omega_{t,m,p,r}| = \binom{cN+b}{a}. \tag{PO2}$$

If ϕ is the restriction of a monic degree-c polynomial and every fiber $\phi^{-1}(y)$ is the complete simple root set in D_0 of $\phi(X) - y$, then their locators satisfy

$$Q_S = Q_{X_S} P_{R(S)} V_{E(S)}(\phi).$$

Proof. Full fibers, partial fibers, and empty fibers are mutually exclusive, so the displayed decomposition is canonical. Choose the t exceptional points, the p partial fibers, and the m full fibers in $\binom{b}{t}\binom{N}{p}\binom{N-p}{m}$ ways. On one partial fiber the weight enumerator of the allowed nonempty proper subsets is $(1+x)^c - 1 - x^c$; hence the coefficient in (PO1) counts the choices having total partial weight r. Summing (PO1) and extracting the coefficient of x^a gives

$$[x^a](1+x)^b \left(1 + x^c + ((1+x)^c - 1 - x^c)\right)^N = [x^a](1+x)^{b+cN} = \binom{b+cN}{a}.$$

Multiplying the linear factors belonging to the three disjoint parts of the support proves the locator identity. ☐

Proposition 12.6 (Partial-occupancy Fourier factorization)**.** *In the setting of Theorem 12.5, let G be a finite abelian group, let $g : D \to G$, and put $\Phi(S) = \sum_{x\in S} g(x)$. For a character $\chi \in \widehat{G}$ and a regular fiber $F_y = \phi^{-1}(y)$, define*

$$A_y(\chi) = \prod_{x\in F_y} \chi(g(x)), \qquad P_y(\chi; z) = \prod_{x\in F_y} (1 + z\chi(g(x))) - 1 - z^c A_y(\chi).$$

Then

$$\sum_{S\in\Omega_{t,m,p,r}} \chi(\Phi(S)) = [u^t v^m \zeta^p z^r] \prod_{x\in X} (1 + u\chi(g(x))) \prod_{y\in Q} (1 + vA_y(\chi) + \zeta P_y(\chi; z)). \tag{PO3}$$

Consequently, for every $s \in G$,

$$|\{S \in \Omega_{t,m,p,r} : \Phi(S) = s\}| = \frac{1}{|G|} \sum_{\chi\in\widehat{G}} \overline{\chi(s)} \sum_{S\in\Omega_{t,m,p,r}} \chi(\Phi(S)). \tag{PO4}$$

At the trivial character, (PO3) *is exactly the support count* (PO1)*. Thus support add-back is unconditional, while a finite flatness theorem is exactly a bound for the nontrivial-character sum in* (PO4)*.*

Proof. For a fixed regular fiber the three terms in the second product encode, respectively, an empty fiber, a full fiber, and a nonempty proper subset. The variables v, ζ, z record the numbers m, p, r, while u records the exceptional weight t. Multiplicativity of χ turns the character of the support sum into the product of its selected-point characters, proving (PO3) by coefficient extraction. Formula (PO4) is character orthogonality on G. ☐

Remark 12.7 (Effective normalization of a partial-occupancy slice). Fix a nonempty slice $\Omega_\lambda = \Omega_{t,m,p,r}$, choose $S_0 \in \Omega_\lambda$, and let

$$G_\lambda = \langle \Phi(S) - \Phi(S_0) : S \in \Omega_\lambda \rangle \le G.$$

The attained boundary values lie in the affine coset $\Phi(S_0) + G_\lambda$, and the effective inversion is

$$\text{(PO5)} \qquad |\{S \in \Omega_\lambda : \Phi(S) = s\}| = \frac{1}{|G_\lambda|} \sum_{\chi \in \widehat{G_\lambda}} \overline{\chi(s - \Phi(S_0))} \sum_{S \in \Omega_\lambda} \chi(\Phi(S) - \Phi(S_0)).$$

Every character of G_λ extends to the ambient finite abelian group, so the inner sum may be evaluated from (PO3) after multiplication by the fixed factor $\overline{\chi(\Phi(S_0))}$. Different extensions agree on support differences. Equivalently, ambient characters group by their restriction to G_λ, and the annihilator multiplicity cancels exactly against the ambient denominator. This identifies the correct Fourier denominator; it does not assert that the realized image fills the affine group.

Remark 12.8 (What partial-occupancy add-back does and does not prove). The exponentially many local proper-subset choices are already contained inside the coarse slices $\Omega_{t,m,p,r}$; they incur no separate profile-count loss in (PO2). The theorem does not bound the nontrivial Fourier terms in (PO4), the size of a realized boundary image, or the projection of a witness cell to distinct slopes. Those are separate analytic and ray payments.

Theorem 12.9 (Canonical partial-occupancy atlas and bounded-boundary closure)**.** *Fix a received line and exact agreement a. For every nonempty canonical slice $\Omega_\lambda = \Omega_{t,m,p,r}$ of Theorem 12.5, let $\mathcal{C}_\lambda$ be the witnesses whose support lies in Ω_λ, and order these cells arbitrarily. Then they form a witness-exhaustive first-match atlas and, with Z_λ° its first-match slope projections,*

$$\text{(PO6)} \qquad |Z_\lambda^\circ| \le |\Omega_\lambda|, \qquad \sum_\lambda |Z_\lambda^\circ| \le \sum_\lambda |\Omega_\lambda| = \binom{n}{a}.$$

More generally, give each slice a boundary map $\Phi_\lambda : \Omega_\lambda \to G_\lambda^{\mathrm{amb}}$ into a finite abelian group of order A_λ, let N_λ be its active-coordinate count, let L_λ be its realized image size, and put $\bar{N}_\lambda = |\Omega_\lambda|/L_\lambda$. The exact finite ray multiplier and challenge-set bound are

$$\text{(PO7)} \qquad |Z_\lambda^\circ| \le L_\lambda \bar{N}_\lambda.$$

$$\text{(PO7a)} \qquad B_{C,\Gamma}^{\mathrm{MCA}}(a) \le \min\left\{ |\Gamma|,\ \sup_r \sum_{\lambda \in \Lambda(r;a)} L_\lambda \bar{N}_\lambda \right\} \le \min\left\{ |\Gamma|, \binom{n}{a} \right\}.$$

Define the coarse partial-occupancy envelope

$$\mathfrak{E}_n^{\mathrm{PO}}(a) = 1 + \sup_r \sum_{\lambda \in \Lambda(r;a)} (1 + \bar{N}_\lambda).$$

If, uniformly over received lines and the polynomially many nonempty canonical slices,

$$\text{(PO8)} \qquad \log A_\lambda = o(n),$$

then this atlas has a closed asymptotic profile-envelope compiler:

$$\text{(PO9)} \qquad B_C^{\mathrm{MCA}}(a) \le e^{o(n)} \mathfrak{E}_n^{\mathrm{PO}}(a).$$

Condition (PO8) *discharges atlas exhaustivity, support add-back, and the direct alternative in* (RC)*, whose exact multiplier is $L_\lambda \le A_\lambda$. If, in addition,*

$$\text{(PO10)} \qquad \log A_\lambda = o(N_\lambda)$$

on every slice to which effective Fourier terminology is applied, then the effective inversion (PO5) *may designate every nontrivial effective character major: its minor sum is empty and its exact major multiplier is at most $A_\lambda - 1$. Thus* (PO10) *verifies the slice analogues of MI and MA rather than assuming them. A concrete specialization is a polynomial-size coefficient field with uniformly bounded boundary depth and $N_\lambda = \Theta(n)$ on every live canonical folding profile. This is a coarse support-paid envelope, not an identity-scale or refined quotient-envelope comparison. It gives a safe target whenever the literal right side of* (PO7a) *is at most B^*; a sharp frontier* *additionally uses the unsafe inequality in* (SB2)*. Condition* (PO8) *alone does not locate a target frontier.*

Proof. The support slices partition $\binom{D}{a}$ by (PO2), hence the corresponding witness cells partition the exact witness incidence. For each first-match slope choose one witness support in its cell. A fixed noncommon support carries at most one slope, so this choice is injective and proves the first inequality in (PO6). Exact add-back gives the equality there, and (PO7) is the same inequality rewritten using $|\Omega_\lambda| = L_\lambda \bar{N}_\lambda$.

The exact challenge bound follows by summing (PO7), taking the maximum over received lines, and using (PO2). Since $L_\lambda \le A_\lambda$, condition (PO8) turns (PO7) into $e^{o(n)}\bar{N}_\lambda$. There are only polynomially many quadruples (t, m, p, r), so their sum proves (PO9). If $A_\lambda \le |\mathbb{B}|^{R_\lambda}$, a polynomial-size $\mathbb{B}$ and bounded R_λ imply (PO8). Finally, the effective difference group in (PO5) has at most A_λ characters. Under (PO10), designating all of its nontrivial characters major makes the minor sum empty and bounds the normalized major aggregate by $A_\lambda - 1 = e^{o(N_\lambda)}$. □

Remark 12.10 (Exact dimension rounding)*.* If the global locator degree is $A = cm + r$, the list-code dimension is K, and $w = A - K$, then the quotient depth in Theorem 12.3 is

$$d = \left\lfloor \frac{A-K}{c} \right\rfloor = m - \left\lceil \frac{K-r}{c} \right\rceil . \tag{QR5b}$$

Thus the descended list dimension is exactly $K_c = \lceil (K-r)/c \rceil$. More explicitly, if $w = cu + s$ with $0 \le s < c$, then

$$|\mathbb{B}_\phi|^{-\lfloor w/c \rfloor} = |\mathbb{B}_\phi|^{s/c} |\mathbb{B}_\phi|^{-w/c}. \tag{QR5c}$$

Thus the exact rounding multiplier is $|\mathbb{B}_\phi|^{s/c}$, lying between one and $|\mathbb{B}_\phi|^{(c-1)/c}$. Retaining the integer $\lfloor w/c \rfloor$ avoids even this comparison; no polynomial or $e^{o(n)}$ placeholder is needed in a finite certificate.

Remark 12.11 (Nonmonic maps and Chebyshev maps)*.* If the leading coefficient of ϕ is $b \ne 0$, replace ϕ by $b^{-1}\phi$ and Q by $b^{-1}Q$. This changes quotient coefficients by fixed nonzero scalar factors and leaves every support and every fiber cardinality unchanged. The theorem therefore applies to the Chebyshev folding map T_c after this normalization. On a circle twin-coset domain on which T_c has complete fibers of size c, it gives the same statement verbatim. The fixed or ramified endpoints form an exceptional set X of cardinality b and are included exactly by the factor $\binom{b}{t}$ in (PO1); the crude total exceptional-point multiplier is at most 2^b, with $b \le 2$ for involution-fixed endpoints. Coincident twin cosets are not a bounded planted set: they form the separate dihedral collapse profile of the circle edge-case audit imported from [Cho26, Sec. 11].

For a declared complete-fiber-plus-Euclidean-remainder profile, the exact normal-form theorem separates algebraic bookkeeping from prefix equidistribution only when the remainder degree is less than c. The exact disintegration of Theorem 12.5 exhausts arbitrary partial-occupancy supports, but when $\deg P_R \ge c$ the quotient and remainder coefficients interlace. Thus that disintegration does not by itself prove prefix flatness on the resulting coarse slices. Let $\mathbb{B}_\phi \subseteq \mathbb{B}$ be a subfield and suppose $Q \subseteq \eta \mathbb{B}_\phi$ for some $\eta \in \mathbb{B}^\times$. Then $v_j(E) \in \eta^j \mathbb{B}_\phi$. Hence, in case (i), the natural average scale of the quotient-prefix fiber with fixed R is

$$\overline{N}_{\phi,R}(w) := \binom{N - |\phi(R)|}{m} |\mathbb{B}_\phi|^{-\lfloor w/c \rfloor}. \tag{QR6}$$

Pigeonholing proves a fiber of size at least $\lceil \overline{N}_{\phi,R}(w) \rceil$. An upper bound of the same order is not a consequence of the normal form: it is precisely the uniform prefix-flatness, or Sidon/Fourier, input on the quotient row.

Proposition 12.12 (Identity–quotient exponent comparison)**.** *Consider a sequence to which Theorem 12.3 applies with fixed $c \ge 2$ and fixed $0 \le r < c$. Suppose*

$$\frac{a}{n} \longrightarrow \alpha \in (0,1), \qquad \log|\mathbb{B}| = o(n), \qquad \log\binom{n}{a} - w\log|\mathbb{B}| = o(n), \tag{QR7}$$

where $a = cm + r$, and suppose $Q \subseteq \eta\mathbb{B}_c$ for a subfield $\mathbb{B}_c \subseteq \mathbb{B}$. Write $\overline{N}_{c,r}(w) = \overline{N}_{\phi,R}(w)$ for the natural scale in (QR6), with this fixed remainder profile. Then $w \to \infty$, so $w \geq r$ eventually, and

$$\log \overline{N}_{c,r}(w) = \frac{1}{c} \log\left(\binom{n}{a}|\mathbb{B}|^{-w}\right) + \frac{w}{c} \log \frac{|\mathbb{B}|}{|\mathbb{B}_c|} + o(n), \tag{QR8}$$

where replacing $N - |\phi(R)|$ by N changes only the $o(n)$ term. Equivalently, if $\lambda_c = \log|\mathbb{B}_c| / \log|\mathbb{B}|$ has a limit, then at the identity crossing

$$\frac{1}{n} \log \overline{N}_{c,r}(w) = \frac{h(\alpha)}{c}(1 - \lambda_c) + o(1), \tag{QR9}$$

using the entropy convention fixed in the Introduction. Thus the natural average scale of this bounded-degree quotient profile is comparable with the identity scale at exponential accuracy exactly when $\log|\mathbb{B}_c| / \log|\mathbb{B}| = 1 - o(1)$. A proper field drop of fixed relative degree produces a positive exponential quotient term.

Proof. Stirling's formula, $a/n \to \alpha$, and fixed c, r give

$$\log\binom{n/c + O(1)}{(a-r)/c} = \frac{1}{c}\log\binom{n}{a} + o(n).$$

Also $\lfloor w/c \rfloor \log|\mathbb{B}_c| = (w/c)\log|\mathbb{B}_c| + o(n)$, because the rounding error is at most $\log|\mathbb{B}_c| \leq \log|\mathbb{B}| = o(n)$. This proves (QR8). Since $\log\binom{n}{a} = nh(\alpha) + o(n)$, (QR7) gives $w \log|\mathbb{B}| = nh(\alpha) + o(n)$. In particular $w \to \infty$ because $\log|\mathbb{B}| = o(n)$. Substitution in (QR8) proves (QR9). □

Remark 12.13 (Unbounded quotient degrees). If $c = c_n \to \infty$, the complete-fiber choice contributes at most $2^{n/c} = e^{o(n)}$. When $w \geq r$, the remainder is recovered exactly and no further remainder multiplicity occurs. When $w < r < c$, the exact formula (QR4) reduces the profile to an ordinary prefix problem for the remainder support, multiplied by $e^{o(n)}$ quotient choices. Thus only bounded quotient degrees can create a new positive-rate field-drop term; large degrees either cost $e^{o(n)}$ or return to an identity/remainder prefix problem. This does not prove flatness of the remaining prefix problem.

13. Effective-image Fourier normalization

There is, however, an exact normalization that must be performed before declaring any character major. Assume $1 \leq m \leq |T| - 1$, as on every fixed-density primitive leaf, put $p = \operatorname{char}\mathbb{B}$, and fix $S_0 \in \binom{T}{m}$ and $t_0 \in T$. Then

$$\operatorname{Span}_{\mathbb{F}_p}\{\Psi(S) - \Psi(S_0) : S \in \tbinom{T}{m}\} = \operatorname{Span}_{\mathbb{F}_p}\{g(t) - g(t_0) : t \in T\}. \tag{EF0}$$

Define the *effective Fourier span*

$$V_g = \operatorname{Span}_{\mathbb{F}_p}\{g(t) - g(t_0) : t \in T\} \subseteq \mathbb{B}^R, \qquad A_{\text{eff}} = |V_g|. \tag{EF1}$$

The fixed-weight sum map takes values in the affine space $mg(t_0) + V_g$. Thus Fourier inversion belongs on the additive group V_g, not automatically on the ambient group $\mathbb{B}^R$.

Lemma 13.1 (Effective-span Fourier inversion). *For $z \in V_g$, let*

$$N_g(z) = \left|\left\{S \in \binom{T}{m} : \sum_{t \in S} g(t) = mg(t_0) + z\right\}\right|.$$

Then

$$N_g(z) = \frac{1}{A_{\text{eff}}} \sum_{\chi \in \widehat{V_g}} \overline{\chi(z)} e_m(\chi(g(t) - g(t_0)) : t \in T). \tag{EF2}$$

In particular,

$$\max_z N_g(z) \le \frac{1}{A_{\mathrm{eff}}}\left(\binom{|T|}{m} + \sum_{1\ne\chi\in\widehat{V_g}} \left|e_m\big(\chi(g(t)-g(t_0)) : t\in T\big)\right|\right). \tag{EF3}$$

Every character of V_g extends to a trace character of $\mathbb{B}^R$, and a nontrivial effective character is not constant on all the generators $g(t)-g(t_0)$. Ambient characters that are trivial on V_g occur with multiplicity $|\mathbb{B}|^R/A_{\mathrm{eff}}$, which cancels exactly against the ambient Fourier normalization.

Proof. The inclusion from left to right in (EF0) follows by expanding a difference of two support sums. Conversely, for distinct $t,u\in T$, choose an $(m-1)$-set $A\subseteq T\setminus\{t,u\}$; such a set exists because $1\le m\le |T|-1$. Then $\Psi(A\cup\{t\})-\Psi(A\cup\{u\}) = g(t)-g(u)$, so the two spans agree. The first formula is character orthogonality on the finite additive group V_g, applied after translating by $mg(t_0)$; the second follows by the triangle inequality. The trace pairing on the $\mathbb{F}_p$-space $\mathbb{B}^R$ is nondegenerate, so every character of its subspace V_g extends to the ambient space. A character trivial on each displayed generator is trivial on their span. Finally, the annihilator $V_g^{\perp}$ has size $|\mathbb{B}|^R/A_{\mathrm{eff}}$; grouping ambient characters by their restriction to V_g proves the cancellation claim. □

Definition 13.2 (Certified effective major/minor partition)**.** *Let* $\mathrm{res} : \widehat{\mathbb{B}^R} \twoheadrightarrow \widehat{V_g}$ *be restriction. A* certified effective Fourier partition *is a disjoint union*

$$\widehat{V_g}\setminus\{1\} = \mathfrak{m}_{\mathrm{eff}} \sqcup \mathfrak{M}_{\mathrm{eff}}$$

such that every $\chi\in\mathfrak{m}_{\mathrm{eff}}$ is supplied with an ambient lift whose rational phase satisfies the character-sum hypotheses used to pay it. Characters without such a certified lift may be placed in $\mathfrak{M}_{\mathrm{eff}}$, but their aggregate must then be paid by (MA)*. This designation is data of the chart; it is not a property of an unspecified ambient lift.*

Definition 13.3 (Effective Fourier payment)**.** *The full slice has an* effective Fourier payment with constant $\kappa\ge 1$ *if*

$$\sum_{1\ne\chi\in\widehat{V_g}} \left|e_m\big(\chi(g(t)-g(t_0)) : t\in T\big)\right| \le (\kappa-1)\binom{|T|}{m}. \tag{EFP}$$

By Lemma 13.1, this implies the exact max-fiber bound

$$\max_z N_g(z) \le \kappa\binom{|T|}{m}/A_{\mathrm{eff}}. \tag{EF4}$$

It also implies that the realized image contains at least A_{eff}/κ points. The ambient instances of (PF) *and* (MA) *are one sufficient way to prove* (EFP)*, but* (EFP) *is phase-dependent and does not pay an ambient annihilator twice.*

Proof of the image-size assertion. The fibers sum to $\binom{|T|}{m}$, and each nonempty fiber is at most $\kappa\binom{|T|}{m}/A_{\mathrm{eff}}$ by (EF4). Hence at least A_{eff}/κ fibers are nonempty. □

Lemma 13.4 (Constant and Artin–Schreier modes are annihilator modes)**.** *Suppose an ambient trace character has phase constant on T. Then its restriction to V_g is trivial. In particular, if an ambient rational phase is Artin–Schreier of the form G^p-G+c and G is defined at the $\mathbb{B}$-rational points of T, then its character is an annihilator mode and disappears from the nontrivial sum in* (EF2)*.*

Proof. The quotient of the character values at $g(t)$ and $g(t_0)$ is one for every t, so the character is trivial on the generators of V_g. For an Artin–Schreier phase, $\mathrm{Tr}_{\mathbb{B}/\mathbb{F}_p}(G(t)^p-G(t))=0$, hence its additive-character value is constant on all points where it is defined. □

Remark 13.5 (Effective normalization does not prove the remaining aggregate)*.* The effective-span reduction removes exactly the constant annihilator multiplicity. A nonconstant character may still factor through a quotient map or have a phase-dependent estimate too weak for (EFP). Such modes require a recursive quotient estimate, the aggregate (MA), or a direct full-slice bound. Thus effective normalization repairs the Fourier denominator but is not a proof of unrestricted deep-prefix flatness.

Definition 13.6 (Major-character aggregate)**.** *Let A_* be the additive target on which Fourier inversion is being performed, let $h : T \to A_*$, and let $\mathfrak{M}_* \subseteq \widehat{A_*} \setminus \{1\}$ be the designated major characters. The chart satisfies* (MA) *on A_* if*

$$\sum_{\chi\in\mathfrak{M}_*} \left|e_m\big(\chi(h(t)) : t \in T\big)\right| \le e^{o(|T|)}\binom{|T|}{m}. \tag{MA}$$

Unless the ambient target is explicitly named, $A_* = V_g$ and $h(t) = g(t) - g(t_0)$. For the ambient target $A_* = \mathbb{B}^R$, the same condition is equivalently

$$|\mathbb{B}|^{-R} \sum_{\alpha\in\mathfrak{M}_{\mathrm{amb}}} \left|e_m\big(\psi(\alpha \cdot g(t)) : t \in T\big)\right| \le e^{o(|T|)}\overline{N}_0.$$

The condition is vacuous when the designated major set is empty. It can be proved by a recursive quotient census or replaced by a direct full-slice max-fiber bound; first-match terminology alone does not imply it.

Lemma 13.7 (Sparse effective major set)**.** *On the effective group V_g, let $\mathfrak{M}_{\mathrm{eff}}$ be any set of nontrivial characters designated major. Its exact normalized Fourier loss is at most $|\mathfrak{M}_{\mathrm{eff}}|$. Consequently $|\mathfrak{M}_{\mathrm{eff}}| = e^{o(|T|)}$ proves the effective form of* (MA)*; for a finite row, the literal cardinality is a valid major-character constant.*

Proof. Every fixed-weight Fourier coefficient is a sum of $\binom{|T|}{m}$ complex numbers of modulus one, and hence has absolute value at most that binomial coefficient. Sum this bound over the designated set. □

Theorem 13.8 (Exact low-boundary-entropy closure)**.** *Let $\Omega = \binom{T}{m}$, let W be a finite additive $\mathbb{F}_p$-space, and let $\Psi : \Omega \to W$ be a fixed-weight additive boundary map. Fix $S_0 \in \Omega$, put $z_0 = \Psi(S_0)$, and define the effective difference span*

$$V = \mathrm{Span}_{\mathbb{F}_p}\{\Psi(S) - \Psi(S_0) : S \in \Omega\} \subseteq W.$$

Then $\Psi(\Omega) \subseteq z_0 + V$. Put

$$A = |V|, \qquad L = |\Psi(\Omega)|, \qquad M = \binom{|T|}{m}, \qquad \bar{N} = M/L.$$

Designate every nontrivial character of V as major and no character as minor. Then the literal finite-row constants are

$$C_{\min} = 0, \qquad C_{\mathrm{maj}} \le A - 1, \qquad \max_z|\Psi^{-1}(z)| \le L\bar{N} = M. \tag{SE1}$$

For every exact-agreement witness cell whose support projection is contained in Ω, its first-match slope set satisfies

$$|Z| \le L\bar{N} = M. \tag{SE2}$$

Consequently, if $\log A = o(|T|)$, effective (MI)*, effective* (MA)*, image-normalized Q, and the direct alternative in* (RC) *all hold with subexponential loss. Taking the support-restricted incidence over Ω as one ordered cell is exhaustive for that restricted incidence; it is exhaustive for the whole exact-agreement incidence when $\Omega = \binom{D}{a}$, or after an exhaustive partition into such slices. For a locator prefix with $|T| = \Theta(n)$, the concrete condition $(a - k - 1)\log|\mathbb{B}| = o(n)$ implies this hypothesis because $A \le |\mathbb{B}|^{a-k-1}$.*

Proof. Fourier inversion is performed on the effective group V. The minor sum is empty. Every fixed-weight Fourier coefficient has modulus at most M, so the normalized major loss is at most $A - 1$. The max-fiber estimate in (SE1) is the trivial bound by the whole support slice, written as the exact image-normalized multiplier L. By exact-agreement reduction, every bad slope in the cell has a noncommon m-support, and a fixed noncommon support carries at most one slope.

Choosing one support for each slope is therefore injective and proves (SE2). Since $L \le A$, all three finite multipliers $A-1, L, L$ are $e^{o(|T|)}$ when $\log A = o(|T|)$. The one-cell atlas is exhaustive for its support-restricted incidence by construction; the stated whole-incidence alternatives give global exhaustivity. If it is refined by a partial-occupancy partition, (PO1)–(PO2) add the same support set back exactly and do not change the conclusion. ☐

Definition 13.9 (Aggregate minor payment)**.** *After effective-span normalization, let* $\mathfrak{m}_{\mathrm{eff}}$ *be the nontrivial characters not designated major. The chart satisfies* (MI) *if*

$$\sum_{\chi \in \mathfrak{m}_{\mathrm{eff}}} |e_m(\chi(g(t)-g(t_0)) : t \in T)| \le e^{o(|T|)} \binom{|T|}{m}. \tag{MI}$$

For a finite row the corresponding exact loss is this sum divided by $\binom{|T|}{m}$*. The pointwise condition* (PF) *is only one sufficient way to prove* (MI)*; phase-dependent estimates may be strictly stronger.*

Proposition 13.10 (Effective MI plus MA)**.** *Partition the nontrivial effective characters as* $\widehat{V_g} \setminus \{1\} = \mathfrak{m}_{\mathrm{eff}} \sqcup \mathfrak{M}_{\mathrm{eff}}$*. Suppose* (MI) *and the effective form of* (MA) *in Definition 13.6 hold. Then the full slice and every first-match residual satisfy*

$$\max_z |\{S : \Psi(S) = z\}| \le e^{o(|T|)} \binom{|T|}{m} / A_{\mathrm{eff}}. \tag{EF5}$$

The realized image has size at least $e^{-o(|T|)} A_{\mathrm{eff}}$*, so the same assertion holds at image normalization. For a finite row the exact multiplier is*

$$1 + \binom{|T|}{m}^{-1} \sum_{1 \ne \chi \in \widehat{V_g}} |e_m(\chi(g(t)-g(t_0)) : t \in T)|. \tag{EF6}$$

Proof. Add the minor and major sums and apply Lemma 13.1 and Definition 13.3. A first-match residual only deletes supports from a full-slice fiber. ☐

Corollary 13.11 (Exact finite minor and major constants)**.** *Put* $M = \binom{|T|}{m}$ *and, for an arbitrary partition of the nontrivial effective dual, define*

$$C_{\min} = M^{-1} \sum_{\chi \in \mathfrak{m}_{\mathrm{eff}}} |E_m(\chi)|, \qquad C_{\mathrm{maj}} = M^{-1} \sum_{\chi \in \mathfrak{M}_{\mathrm{eff}}} |E_m(\chi)|,$$

where $E_m(\chi) = e_m(\chi(g(t)-g(t_0)) : t \in T)$*. Then*

$$\max_z N_g(z) \le \frac{M}{A_{\mathrm{eff}}}(1 + C_{\min} + C_{\mathrm{maj}}). \tag{EF7}$$

If every minor character has power sums bounded by Λ *through order* m*, then*

$$C_{\min} \le \frac{|\mathfrak{m}_{\mathrm{eff}}|}{M} \binom{\lceil \Lambda \rceil + m - 1}{m}. \tag{EF8}$$

Thus (EF7)*, with the two actual finite sums, records every finite Fourier loss in an adjacent-row certificate exactly.*

Proof. Formula (EF7) is (EF3) after splitting the nontrivial characters. The cycle-index coefficient bound gives $|E_m(\chi)| \le \binom{\lceil \Lambda \rceil + m - 1}{m}$, proving (EF8). ☐

Definition 13.12 (Pointwise minor-arc range)**.** *A primitive weighted prefix chart satisfies the pointwise minor-arc range if the characteristic is larger than* m*, every certified minor lift has a rational phase of total pole-degree at most* Δ_{pole} *and satisfies the nondegeneracy hypotheses of Proposition 18.3, and, with*

$$\Lambda = C_0(\Delta_{\mathrm{pole}} + 1)\sqrt{|\mathbb{B}|} + |P|$$

for the absolute Weil constant C_0 *and planted exceptional set* P*, one has*

$$R \log |\mathbb{B}| + \log \binom{\Lambda + m - 1}{m} - \log \binom{|T|}{m} \le o(|T|). \tag{PF}$$

For circle twin-cosets the same definition is used with $2C_0(\Delta_{\text{pole}}+1)\sqrt{|\mathbb{B}|}$ in place of $C_0(\Delta_{\text{pole}}+1)\sqrt{|\mathbb{B}|}$, accounting for the two inverse-coset lifts on the torus before the involution quotient.

Part 3. Primitive Q and the residual compiler

14. The primitive prefix leaf

After first-match removal of paid cells, a residual prefix-boundary chart has a fixed set of active coordinates T, $|T| = N$, and a fixed support size m. The full profile slice is

$$\Omega_{T,m} := \{x \in \{0,1\}^T : \sum_{t\in T} x_t = m\},$$

and the primitive residual is a subset $\Omega^\circ \subseteq \Omega_{T,m}$. The superscript circle records a first-match residual; no Zariski openness is asserted. The normalization is taken from the full profile slice, not from the possibly much smaller residual mass. It is nevertheless the *realized image*, rather than the formal codomain, that supplies the default number of target fibers.

The three words in *primitive first-match residual* have separate content. "First match" means that a slope is removed as soon as one of its witnesses occurs in an earlier ordered cell; "residual" means that only witnesses surviving those removals remain; and "primitive" means that the row-specific quotient, planted, field-descent, rank, and ray-saturation certificates named by the atlas have all been applied. Primitivity does not mean merely trivial setwise stabilizer, and it does not assume the Fourier, Sidon, max-fiber, or ray estimates that will be imposed below.

The prefix syndrome map is a linear map $\Phi : \mathbb{Z}^T \to \mathbb{B}^R$, restricted first to the full slice and then to Ω°. In the unprofiled locator chart, when $R < \operatorname{char}\mathbb{B}$, it is equivalent through Newton's triangular change of coordinates to

$$\Phi(x) = \sum_{t\in T} x_t(t, t^2, \dots, t^R). \tag{14.1}$$

Equivalently, the augmented columns $\widetilde{v}_t = (1, t, \dots, t^R)$ have constant first coordinate $\sum_t x_t = m$, and the other R coordinates are Φ. A residual quotient or rational chart is allowed to replace these columns by $v_t = \rho(t)(1, t, \dots, t^{R-1})$ only when that genuine weighted Vandermonde representation, with $\rho(t) \neq 0$, has been proved for the chart. It is not a consequence merely of constructibility or of deleting a Jacobian locus. The number R is the number of independent prefix equations; in the unprofiled MCA boundary leaf $R = w = a - K$.

Put $M = |\Omega_{T,m}|$, and write

$$\mathcal{S} = \Phi(\Omega_{T,m}), \qquad L = |\mathcal{S}|, \qquad A = |\mathbb{B}|^R, \qquad \overline{N}^{\text{img}} = M/L, \qquad \overline{N}^{\text{amb}} = M/A, \tag{14.2}$$

and for $s \in \mathcal{S}$ put $F_s = \Omega^\circ \cap \Phi^{-1}(s)$ and $f_s = |F_s|$. Unless an ambient full-image certificate has been proved, we set $\overline{N} = \overline{N}^{\text{img}}$. The certificate is

$$L \geq e^{-o(N)} A. \tag{FI}$$

The set $\mathcal{S}$ is the *effective image*: it consists of the prefix targets actually attained by the full profile slice. The formal ambient target space $\mathbb{B}^R$ can be much larger. Condition (FI), the *full-image certificate*, says that these two target counts differ by at most a subexponential factor. An *effective-image collapse* is the failure of this comparison; it must be retained as an effective-image rank-collapse profile, or handled throughout with image normalization. Under (FI), the two scales differ by only $e^{o(N)}$, so either may be used at exponential accuracy. If the certificate fails, effective-image collapse must be retained as an effective-image rank-collapse profile, or all estimates on this leaf must stay at image scale. Deleting earlier cells can only decrease every fiber.

The image-scale frontier condition is $\log M - \log L = o(N)$. The familiar ambient equation $\log\binom{N}{m} - R\log|\mathbb{B}| = o(N)$ is interchangeable with it only under (FI). This distinction is immaterial for a surjective prefix map but essential for a collapsed primitive image.

Lemma 14.1 (Exact image–ambient moment conversion)**.** *For $q \ge 1$, extend f_s by zero from $\mathcal{S}$ to $\mathbb{B}^R$ and define*

$$\Gamma_q^{\mathrm{img}} = \frac{1}{L}\sum_{s\in\mathcal{S}}\left(\frac{f_s}{\overline{N}^{\mathrm{img}}}\right)^q, \qquad \Gamma_q^{\mathrm{amb}} = \frac{1}{A}\sum_{s\in\mathbb{B}^R}\left(\frac{f_s}{\overline{N}^{\mathrm{amb}}}\right)^q.$$

Then

$$\Gamma_q^{\mathrm{amb}} = \left(\frac{A}{L}\right)^{q-1}\Gamma_q^{\mathrm{img}}. \tag{14.3}$$

Thus an ambient-normalized moment bound implies the corresponding image-normalized bound. The converse is valid with subexponential loss at logarithmic order only under (FI)*.*

Proof. Since $\overline{N}^{\mathrm{amb}} = M/A$ and $\overline{N}^{\mathrm{img}} = M/L$, zero extension and one rearrangement give

$$\Gamma_q^{\mathrm{amb}} = \frac{1}{A}\sum_{s\in\mathcal{S}}\left(\frac{Af_s}{M}\right)^q = \left(\frac{A}{L}\right)^{q-1}\frac{1}{L}\sum_{s\in\mathcal{S}}\left(\frac{Lf_s}{M}\right)^q.$$

Because $L \le A$, the asserted one-way implication follows. Under (FI), $(q-1)\log(A/L) = o(Nq)$, which proves the reverse transfer at the scale used here. ☐

Remark 14.2 (Full-slice flatness certifies the image size)*.* If a Fourier argument proves $\max_s|\Omega_{T,m}\cap \Phi^{-1}(s)| \le e^{o(N)}\overline{N}^{\mathrm{amb}}$, then averaging over the nonempty image fibers gives $A/L \le e^{o(N)}$. Hence such an ambient flatness theorem itself proves (FI); image collapse cannot be silently assumed away before that estimate has been established.

Definition 14.3 (Primitive Q)**.** *The* Q branch *is the primitive max-fiber problem: it asks whether one effective prefix target contains substantially more residual supports than the image-normalized average. A primitive prefix leaf satisfies Q with loss $e^{o(N)}$ if*

$$\max_{s\in\mathcal{S}} f_s \le e^{o(N)}\overline{N}^{\mathrm{img}}.$$

A sequence of primitive leaves satisfies logarithmic-moment Q if, at every logarithmic order $q = q_N \to \infty$, $q \le (\log N)^C$, used by the argument,

$$\Gamma_q^{\mathrm{ord}} := \Gamma_q^{\mathrm{img}} = L^{-1}\sum_{s\in\mathcal{S}}\left(\frac{f_s}{\overline{N}^{\mathrm{img}}}\right)^q \le e^{o(Nq)}.$$

Here a logarithmic moment order *means $q = q_N \to \infty$ with $q \le (\log N)^C$ for a fixed C. The displayed normalized q-th moment is a Rényi-type probe of the fiber distribution: letting q grow slowly makes it approximate the largest normalized fiber without introducing an exponential moment-order cost. An ambient formulation is permitted only when it is proved directly or when* (FI) *has already been certified.*

The next lemma is the primitive logarithmic moment equivalence used in the earlier RS–MCA reductions. It is included because it is the exact interface between moment estimates and max-fiber Q.

Lemma 14.4 (Logarithmic moments and primitive Q)**.** *Assume $q = q_N \to \infty$ and $\log L/q = o(N)$. Then $\Gamma_q^{\mathrm{ord}} \le e^{o(Nq)}$ implies primitive Q. Conversely, primitive Q implies $\Gamma_q^{\mathrm{ord}} \le e^{o(Nq)}$.*

Proof. Let $M_* = \max_s f_s/\overline{N}^{\mathrm{img}}$. Since $M_*^q \le \sum_s (f_s/\overline{N}^{\mathrm{img}})^q = L\Gamma_q^{\mathrm{ord}}$, the moment bound gives $\log M_* \le (\log L + \log\Gamma_q^{\mathrm{ord}})/q = o(N)$. Conversely, $\sum_s f_s \le |\Omega_{T,m}| = L\overline{N}^{\mathrm{img}}$, so $\Gamma_q^{\mathrm{ord}} \le M_*^{q-1}$. Primitive Q gives $M_* = e^{o(N)}$, hence $\Gamma_q^{\mathrm{ord}} = e^{o(Nq)}$. ☐

The genuine Vandermonde condition removes the ordinary rank-defect alternative.

Lemma 14.5 (Augmented Vandermonde independence)**.** *Let $t_1, \dots, t_r \in \mathbb{B}$ be distinct and $r \le R+1$. Then the columns $\widetilde{v}_{t_i} = (1, t_i, \dots, t_i^R)$ are linearly independent. The same holds for $\rho(t_i)(1, t_i, \dots, t_i^{R-1})$ when $r \le R$ and every $\rho(t_i) \ne 0$.*

Proof. In either case the leading $r \times r$ minor, after removing the nonzero column weights when present, is $(t_i^j)_{0 \le j < r, 1 \le i \le r}$. Its determinant is $\prod_{i<j}(t_j - t_i) \neq 0$. □

In the high-energy argument below the Vandermonde lemma is not used: Boolean-cube growth gives the contradiction directly. The lemma certifies only the exact locator chart, or a residual chart for which the displayed weighted form has separately been verified; it does not classify arbitrary rank defects as paid cells.

15. Prefix coordinates and the Vandermonde map

This section explains why the primitive residual has the linear Vandermonde form used in Section 14. Let $S \subseteq D$ have size m, and write its monic locator as

$$Q_S(X) = \prod_{t \in S}(X - t) = X^m + q_1(S)X^{m-1} + \cdots + q_m(S).$$

The first R boundary coordinates may be taken either as elementary symmetric coefficients $q_1, \ldots, q_R$ or, when $R < \operatorname{char} \mathbb{B}$, as power sums $p_j(S) = \sum_{t \in S} t^j$, $1 \le j \le R$. Newton identities give triangular polynomial changes of variables between these two coordinate systems; the diagonal entries are $1, 2, \ldots, R$, so the change is invertible in characteristic greater than R.

For the unprofiled locator chart, fixing the support size makes the zeroth power sum constant and Newton's identities identify the first R locator coefficients with the power sums $\sum_t x_t t^j$, $1 \le j \le R$. Thus (14.1) is exact. For a general residual chart one first fixes the planted roots, quotient data, common factors, and all coordinates charged to earlier cells. The fixed part contributes a constant syndrome s_0. To use the primitive theorem one must then exhibit the remaining map explicitly, either as the same power-sum map, as a genuine common-weight Vandermonde map, or as a map for which the required Sidon estimate has been proved directly. Coordinate-dependent rational weights do not become a Vandermonde system merely because they are nonzero. Here "common-weight" means that one scalar $\rho(t)$ multiplies every monomial coordinate of the column at t, rather than using a different weight for each coordinate. This verification is part of the primitive-map requirement in Definition 11.1.

The fixed-density condition is also intrinsic. In a prefix leaf the total support size is fixed by the agreement threshold, and all paid planted blocks have fixed size. Thus the number of remaining choices is fixed. If a profile leaves several colors of coordinates, one obtains a product of fixed-density slices; the proof applies to each color after encoding the colors in a bounded alphabet. For the main statement we present the Boolean case because it is the MCA support case and because the Boolean-cube difference bound applies directly.

Lemma 15.1 (Newton-coordinate fiber equivalence)**.** *Assume $R < \operatorname{char} \mathbb{B}$. For supports of fixed size, the fibers of the first R elementary locator coefficients are exactly the fibers of the first R power sums. Consequently the Q max-fiber problem may be stated using either coordinate system.*

Proof. Newton identities give, for $1 \le j \le R$,

$$p_j + q_1 p_{j-1} + \cdots + q_{j-1}p_1 + jq_j = 0.$$

Since j is invertible in $\mathbb{B}$, q_j is determined by $p_1, \ldots, p_j$, and conversely p_j is determined by $q_1, \ldots, q_j$. The transformations are triangular inverse polynomial maps on the first R coordinates. □

The *description entropy* of a profile family is the logarithm of the number of allowed auxiliary/profile descriptions. It measures the census of cells and is distinct from the algebraic description complexity of one cell, which measures the number and degrees of its defining equations.

Lemma 15.2 (Primitive profiles have subexponential multiplicity)**.** *If the cell ledger fixes only bounded-complexity quotient, planted, tangent, extension, and split-pencil profiles, and the total description entropy of all data outside the moving support is $o(n)$ bits, then the number of primitive profiles is $e^{o(n)}$.*

Proof. There are at most $2^{o(n)} = e^{o(n)}$ descriptions by hypothesis. In particular, a planted or common block of size $o(n)$ has $\sum_{j=o(n)} \binom{n}{j} = e^{o(n)}$ possible supports, but recording $s = o(n)$ freely chosen field values costs $s \log_2 |\mathbb{B}|$ bits and is subexponential only when that product is $o(n)$. Positive-density data requires its own profile census and entropy payment; it cannot be discarded as a bounded-complexity label. ☐

Remark 15.3. The subexponential profile lemma is the asymptotic counterpart of the finite ledger census. It is the reason a row proof must distinguish between a genuine cell payment and a mere change of variables. A change of variables with exponentially many charts would consume the whole frontier budget.

16. The Sidon split

The Q moment can be large for two opposite reasons. A fiber may contain many repeated additive differences, which is the high-energy situation seen by inverse additive combinatorics, or it may be large while its differences almost never repeat, which is the Sidon-like situation. The split below separates these mechanisms before either one is estimated.

Every support is identified with its incidence vector in the torsion-free group $\mathbb{Z}^T$. In particular, all sums and differences in this section are integer operations, even when the Reed–Solomon coefficient field has characteristic two. For a finite set $F \subseteq \{0,1\}^T$, define its additive energy

$$E(F) = |\{(a,b,c,d) \in F^4 : a - b = c - d\}|.$$

Let X, X' be independent uniform points of F. Then $E(F) = |F|^4 \sum_z \mathbb{P}(X - X' = z)^2$. It is convenient to normalize by

$$\Delta(F) = \frac{E(F)}{|F|^3}.$$

The quantity $\Delta(F)$ is an additive closure probability:

$$\Delta(F) = \mathbb{P}_{a,b,c \in F}(a - b + c \in F).$$

Indeed, once a, b, c are chosen, the fourth point in an energy quadruple is forced to be $d = a-b+c$.

Thus additive energy counts repeated differences, and $\Delta(F)$ is their normalized density. Values $\Delta(F) = e^{-o(N)}$ indicate substantial additive closure at exponential scale, whereas $\Delta(F) \le e^{-\sigma N}$ is the low-energy, Sidon-like regime. The logarithmic moments below are the slowly growing normalized moments from Definition 14.3; they measure how much of the Q obstruction is carried by fibers of either type. Throughout these sections, *positive rate* means a fixed positive exponential rate along a subsequence: for a fiber it means an excess of the form $e^{cN - o(N)}$, and for a logarithmic moment it means an excess of the form $e^{cNq - o(Nq)}$, for some constant $c > 0$.

For the fiber F_s, write $\Delta_s = E(F_s)/f_s^3$, with $\Delta_s = 0$ if $f_s = 0$.

Definition 16.1 (Sidon-heavy logarithmic moment)**.** *For $\sigma > 0$, the Sidon-heavy part of the normalized moment is*

$$\Gamma^{\mathrm{sid}}_{q,\sigma} = L^{-1} \sum_{\Delta_s \le e^{-\sigma N}} \left(\frac{f_s}{\overline{N}}\right)^q.$$

A sequence has no positive-rate Sidon-heavy obstruction if $\Gamma^{\mathrm{sid}}_{q,\sigma} \le e^{o(Nq)}$ for every fixed $\sigma > 0$ and every logarithmic q. Here "Sidon" refers to the exponentially small normalized energy Δ_s, while "heavy" refers to the positive-exponential contribution of those fibers to the normalized logarithmic moment; it does not assert that every low-energy fiber is individually large.

The following dichotomy is the correct replacement for the naive assertion that collision excess directly creates small doubling.

Proposition 16.2 (Ordinary moment split)**.** *Fix $\eta > 0$ and $\sigma > 0$. If $\Gamma^{\mathrm{ord}}_q \ge e^{\eta N q}$ and $\Gamma^{\mathrm{sid}}_{q,\sigma} \le \frac{1}{2} e^{\eta N q}$, then there is a syndrome s such that*

$$f_s \ge e^{\eta N - o(N)} \overline{N}, \qquad \Delta_s > e^{-\sigma N}.$$

If no positive-rate Sidon-heavy obstruction is present, then any positive-rate moment failure gives, after taking $\sigma \to 0$ slowly, a fiber with $f_s \ge e^{\eta N - o(N)} \overline{N}$ and $E(F_s) \ge f_s^3 e^{-o(N)}$.

Proof. The high-energy part of the moment is at least $\frac{1}{2}e^{\eta Nq}$. Since there are L syndromes, some high-energy syndrome satisfies

$$\left(\frac{f_s}{\overline{N}}\right)^q \geq \frac{L}{2}\, e^{\eta Nq} \cdot L^{-1} = \frac{1}{2}e^{\eta Nq}.$$

Thus $f_s \geq e^{\eta N - o(N)}\overline{N}$, and by construction $\Delta_s > e^{-\sigma N}$. If $\Gamma^{\mathrm{sid}}_{q,\sigma} \leq e^{o(Nq)}$ for every fixed σ, apply the first part along a sequence $\sigma_j \downarrow 0$ and diagonalize; the normalization $\Delta_s = E(F_s)/f_s^3$ gives the energy conclusion. ☐

The proof also explains the no-go point. Suppose a fiber is heavy but $\Delta_s \leq e^{-\sigma N}$. A closure-rich diagram using tests $a - b + c \in F_s$ pays a factor Δ_s for each nonredundant closure. Such diagrams cannot account for the ordinary contribution f_s^q of q independent choices in F_s. Therefore Sidon-heavy fibers must be removed by a separate cell; they cannot be forced into the high-energy inverse branch.

Definition 16.3 (Sidon moment payment)**.** *A primitive prefix leaf has a Sidon moment payment if $\Gamma^{\mathrm{sid}}_{q,\sigma} = e^{o(Nq)}$ for every fixed $\sigma > 0$ and every logarithmic moment order used by the primitive-Q argument, uniformly over profiles. This is an analytic support-fiber statement. It is not equivalent to a first-match distinct-slope payment; the latter still requires* (RC) *or a direct ray bound.*

The equivalence in the definition follows from Lemma 14.4 restricted to the low-energy set of syndromes. Operationally, one proves payment either directly by bounding these fibers or indirectly by proving Fourier flatness on the primitive complement of algebraic major arcs.

17. Why the Sidon cell is necessary

It is tempting to try to prove primitive Q by expanding the logarithmic moment into diagrams and then showing that every primitive diagram contains many additive closure tests. That strategy is correct only for a closure-rich submoment. It cannot capture the ordinary moment on a heavy low-energy fiber. The obstruction is simple enough to record as a lemma.

Let $F = F_s$ be a fiber of size f. A closure test is an equation of the form $a - b + c \in F$, with $a, b, c \in F$ already chosen. Its success probability is $\Delta(F) = E(F)/f^3$. A closure-rich diagram with ℓ nonredundant closure tests and r free choices contributes, up to a subexponential diagram factor,

$$f^r \Delta(F)^\ell.$$

After the free coordinates have been chosen, in the image normalization of a primitive prefix leaf, the remaining $q - r$ coordinates are supplied only at the random scale $\overline{N}$. Thus its contribution is at most

$$e^{o(Nq)}\overline{N}^{q-r} f^r \Delta(F)^\ell.$$

Lemma 17.1 (No-go for ordinary moment capture)**.** *Assume that $f \geq e^{\eta N}\overline{N}$, $\Delta(F) \leq e^{-\sigma N}$, and that a family of diagrams has $r = o(q)$ free fiber choices and $\ell \geq cq$ nonredundant closure tests. Then the total contribution of these diagrams is smaller than the ordinary fiber contribution f^q by a factor*

$$\exp(-(\eta + c\sigma)Nq + o(Nq))$$

up to changing η by $o(1)$.

Proof. For one diagram the ratio to f^q is at most

$$e^{o(Nq)} \left(\frac{\overline{N}}{f}\right)^{q-r} \Delta(F)^\ell \leq e^{o(Nq)} \exp(-\eta N(q - r) - c\sigma Nq).$$

Since $r = o(q)$, this is $\exp(-(\eta + c\sigma)Nq + o(Nq))$. The number of bounded-complexity logarithmic diagrams is $e^{O(q \log q)} = e^{o(Nq)}$, so summing over them does not change the exponential rate. ☐

The lemma has an important conceptual consequence. A large Sidon-like fiber is not a hidden approximate group. It is a different phenomenon: many independent choices land in the same syndrome, but their pairwise differences almost never repeat. Additive-combinatorics inverse tools are designed for the opposite situation, where repeated differences are abundant. Therefore the first-match ledger must contain a Sidon/Fourier cell before the primitive inverse argument begins.

The valid implication is exactly Proposition 16.2. Ordinary moment mass splits into a low-energy part, which needs a separately proved Sidon moment estimate, and a high-energy part, which is dispatched by Proposition 21.1.

This mechanism does not replace a prefix-flatness or Sidon-payment theorem; it identifies exactly where one is needed. The first-match order is: algebraic major arcs first, then a separately certified Sidon/Fourier cell, and only then the high-energy primitive inverse step. Naming a large low-energy fiber a cell does not pay it.

18. Fourier flatness as a Sidon payment

This section records the general weighted form of the Fourier/Sidon payment. It is phrased for an arbitrary finite field $\mathbb{B}$, because the primitive residual charts that come from the moduli ledger need not have unweighted power-sum coordinates. Let T be a finite set, $|T| = N$, let $g : T \to \mathbb{B}^R$ be any function, and set

$$\Omega_{T,m} = \{x \in \{0,1\}^T : \sum_{t\in T} x_t = m\}, \qquad \Psi(x) = \sum_{t\in T} x_t g(t).$$

In applications $g(t) = (\rho_0(t), \rho_1(t)t, \ldots, \rho_{R-1}(t)t^{R-1})$ with nonzero rational weights ρ_i on the primitive chart.

Fourier inversion has three pieces. The zero character produces the average-fiber main term. On the effective dual, the chart-specific partition of Definition 13.2 designates certified minor characters and the complementary major characters. The ambient algebraic exceptional locus of Definition 18.5 helps construct that partition but does not canonically determine it, because an effective character has many ambient lifts. The pointwise condition (PF) controls every certified minor lift, while the aggregate condition (MA) controls the sum of all designated major characters. Neither estimate substitutes for the other.

The condition (PF') below is the raw power-sum flatness inequality when one has a uniform bound for every nonzero character. Condition (PF) is its row-specific Weil specialization for minor characters after substituting the pole-degree bound; the omitted major characters are then handled by (MA).

Theorem 18.1 (Prefix flatness from power-sum control)**.** *Fix a nontrivial additive character ψ of $\mathbb{B}$. For $\alpha \in \mathbb{B}^R \setminus \{0\}$ and $1 \le j \le m$, put*

$$P_j(\alpha) = \sum_{t\in T} \psi(j\alpha \cdot g(t)).$$

Assume $|P_j(\alpha)| \le \Lambda$ for every $\alpha \ne 0$ and every $1 \le j \le m$. Then, for every $z \in \mathbb{B}^R$,

$$|\Psi^{-1}(z)| \le |\mathbb{B}|^{-R}\binom{N}{m} + \binom{\Lambda + m - 1}{m}.$$

Consequently, if

$$\text{(PF')} \qquad R\log|\mathbb{B}| + \log\binom{\Lambda + m - 1}{m} - \log\binom{N}{m} \le o(N),$$

then $\max_z|\Psi^{-1}(z)| \le e^{o(N)}|\mathbb{B}|^{-R}\binom{N}{m}$.

Proof. By orthogonality,

$$|\Psi^{-1}(z)| = |\mathbb{B}|^{-R} \sum_{\alpha\in\mathbb{B}^R} \psi(-\alpha \cdot z)e_m(\psi(\alpha \cdot g(t)) : t \in T),$$

where e_m denotes the m-th elementary symmetric coefficient. The term $\alpha = 0$ is the main term $|\mathbb{B}|^{-R}\binom{N}{m}$. For $\alpha \neq 0$, write $a_t = \psi(\alpha \cdot g(t))$. The generating identity $\sum_{r\geq 0} e_r(a_t)u^r = \exp(\sum_{j\geq 1}(-1)^{j-1}(\sum_t a_t^j)u^j/j)$ and the hypothesis $|\sum_t a_t^j| \leq \Lambda$ for $j \leq m$ give $|e_m(a_t : t \in T)| \leq [u^m](1-u)^{-\Lambda} = \binom{\Lambda+m-1}{m}$. Summing the nontrivial characters and using the outside factor $|\mathbb{B}|^{-R}$ gives the first bound. The displayed condition (PF') makes the error term subexponential relative to the random-fiber main term. ☐

Remark 18.2 (Small characteristic cycles). If some integers $j \leq m$ vanish in the characteristic, the formal Li–Wan cycle index [LW10] separates the cycles divisible by $p = \operatorname{char} \mathbb{B}$; the corresponding coefficient majorant is $[u^m](1-u)^{-\Lambda}(1-u^p)^{-(N-\Lambda)/p}$. This bookkeeping identity does not supply the missing character-sum bounds for those cycles, exclude Artin–Schreier phases, or prove a uniform Sidon payment. The pointwise theorem above is therefore deliberately stated in the large-characteristic range. A small-characteristic leaf needs a direct cycle-by-cycle certificate or a separate image-normalized Q/Sidon theorem.

Proposition 18.3 (Weighted Weil minor arcs). *Let T be either a multiplicative coset in $\mathbb{B}^\times$ or the x-coordinate of a torus twin coset. Suppose that every certified minor lift $\alpha \in \mathbb{B}^R$ has phase $R_\alpha(t) = \alpha \cdot g(t)$ which is a nonconstant rational function on the corresponding genus-zero curve, is not of Artin–Schreier form, and has total pole-degree at most Δ_{pole}. If $m < \operatorname{char} \mathbb{B}$, then the stated power-sum bound holds for every certified minor lift, with*

$$\Lambda = C_0(\Delta_{\text{pole}} + 1)\sqrt{|\mathbb{B}|} + |P|$$

for multiplicative cosets, and with $2C_0(\Delta_{\text{pole}} + 1)\sqrt{|\mathbb{B}|} + |P|$ for circle twin-cosets. Here P is the planted exceptional set and C_0 is an absolute Weil constant.

Proof. For a multiplicative coset $\theta H \subseteq \mathbb{B}^\times$, write the coset indicator as the average of the multiplicative characters of $\mathbb{B}^\times/H$. Each nontrivial power sum becomes an average of mixed multiplicative-additive character sums $\sum_x \chi(x)\psi(jR_\alpha(\theta x))$. Since $1 \leq j < m < \operatorname{char} \mathbb{B}$, multiplication by j does not make the rational function constant or Artin–Schreier. Weil's bound for one-variable mixed character sums [Wei48] gives $C_0(\Delta_{\text{pole}} + 1)\sqrt{|\mathbb{B}|}$. Removing planted exceptional points changes the sum by at most $|P|$. The circle case is the same after pulling back by $x = (u + u^{-1})/2$ to the norm-one torus; the two torus branches give the factor two. ☐

Theorem 18.4 (Unconditional shallow-prefix Fourier flatness). *Let $p \to \infty$ through odd primes. For each p, let T_p be either a multiplicative coset in $\mathbb{F}_p^\times$ or a Chebyshev twin-coset x-domain over $\mathbb{F}_p$, in the latter case with $p \equiv 3 \pmod 4$, and put $N_p = |T_p|$. Fix $0 < \alpha < 1/2$, choose integers m_p, R_p with*

$$\text{(SFM1)} \qquad 1 \leq R_p, \qquad \alpha \leq \frac{m_p}{N_p} \leq 1-\alpha, \qquad g_p(t) = (t, t^2, \ldots, t^{R_p}), \qquad R_p\sqrt{p} = o(N_p),$$

and let $\Psi_p(S) = \sum_{t\in S} g_p(t)$ on $\binom{T_p}{m_p}$. Then $V_{g_p} = \mathbb{F}_p^{R_p}$, and the certified effective partition may take every nontrivial effective character to be minor and no character to be major. For some $c_\alpha > 0$,

$$\text{(SFM2)} \qquad \sum_{1\neq\chi\in\widehat{V_{g_p}}} |e_{m_p}(\chi(g_p(t) - g_p(t_0)) : t \in T_p)| \leq e^{-c_\alpha N_p}\binom{N_p}{m_p}.$$

Consequently effective (MI) *holds,* (MA) *is vacuous, and*

$$\text{(SFM3)} \qquad \max_z |\Psi_p^{-1}(z)| \leq (1 + e^{-c_\alpha N_p})p^{-R_p}\binom{N_p}{m_p}.$$

Indeed every $z \in \mathbb{F}_p^{R_p}$ is attained for all sufficiently large rows, and uniformly in z,

$$\text{(SFM4)} \qquad |\Psi_p^{-1}(z)| = p^{-R_p}\binom{N_p}{m_p}\left(1 + O(e^{-c_\alpha N_p})\right).$$

Thus these smooth and circle prefix charts have unconditional effective Fourier payment and image coverage in the shallow range (SFM1)*.*

Proof. Condition (SFM1) implies $R_p < p$, $R_p < N_p$, and $R_p \log p = o(N_p)$. Moreover $N_p \le p+1$, so the fixed upper density bound gives $m_p < p$ for every sufficiently large p. For a nonzero ambient parameter, let $d = \max\{j : a_j \neq 0\}$. Then $1 \le d \le R_p < p$, and $P(X) = \sum_{j=1}^{R_p} a_j X^j$ has a pole of order d at infinity. Every pole of a nonconstant Artin–Schreier function $G^p - G + c$ has order divisible by p, so P is non-Artin–Schreier. In the circle case $P((u+u^{-1})/2)$ has poles of the same order $d < p$ at the two ends of the torus compactification and is likewise non-Artin–Schreier.

For a multiplicative coset, the character expansion of its indicator and the mixed Weil bound give, uniformly for $1 \le j \le m_p < p$,

$$\left|\sum_{t\in T_p} \psi(jP(t))\right| = O(R_p\sqrt{p}) = o(N_p).$$

For a circle domain, pullback by $x = (u+u^{-1})/2$ gives the same conclusion with an inessential factor two. Put $\Lambda_p = O(R_p\sqrt{p}) = o(N_p)$. The cycle-index estimate therefore bounds every nontrivial fixed-weight Fourier coefficient by

$$\binom{\lceil \Lambda_p\rceil + m_p - 1}{m_p} = e^{o(N_p)}.$$

If a nonzero ambient character annihilated V_{g_p}, its character value would be constant on T_p, making the power sum have modulus N_p, contrary to the preceding $o(N_p)$ bound. Thus the annihilator is zero and $V_{g_p} = \mathbb{F}_p^{R_p}$. There are $p^{R_p} = e^{o(N_p)}$ effective characters, whereas Stirling's formula and the density interval give $\binom{N_p}{m_p} \ge e^{(h(\alpha)+o(1))N_p}$. Summing the coefficient bounds proves (SFM2), after decreasing the positive constant c_α. All nontrivial effective characters have just received a certified minor lift, so the major set is empty. Finally Fourier inversion shows that the absolute nonzero-character error in each fiber is $e^{o(N_p)}$, whereas its main term is $p^{-R_p}\binom{N_p}{m_p} = e^{(h(m_p/N_p)+o(1))N_p}$. This proves (SFM3)–(SFM4) and eventual surjectivity. ☐

Definition 18.5 (Ambient algebraic exceptional locus)**.** *An ambient phase parameter is* algebraically exceptional *if its weighted phase is constant on a routed quotient fiber, is Artin–Schreier degenerate, has a pole or zero supported on a positive-density moving divisor, or factors through a proper quotient, Chebyshev map, field-descent locus, tangent rank drop, differential-locator defect, or ray-saturation collapse. On an effective chart this locus is used only to construct the certified partition of Definition 13.2; it is not itself a numerical estimate. In particular, an Artin–Schreier phase defined on the whole active set restricts to the trivial effective character by Lemma 13.4.*

Proposition 18.6 (Major-arc algebraic interface)**.** *Each condition in Definition 18.5 is a constructible condition on the dual phase parameters and points to a quotient, Chebyshev, field-descent, planted-divisor, differential, or rank-loss profile. This classification does not imply that the character vanishes after primal first-match deletion and does not prove* (MA)*. A flatness argument must also bound* (MA)*, recursively or directly.*

Proof. Constancy on a quotient fiber, Chebyshev factorization, Artin–Schreier degeneracy, a positive-density pole divisor, and a rank drop are respectively coefficient, Frobenius, resultant, or determinantal conditions. They therefore identify constructible dual loci on which a recursive estimate can be attempted. Fourier inversion sums characters, not primal witnesses, so no support-level payment follows from this fact. ☐

Proposition 18.7 (Frontier Weil separation)**.** *Let* $N = |T|$*, let* $m = \theta N + o(N)$ *with* $0 < \theta < 1$*, and suppose a primitive smooth or circle chart has minor-arc power-sum parameter* $\Lambda = o(N)$*. If the ambient random-fiber scale* $\overline{N} = |\mathbb{B}|^{-R}\binom{N}{m}$ *satisfies* $\log \overline{N} = o(N)$*, or more generally* $\log \overline{N} \ge -o(N)$*, then the prefix-flat inequality* (PF') *holds. In particular, minor arcs alone* *cannot create a positive-rate max-fiber or Sidon-heavy moment at this near-zero ambient entropy scale.*

Proof. The elementary estimate $\binom{\Lambda+m-1}{m} = \binom{\Lambda+m-1}{\Lambda-1} \le (e(m+\Lambda)/\Lambda)^\Lambda$ gives $\log\binom{\Lambda+m-1}{m} = o(N)$, since $\Lambda = o(N)$. The relative minor-arc error exponent is

$$\log\binom{\Lambda+m-1}{m} - \log\overline{N} = R\log|\mathbb{B}| + \log\binom{\Lambda+m-1}{m} - \log\binom{N}{m}.$$

The assumed frontier inequality makes this at most $o(N)$; it may be negative by a linear amount, which is only stronger. Thus the minor-arc contribution to every full-profile fiber is at most $e^{o(N)}\overline{N}$. This statement concerns only minor characters; a full flatness conclusion also requires (MA). ☐

Theorem 18.8 (Sidon payment in the pointwise flat range)**.** *Let $\Omega^\circ \subseteq \Omega_{T,m}$ be a primitive first-match residual of a smooth or circle chart, with weighted prefix map Ψ and ambient scale $\overline{N}^{\mathrm{amb}} = |\mathbb{B}|^{-R}\binom{N}{m}$. Put $A_{\mathrm{amb}} = |\mathbb{B}|^R$ and $L = |\Psi(\Omega_{T,m})|$. Suppose the pointwise minor-arc condition* (PF) *and the ambient instance of the aggregate condition* (MA) *hold. Then, for every fixed $\sigma > 0$ and every logarithmic $q \le (\log N)^C$, for a fixed C,*

$$\Gamma^{\mathrm{sid}}_{q,\sigma}(\Omega^\circ) \le e^{o(Nq)},$$

where Γ^{sid} uses the image normalization of Definition 16.1. Equivalently, no such pointwise-flat residual leaf supports positive-rate low-energy heavy fibers.

Proof. It is enough to bound every residual fiber, and a residual is contained in the full slice. Fourier inversion on that slice splits into the zero, minor, and major characters. The zero character contributes $\overline{N}^{\mathrm{amb}}$. The cycle-index estimate and Proposition 18.3, together with (PF), bound the complete minor contribution by $e^{o(N)}\overline{N}^{\mathrm{amb}}$. Condition (MA) bounds the complete major contribution by the same quantity. Thus every full-slice, and hence every residual, fiber is $e^{o(N)}\overline{N}^{\mathrm{amb}}$. This proves the full-image certificate by Remark 14.2, and Lemma 14.1 transfers the ambient moment bound to image normalization. ☐

Remark 18.9 (Role of the Sidon-heavy cell)*.* The theorem implements "pay structure, flatten Sidon, inverse high energy" only in the range where both (PF) and (MA) are verified. At entropy depth a phase can have degree up to R, and the raw Weil parameter is of order $R\sqrt{|\mathbb{B}|}$; this is not automatically $o(N)$ when $R \asymp N/\log|\mathbb{B}|$. Outside the pointwise-flat range a different Sidon payment remains necessary.

19. External additive combinatorics

Only two external additive-combinatorics inputs are used in the primitive high-energy proof.

Theorem 19.1 (Balog–Szemeredi–Gowers, energy form)**.** *There is an absolute constant C such that if A is a finite subset of an abelian group and $E(A) \ge |A|^3/K$, then there is $A' \subseteq A$ with $|A'| \ge K^{-C}|A|$ and $|A' - A'| \le K^C|A'|$.*

This is the usual energy form of the Balog–Szemeredi–Gowers theorem; polynomial dependence on K follows from Gowers's refinement of the Balog–Szemeredi theorem [BS94, Gow98]. The difference-set form follows from the sumset form by replacing A by $-A$ and using standard Ruzsa calculus [TV06], or directly by applying the graph form to differences.

Theorem 19.2 (Boolean-cube difference growth)**.** *If $U \subseteq \{0,1\}^d$, then*

$$|U - U| \ge |U|^{3/2}.$$

This is the Boolean-cube specialization of a broader sumset inequality of Green–Matolcsi–Ruzsa–Shakan–Zhelezov [GMR+22]. It also follows from the induced-doubling identity of Matolcsi–Ruzsa–Shakan–Zhelezov [MRSZ22]. Only the displayed Boolean statement is used below.

Remark 19.3. Entropy inverse theorems of Tao and the entropy-doubling theory of Green–Manners–Tao are compatible with this proof and are often useful after one has produced small-doubling trades [Tao10, GMT23]. They are not needed as hypotheses here. The Boolean-cube growth theorem gives a direct contradiction to a large small-doubling subset of a primitive Boolean support fiber.

20. Entropy inverse theory and its role

The terminology "Tao–Gowers method" can refer to two related but distinct steps. The first is BSG, used directly in Proposition 21.1. The second is Freiman-type structural theory, meaning inverse theorems that model a set with small sumset or difference set by a generalized arithmetic progression or a coset progression. It is not needed for the Boolean contradiction, but it can suggest candidate geometric strata in more general bounded-alphabet residuals; those strata still require direct payment proofs.

A *bounded integer trade* is a vector $Y \in \mathbb{Z}^T$ whose coordinates are bounded independently of N, with $\Phi(Y) = 0$ and, on a fixed-weight slice, $\sum_t Y_t = 0$. If Y is random, Y' is an independent copy, and $\mathrm{H}(Y - Y') \le \mathrm{H}(Y) + o(N)$, then entropy BSG and entropy Freiman theory produce, after conditioning and losing $o(N)$ entropy, a set model with doubling $e^{o(N)}$. Tao's entropy inverse theorem gives this in *coset-progression* language—a subgroup coset plus a generalized arithmetic progression—for Shannon entropy [Tao10]; Green–Manners–Tao recast the theory in terms of entropic doubling and Ruzsa distance [GMT23]; Goh formulates an entropic analogue of additive energy and the entropic BSG connection [Goh24]. In a torsion model one may also use polynomial Freiman–Ruzsa technology after reducing bounded integer trades modulo a large prime with no wraparound.

In the present paper the Boolean-cube difference theorem makes this structural compression unnecessary for the primitive Q step. Once BSG finds $A' \subseteq F_s \subseteq \{0,1\}^T$ with $|A'-A'| \le e^{o(N)}|A'|$, Boolean-cube difference growth already forces $|A'| = e^{o(N)}$. This contradicts the exponential size supplied by moment excess. Therefore there is no separate entropy-Freiman hypothesis in the proof.

For non-Boolean residual trades, entropy inverse theory can suggest coset-progression incidence strata or rank-defect tests. This is a useful heuristic for constructing a row-specific atlas, not an exhaustive dichotomy: each suggested stratum still needs the explicit first-match census and slope-projection payment required by Definition 11.1, and arbitrary weighted charts are not covered by Lemma 14.5. No such heuristic is used in the proof of the Boolean primitive theorem.

21. Primitive Q after the Sidon cut

The key primitive theorem is now short and self-contained modulo Theorems 19.1 and 19.2. It also addresses the main obstruction: ordinary Rényi mass can be Sidon-heavy, but after that cell is removed it must be high-energy, and high-energy is impossible in a large Boolean fiber.

Proposition 21.1 (High-energy Boolean fibers are small)**.** *Let $F \subseteq \{0,1\}^T$, $|T| = N$. If $E(F) \ge |F|^3/K$, then $|F| \le K^{O(1)}$. Consequently an exponential set $|F| \ge e^{cN-o(N)}$ cannot have $E(F) \ge |F|^3/e^{o(N)}$.*

Proof. Apply Theorem 19.1. There is $F' \subseteq F$ with $|F'| \ge K^{-C}|F|$ and $|F' - F'| \le K^C|F'|$. Since $F' \subseteq \{0,1\}^T$, Theorem 19.2 gives $|F' - F'| \ge |F'|^{3/2}$. Hence $|F'|^{1/2} \le K^C$, so $|F| \le K^{3C}$ after increasing C. Taking $K = e^{o(N)}$ rules out $|F| \ge e^{cN-o(N)}$. □

Theorem 21.2 (Primitive Q after a Sidon moment payment)**.** *Let $\Omega_N^0 \subseteq \{0,1\}^{T_N}$ be fixed-density full profile slices, with $|T_N| = N$ and $m_N/N \in [\alpha, 1-\alpha]$, and let $\Omega_N^\circ \subseteq \Omega_N^0$ be primitive first-match residuals. Suppose there is a logarithmic moment order $q_N \to \infty$ with $\log L_N/q_N = o(N)$, where $L_N = |\Phi_N(\Omega_N^0)|$ and $\overline{N}_N = |\Omega_N^0|/L_N$, for which $\Gamma^{\mathrm{sid}}_{q_N,\sigma} \le e^{o(Nq_N)}$ for every fixed $\sigma > 0$. Then the leaves satisfy primitive Q:*

$$\max_s |\Omega_N^\circ \cap \Phi_N^{-1}(s)| \le e^{o(N)}\overline{N}_N.$$

Proof. Assume primitive Q fails at positive exponential rate. The lower half of the moment sandwich gives $\Gamma^{\mathrm{ord}}_{q_N} \geq e^{\eta N q_N - o(N q_N)}$ along a subsequence. The Sidon moment hypothesis and Proposition 16.2 then give a fiber F_s with $f_s \geq e^{\eta N - o(N)} \overline{N}_N$ and $E(F_s) \geq f_s^3 e^{-o(N)}$. Since $\overline{N}_N = |\Omega^0_N|/L_N \geq 1$, this fiber has size $e^{\eta N - o(N)}$, contradicting Proposition 21.1 with $K = e^{o(N)}$. Thus no positive-rate max-fiber failure occurs. □

22. Q, SP, and the MCA compiler

This section uses the prefix rigidity, exact second moment, and slope-elimination results of [Cho26, Secs. 19–21 and Apps. E–F] as imported inputs. The new content is the occupancy/ray compiler and the precise hypotheses under which Q controls MCA slopes.

The *Q branch* is the maximum-prefix-fiber problem. The *SP branch* (support-pair, or shift-pair, branch) is its second-moment problem: it counts ordered pairs of supports in one prefix fiber. These names label two counting interfaces and carry no assertion that either bound has already been proved. Neither count is automatically the MCA numerator: a bad slope begins with one support, while converting support pairs to projective slope directions requires a proved multiplicity or incidence bound. We first record the exact moment identities and then state that separate ray compiler.

Definition 22.1 (Q and SP ledgers)**.** *For a first-match residual profile λ, let Ω^0_λ be its full profile slice, let $\Omega^\circ_\lambda \subseteq \Omega^0_\lambda$ be the residual, and put*

$$\mathcal{S}_\lambda = \Phi_\lambda(\Omega^0_\lambda), \qquad L_\lambda = |\mathcal{S}_\lambda|, \qquad \overline{N}_\lambda = |\Omega^0_\lambda|/L_\lambda, \qquad Q_\lambda(s) = |\Omega^\circ_\lambda \cap \Phi^{-1}_\lambda(s)|.$$

The Q ledger is discharged if $\max_s Q_\lambda(s) \leq e^{o(n)} \overline{N}_\lambda$ *for every primitive profile. The support-pair SP statistic is* $\sum_s Q_\lambda(s)^2$*. A distinct-ray ledger is discharged only by a direct slope bound or by* (RC) *below. If the ambient target count* $|\mathbb{B}_\lambda|^{R_\lambda}$ *is used in place of* L_λ*, the ledger must also record the full-image certificate* (FI) *or a direct ambient theorem.*

The compiler uses two elementary inequalities repeatedly. They are included to make explicit how the logarithmic moment, max-fiber Q, and shift-pair second moment interact.
Imported moment inequality. For fiber sizes $N(s)$, mean $\bar{N}$, ratios $R(s) = N(s)/\bar{N}$, and $\Gamma_r = L^{-1} \sum_s R(s)^r$, CAP25 v13.2 [Cho26, Sec. 19 and Apps. E–F] gives the elementary bounds

$$\text{(M)} \qquad \max_s R(s) \leq (L\Gamma_r)^{1/r}, \qquad \Gamma_r \leq (\max_s R(s))^{r-1}.$$

Corollary 22.2 (Finite moment floor)**.** *Suppose a finite row has remaining Q bit margin Δ, meaning that one needs* $\max_s R(s) \leq 2^\Delta$*. A moment proof of order r can certify this only if*

$$\log_2 \Gamma_r + \log_2 L \leq r\Delta,$$

where L is the number of realized targets in the selected normalization. In an ambient prefix proof this specializes to $\log_2 \Gamma^{\mathrm{amb}}_r + R \log_2 |\mathbb{B}| \leq r\Delta$*; transferring an image-normalized moment to that statement additionally costs* $(r-1)\log_2(|\mathbb{B}|^R/L)$*.*

Proof. The moment sandwich gives $\log_2 \max_s R(s) \leq (\log_2 L + \log_2 \Gamma_r)/r$. To make this at most Δ one needs the displayed inequality. □

The finite corollary explains why a small adjacent margin cannot be closed by a fixed-order moment when $R \log|\mathbb{B}|$ is large.

Definition 22.3 (Explanation image and retained-support occupancy)**.** *Fix $a \geq k+1$, a received pair $r = (r_0, r_1)$, and a cell $\mathcal{C} \subseteq \mathcal{W}_a(r)$. Its* explanation image *and* slope image *are*

$$\mathcal{R}(\mathcal{C}) = \{(\gamma, h) : \exists S \ (\gamma, S, h) \in \mathcal{C}\}, \qquad Z(\mathcal{C}) = \{\gamma : \exists h \ (\gamma, h) \in \mathcal{R}(\mathcal{C})\}.$$

For $\rho = (\gamma, h) \in \mathcal{R}(\mathcal{C})$, its retained-support occupancy *is*

$$\nu_{\mathcal{C}}(\rho) = \left|\left\{S \in \binom{D}{a} : (\gamma, S, h) \in \mathcal{C}\right\}\right|.$$

Thus the witness projection factors canonically as

$$\mathcal{C} \longrightarrow \mathcal{R}(\mathcal{C}) \longrightarrow Z(\mathcal{C}), \qquad (\gamma, S, h) \longmapsto (\gamma, h) \longmapsto \gamma.$$

The first map forgets an exact-agreement support; the second forgets the explaining polynomial. Distinct explanation states may have the same slope, so neither map is silently treated as a bijection. A ray compiler *means a proved estimate for the final slope image, obtained from these actual fibers or from a direct incidence bound.*

Lemma 22.4 (Exact occupancy compiler)**.** *With the notation above,*

$$|\mathcal{C}| = \sum_{\rho\in\mathcal{R}(\mathcal{C})} \nu_{\mathcal{C}}(\rho), \qquad |\mathcal{R}(\mathcal{C})| = \sum_{w\in\mathcal{C}} \frac{1}{\nu_{\mathcal{C}}(\pi_{\exp}(w))}, \qquad |Z(\mathcal{C})| \le |\mathcal{R}(\mathcal{C})|. \tag{RC$_{\text{occ}}$}$$

Here $\pi_{\exp}(\gamma, S, h) = (\gamma, h)$. In particular, if every retained explanation state has occupancy at least $H \ge 1$, then

$$|Z(\mathcal{C})| \le \left\lfloor \frac{|\mathcal{C}|}{H} \right\rfloor. \tag{RC1}$$

Also, because one noncommon exact-a support carries at most one slope and the explaining polynomial on $a \ge k+1$ points is unique,

$$|Z(\mathcal{C})| \le |\{S : \exists\gamma, h\ (\gamma, S, h) \in \mathcal{C}\}|. \tag{RC2}$$

All these statements are exact finite-row identities or inequalities.

Proof. Partition $\mathcal{C}$ by the fibers of $\pi_{\exp}$. This gives the first identity; summing the reciprocal of the fiber size over each fiber gives the second. Projection $(\gamma, h) \mapsto \gamma$ can only decrease cardinality. If every first fiber has size at least H, (RC1) follows. Finally, if the same noncommon support carried two distinct slopes, slope elimination would simultaneously explain the received pair on that support. For fixed support and slope, uniqueness of interpolation gives at most one h. This proves (RC2). □

Lemma 22.5 (Max-occupancy add-back)**.** *Let $\mathcal{C} \subseteq \mathcal{W}_a(r)$ be partitioned into profile cells $\mathcal{C} = \coprod_{\lambda\in\Lambda} \mathcal{C}_\lambda$, and put $P = |\Lambda|$. For every explanation state $\rho = (\gamma, h) \in \mathcal{R}(\mathcal{C})$, let*

$$M(\rho) = |\{S : (\gamma, S, h) \in \mathcal{C}\}|.$$

Assign ρ to a profile $\lambda(\rho)$ on which $\nu_{\mathcal{C}_\lambda}(\rho)$ is maximal, and let $\mathcal{R}^\star_\lambda$ be the assigned states. Then

$$\nu_{\mathcal{C}_\lambda}(\rho) \ge \lceil M(\rho)/P \rceil \qquad (\rho \in \mathcal{R}^\star_\lambda). \tag{RC$_{\max}$1}$$

Consequently, if $M(\rho) \ge H$ for every retained state, then

$$|Z(\mathcal{C})| \le \sum_{\lambda\in\Lambda} |\mathcal{R}^\star_\lambda| \le \sum_{\lambda\in\Lambda} \left\lfloor \frac{|\mathcal{C}^\star_\lambda|}{\lceil H/P \rceil} \right\rfloor, \tag{RC$_{\max}$2}$$

where $\mathcal{C}^\star_\lambda$ consists of the witnesses in $\mathcal{C}_\lambda$ lying above assigned states. Call $\mathcal{C}$ explanation-state complete *if, whenever $(\gamma, h) \in \mathcal{R}(\mathcal{C})$, every exact-a noncommon witness $(\gamma, S, h) \in \mathcal{W}_a(r)$ also belongs to $\mathcal{C}$. If $\mathcal{C}$ is explanation-state complete, every retained state has exact agreement $b \ge a$, and the received pair is not simultaneously explained on any a-subset of that agreement set, then one may take $H = \binom{b}{a}$.*

Proof. The P profile occupancies of a fixed state sum to $M(\rho)$, so their maximum is at least $\lceil M(\rho)/P \rceil$. Apply Lemma 22.4 on each assigned cell and sum. Under the final hypotheses, explanation-state completeness retains every a-subset of the b-point agreement set, and every such subset is noncommon. Hence $M(\rho) = \binom{b}{a}$. □

Definition 22.6 (Support interpolation functionals)**.** *Let $T \subseteq D$, $|T| = m \ge k$, put $w = m - k$, and let $Q_T(X) = \prod_{x\in T}(X - x)$. For a word u, define*

$$\mathcal{A}_T(u) = \left(\sum_{x\in T} \frac{u(x)x^r}{Q'_T(x)} \right)_{0\le r<w} \in \mathbb{F}^w. \tag{22.1}$$

Imported slope interface. CAP25 v13.2 [Cho26, Sec. 19 and Apps. E–F] proves that a word u is explained by $\mathrm{RS}_{\mathbb{F}}(D,k)$ on T exactly when $\mathcal{A}_T(u)=0$. Hence

$$\mathcal{A}_T(u_0)+\gamma\mathcal{A}_T(u_1)=0, \tag{SE}$$

and every noncommon support carries at most one finite slope. We use this only as an imported incidence input to the occupancy compiler above.
Imported support-pair identity. For every residual prefix profile λ, the foundational paper [Cho26, Sec. 19 and Apps. E–F] identifies the ordered same-prefix support-pair count as

$$\sum_s Q_\lambda(s)^2 \le (\max_s Q_\lambda(s)) \sum_s Q_\lambda(s). \tag{SP$_0$}$$

Q therefore controls a support-pair statistic. It does not justify dividing that statistic by $|\Omega^0_\lambda|$: a ray can have a single representative at exact agreement.

Proposition 22.7 (Q and SP do not determine rays)**.** *Let $\Phi:\Omega\to\Sigma$ be any support-profile map with $\Omega\neq\varnothing$, put $Q(s)=|\Phi^{-1}(s)|$, and $\mathrm{SP}=\sum_s Q(s)^2$. Regard Ω as an abstract set of candidate noncommon supports, and suppose the only retained slope-incidence axiom is that each support carries at most one slope. For every nonempty finite slope set Γ and every $1\le M\le\min\{|\Omega|,|\Gamma|\}$, within the class of abstract incidence structures obeying these retained data there is a partial support-to-slope map having exactly M distinct values. There is also a one-value map. Hence no ray bound can be deduced from Q, SP, and that axiom alone; an RS-specific bound must use additional algebraic incidence information.*

Proof. Choose M distinct supports and M distinct slopes and map them bijectively, leaving all other supports unlabeled. This changes neither Φ, its fibers, nor SP, and it respects one-slope-per-support. Mapping the same chosen supports to one slope gives identical Q and SP data with a one-ray image. The missing datum is the image incidence, not another moment of the prefix fibers. □

Proposition 22.8 (Exact pair-to-ray multiplicity requirement)**.** *Let $\mathcal{P}=\{(A,B)\in\Omega^2:\Phi(A)=\Phi(B)\}$, so $|\mathcal{P}|=\mathrm{SP}$. Let Z be a bad-slope set and $I\subseteq Z\times\mathcal{P}$ an incidence relation, and let $H,J\ge 1$. If every $\gamma\in Z$ is incident with at least H pairs and every pair is incident with at most J slopes, then*

$$|Z|\le\frac{J\,\mathrm{SP}}{H}\le\frac{J(\max_s Q(s))|\Omega|}{H}. \tag{22.2}$$

Consequently this pair-count argument yields a ray estimate at the same profile scale provided $J|\Omega|/H=e^{o(n)}$; alternatively one may prove a direct transverse-secant estimate. Merely proving that every ray has one representing pair is insufficient.

Proof. Double counting gives $H|Z|\le|I|\le J|\mathcal{P}|=J\,\mathrm{SP}$. The second inequality follows from $\mathrm{SP}\le(\max_s Q(s))\sum_s Q(s)=(\max_s Q(s))|\Omega|$. □

Corollary 22.9 (Exact finite pair-to-ray constant)**.** *For a residual profile λ, put $m_\lambda=\sum_s Q_\lambda(s)$, $L_\lambda=|\Phi_\lambda(\Omega^0_\lambda)|$, and $\overline{N}_\lambda=|\Omega^0_\lambda|/L_\lambda$. For an integer $q\ge 2$, define*

$$\Gamma_{q,\lambda}=L_\lambda^{-1}\sum_s\left(\frac{Q_\lambda(s)}{\overline{N}_\lambda}\right)^q.$$

If the pair-to-ray incidence has lower ray degree H_λ and upper pair degree J_λ, then

$$|Z^\circ_\lambda|\le\left\lfloor\frac{J_\lambda m_\lambda}{H_\lambda}\overline{N}_\lambda(L_\lambda\Gamma_{q,\lambda})^{1/q}\right\rfloor. \tag{RC$_{\mathrm{pair}}$}$$

Thus the finite compiler uses the actual residual mass and literal incidence degrees; it contains no unnamed polynomial or asymptotic loss.

Proof. The moment sandwich gives $\max_s Q_\lambda(s)\le\overline{N}_\lambda(L_\lambda\Gamma_{q,\lambda})^{1/q}$. Hence $\sum_s Q_\lambda(s)^2\le m_\lambda\max_s Q_\lambda(s)$. Apply Proposition 22.8 and take the integer floor. □

Proposition 22.10 (Curve-degree ray compiler)**.** *Let V be a finite-dimensional space of locator polynomials and, for $x \in D$, let $H_x = \{[L] \in \mathbb{P}(V) : L(x) = 0\}$ be its* evaluation hyperplane*. Suppose a residual slope set Z admits an injective assignment into a disjoint union of projective integral curves*

$$Z \hookrightarrow \coprod_{i=1}^{s} X_i(\mathbb{F}),$$

where "integral curve" means an irreducible reduced one-dimensional projective variety, together with locator morphisms $\psi_i : X_i \to \mathbb{P}(V)$. Put

$$\delta_i = \deg \psi_i^* \mathcal{O}_{\mathbb{P}(V)}(1),$$

the degree on X_i of the pullback of a projective hyperplane. For every $\gamma \in Z$, if its assigned point is $y \in X_i(\mathbb{F})$, require $\psi_i(y) = [L_{\gamma,y}]$, where $L_{\gamma,y}$ is a locator from an actual residual witness to γ. For each i, put

$$G_i = \{x \in D : \psi_i(X_i) \subseteq H_x\}, \qquad D_i^{\mathrm{mov}} = D \setminus G_i.$$

Assume the locator attached to every chosen representative on X_i has at least $h_i \geq 1$ roots in D_i^{mov}. Then

$$|Z| \leq \left\lfloor \sum_{i=1}^{s} \frac{|D_i^{\mathrm{mov}}|\delta_i}{h_i} \right\rfloor \leq \left\lfloor n \sum_{i=1}^{s} \frac{\delta_i}{h_i} \right\rfloor. \tag{RC$_{\mathrm{curve}}$}$$

In particular, a cover of subexponential total locator degree gives a subexponential ray budget when the h_i have no exponential loss. The standard degree-one pencil $[s : t] \mapsto [sA + tB]$ is the special case $X = \mathbb{P}^1$ and $\delta = 1$.

Proof. Choose one representative point for each slope. For fixed i and $x \in D_i^{\mathrm{mov}}$, the pullback of H_x is a nonzero effective divisor of degree δ_i, and hence contains at most δ_i distinct rational points. Each chosen point on X_i supplies at least h_i moving-root incidences. Thus the number assigned to X_i is at most $|D_i^{\mathrm{mov}}|\delta_i/h_i$. Sum over i and take the integer floor. □

Corollary 22.11 (Automatic exact-support locator-curve compiler)**.** *Fix a received line and exact agreement $a \geq k + 1$. For each slope in a residual set Z, choose one noncommon witness support $S_\gamma \in \binom{D}{a}$ and its locator Q_{S_γ}. Suppose the locator points are covered by $\psi_i(X_i(\mathbb{F}))$, where $\psi_i : X_i \to \mathbb{P}(V)$ are nonconstant morphisms from projective integral curves. Put*

$$\delta_i = \deg \psi_i^* \mathcal{O}(1), \qquad G_i = \{x \in D : \psi_i(X_i) \subseteq H_x\}, \qquad g_i = |G_i|.$$

Then

$$|Z| \leq \#\{i : g_i = a\} + \left\lfloor \sum_{i:g_i<a} \frac{(n - g_i)\delta_i}{a - g_i} \right\rfloor. \tag{RC$_{\mathrm{loc}}$}$$

No separate slope-to-curve injectivity hypothesis is required. A component with $g_i = a$ contains at most one selected degree-a locator, because its a common roots determine the monic support locator uniquely; this accounts for the first term. Components with $g_i > a$ contain no selected degree-a locator and are discarded.

Proof. Two distinct slopes cannot use the same chosen noncommon support. Since a monic squarefree locator determines its root set, their projective locator points are distinct as well. Assign each locator to the first covering component and choose one rational preimage; this is the injective assignment required by Proposition 22.10. On component i, every selected degree-a locator has the g_i common roots and at least $a - g_i$ roots outside them. If $g_i = a$, there is only one such monic locator. On the other components apply (RC$_{\mathrm{curve}}$) with $h_i = a - g_i$. □

Corollary 22.12 (Automatic exact-support one-pencil compiler)**.** *In the setting above, suppose all selected locators lie in the projective pencil spanned by A, B, and let $g < a$ be the number of common roots of A, B in D. Then*

$$|Z| \leq \lfloor (n - g)/(a - g) \rfloor. \tag{RC$_{\mathrm{pen}}$}$$

The same argument applies to residual locators of degree e whose roots lie in a fixed active domain $T \subseteq D$, provided the fixed profile data together with the residual locator determine the exact support. If $N = |T|$ and g_T is the number of common roots of A, B in T, then

$$|Z| \leq \lfloor (N - g_T)/(e - g_T) \rfloor .$$

Proof. The exact-support locator assignment is injective by the preceding proof. Apply Lemma 23.2, or the curve corollary with $X = \mathbb{P}^1$ and $\delta = 1$. □

By Theorem 4.2, the genuinely missing invariant can be taken to be $\Theta_t(y_0, y_1)$. Equivalently one may construct a higher-dimensional split-pencil *eliminant*, meaning a polynomial in the slope parameter obtained after eliminating locator and support variables, whose degree bounds that transverse secant incidence. By Lemma 4.3, only dependent support unions can support more than $t + 1$ rays. Those unions are no longer untreated: Theorems 4.5 and 6.27 bound every fixed union, and Proposition 23.16 gives the exact active low-nullity ranges. What remains is an uncontrolled family of many such unions, a high-nullity chart, or a direction whose rank defect exceeds the quantitative threshold of Corollary 6.29; a higher-dimensional balanced core must still be routed into one of those certified cells or paid by its own eliminant.

Here an *MDS-circuit stratum* is a chart on which the relevant union of generalized Vandermonde parity-check columns has a minimal linear dependence. Equivalently, deleting any one column restores independence. The name identifies the only dependence pattern left by the MDS property; it does not by itself place all witnesses in one fixed circuit or provide a count for an unrestricted union of such strata.

The next hypothesis packages exactly this missing conversion. Its number H_λ is a lower bound on how many certified support pairs represent one bad ray, while J_λ is an upper bound on how many bad rays one pair can represent. The ratio in (RC) is what makes the resulting double count stay at the profile scale.

Hypothesis 22.13 (Ray compiler)**.** *For every received line and primitive residual profile λ, let Z°_λ be its actual first-match set of unpaid bad slopes and put*

$$\mathcal{P}_\lambda = \{(A, B) \in (\Omega^\circ_\lambda)^2 : \Phi_\lambda(A) = \Phi_\lambda(B)\}.$$

Put $m_\lambda = |\Omega^\circ_\lambda| = \sum_s Q_\lambda(s)$. The row satisfies (RC) *on this profile if either it has the direct bound $|Z^\circ_\lambda| \leq e^{o(n)}(1 + \overline{N}_\lambda)$, or there are an incidence $I_\lambda \subseteq Z^\circ_\lambda \times \mathcal{P}_\lambda$ and numbers $H_\lambda, J_\lambda \geq 1$ such that*

$$\text{(RC)} \qquad \deg_{I_\lambda}(\gamma) \geq H_\lambda, \qquad \deg_{I_\lambda}(A, B) \leq J_\lambda, \qquad \frac{J_\lambda m_\lambda}{H_\lambda} = e^{o(n)}.$$

All bounds are uniform in the received line and realized profile. The incidence must arise from the RS witness geometry; its existence is not a consequence of Q or SP. In particular, one representing pair per ray does not supply the required lower degree.

Proposition 22.14 (Q plus RC implies rays)**.** *If primitive Q is discharged and* (RC) *holds on every residual profile, then all primitive ray ledgers are discharged at total cost $e^{o(n)}\mathfrak{E}_n$.*

Proof. The boundary prefix fibers are bounded by Q. For every remaining unpaid slope choose one noncommon support, as permitted by the imported slope interface (SE); (RC) bounds their distinct slope set by $e^{o(n)}\overline{N}_\lambda$. Sum over the subexponential profile set and use (11.1). □

The proof of Theorem 11.2 is now complete: the Sidon payment and Theorem 21.2 pay primitive Q, while Proposition 22.14 and (RC) convert the remaining support bounds into distinct-ray bounds.

Part 4. Finite deployment: Q, one-parameter BC, and the exact ledger

23. Finite max-fiber calibration

We import the four unsafe endpoints and challenge budgets from CAP25 v13.2 [Cho26, Cor. 14.3 and Def. 21.1]. Only the adjacent-row quantities consumed by the new upper compiler are printed here.

Table 2. Imported active frontier rows and the adjacent safe-side calibration.

row	a_0	$a_+ = a_0 + 1$	$w = a_+ - K$	B^*
KoalaBear MCA	1116047	1116048	67471	274980728111395087
KoalaBear list	1116046	1116047	67471	274980728111395087
Mersenne-31 MCA	1116023	1116024	67447	16777215
Mersenne-31 list	1116022	1116023	67447	16777215

Here $n = 2^{21}$, $K = 2^{20} + 1$ on the MCA route, and $K = 2^{20}$ on the list route. Put

$$\overline{F} = \binom{n}{a_+} |\mathbb{B}|^{-w}.$$

The exact ceilings of this average are, in table order,

$$57198030366, \quad 65065153468, \quad 1752700, \quad 1993678. \tag{23.1}$$

Hence the full-budget Q overhead ceilings are respectively

$$4807520.9295, \quad 4226236.5253, \quad 9.5722, \quad 8.4152. \tag{23.2}$$

Any payment made to another cell lowers these values.

The exact moment interface imported from the foundation is

$$\max_z R(z) \le (|\mathbb{B}|^w \Gamma_r)^{1/r}, \qquad R(z) = \frac{|\operatorname{Fib}_w(z)|}{\binom{n}{a_+} |\mathbb{B}|^{-w}}. \tag{23.3}$$

The exact second-moment decomposition is written

$$\sum_z N_w(z)^2 = \binom{n}{a_+} + \sum_e P_e. \tag{23.4}$$

Theorem 23.1 (Finite moment criterion for Q)**.** *If $\Gamma_r \le e^A$, then the normalized maximum fiber satisfies*

$$R_Q^{\max} \le \exp\left(\frac{A + w \log |\mathbb{B}|}{r}\right).$$

Thus a finite row with remaining bit margin Δ can be proved by this route only if

$$\log_2 \Gamma_r + w \log_2 |\mathbb{B}| \le r\Delta. \tag{23.5}$$

Proof. Take logarithms in (23.3). □

Lemma 23.2 (Pencil payment)**.** *Let $L_{[s:t]} = sA + tB$ be a projective polynomial pencil and let T be a finite set. If every counted member has at least h roots in T that are not common roots of A and B, then at most $\lfloor |T|/h \rfloor$ projective parameters are counted.*

Proof. A point $x \in T$ that is not a common root solves the homogeneous linear equation $sA(x) + tB(x) = 0$ for exactly one projective parameter. Count moving-root incidences. □

Definition 23.3 (projective locator pencil and its common D-part)**.** *Let $D \subseteq \mathbb{F}$ be finite, put $\Lambda_D = \prod_{x \in D}(X - x)$, and let $A, B \in \mathbb{F}[X]$ be not both zero. The projective locator pencil generated by A, B is*

$$L_{[s:t]}(X) = sA(X) + tB(X), \qquad [s:t] \in \mathbb{P}^1(\mathbb{F}).$$

Its fixed D-part is

$$G_D(A, B) = \gcd(A, B, \Lambda_D), \qquad g_D(A, B) = \deg G_D(A, B).$$

A root of $G_D(A, B)$ is a common root of every member of the pencil; these are precisely the roots assigned to the common-GCD branch in the first-match ledger.

Theorem 23.4 (moving-root bound for one-parameter split pencils)**.** *Let $D \subseteq \mathbb{F}$ have size n, and let $L_{[s:t]} = sA + tB$ be a projective locator pencil as in Definition 23.3. Put $G = G_D(A, B)$ and $g = \deg G$. Let $\mathcal{Z} \subseteq \mathbb{P}^1(\mathbb{F})$, and suppose that for every $\lambda = [s : t] \in \mathcal{Z}$, the polynomial L_λ/G has at least $h \geq 1$ distinct roots in $D \setminus Z(G)$. Then*

$$|\mathcal{Z}| \leq \left\lfloor \frac{n-g}{h} \right\rfloor .$$

In particular, if every counted member has degree j, has fixed D-part exactly G, and is D-split, then

$$|\mathcal{Z}| \leq \left\lfloor \frac{n-g}{j-g} \right\rfloor , \qquad j > g.$$

The case $j = g$ is not a moving split-pencil cell: all D-roots are fixed and the member belongs to the common-GCD branch.

Proof. Count incidences

$$I = \{(\lambda, x) : \lambda \in \mathcal{Z},\ x \in D \setminus Z(G),\ L_\lambda(x) = 0\}.$$

For a fixed moving root $x \in D \setminus Z(G)$, the pair $(A(x), B(x))$ is not $(0, 0)$. Hence the homogeneous linear equation

$$sA(x) + tB(x) = 0$$

has exactly one solution $[s : t] \in \mathbb{P}^1(\mathbb{F})$. Thus each moving domain point contributes to at most one incidence, and $|I| \leq n - g$. On the other hand, every $\lambda \in \mathcal{Z}$ contributes at least h incidences by hypothesis, so $|I| \geq h|\mathcal{Z}|$. This proves the first bound.

If a counted member is D-split of degree j and has fixed D-part exactly G, then exactly $j - g$ of its D-roots are moving roots in $D \setminus Z(G)$. Applying the first bound with $h = j - g$ gives the displayed inequality. If $j = g$, no moving root remains; the split condition has already been paid by the common-GCD branch rather than by a primitive moving pencil. □

Corollary 23.5 (BC is settled on primitive one-parameter locator pencils)**.** *Consider an MCA split-pencil chart at agreement m, and write $\omega = n - m$. Suppose that after the first-match tangent, common-support, quotient, extension, degree-drop, and common-GCD branches have been removed, the residual candidate error locators in this chart are represented by a projective locator pencil $L_\lambda = sA + tB$, with the slope parameter injecting into λ, and that every residual bad slope in the chart has at least one degree-ω D-split locator in the pencil. Then the number of residual bad finite slopes in this chart is at most*

$$\left\lfloor \frac{n-g}{\omega-g} \right\rfloor , \qquad g = g_D(A, B) < \omega.$$

In the primitive case $g = 0$, this is $\lfloor n/(n - m) \rfloor$. For the two active MCA adjacent rows in Table 2,

row	$m = a_+$	$\omega = n - m$	$\lfloor n/\omega \rfloor$
KoalaBear MCA	1116048	981104	2
Mersenne-31 MCA	1116024	981128	2

so every primitive one-parameter BC pencil contributes at most two finite bad slopes.

Proof. The first-match removals ensure that the chart is counted only through moving roots of the displayed pencil. A residual bad slope supplies at least one degree-ω split locator, and the slope-to-pencil parameter is injective by hypothesis; hence the slope count is bounded by the number of split parameters. Apply Theorem 23.4 with $j = \omega$. The displayed row values use $n = 2^{21}$, $a_+ = 1116048$ for KoalaBear MCA, and $a_+ = 1116024$ for Mersenne-31 MCA. □

Definition 23.6 (saturated codeword rays and support census)**.** *Let $C = \mathrm{RS}[\mathbb{F}, D, K]$, let $U : D \to \mathbb{F}$, and put*

$$S_c(U) = \{x \in D : U(x) = c(x)\}, \qquad s_c(U) = |S_c(U)|$$

for $c \in C$. The saturated ray set at agreement m is

$$\mathrm{Ray}(U; m) = \{c \in C : s_c(U) \geq m\}.$$

Let $\omega = n - m$, and define the support-locator census

$$\begin{aligned}\mathrm{Cen}(U;m) = \#\{(W,N) \in \mathbb{F}[X]^2 : W \text{ monic},\ \deg W = \omega,\ W \mid \ell_D,\\ WU = N \text{ on } D,\quad W \mid N,\quad \deg(N/W) < K\}.\end{aligned}$$

Equivalently, $W = \ell_{D\setminus T}$ for an agreement support $T \subseteq D$ of size m.

Theorem 23.7 (exact saturation identity)**.** *With $C = \mathrm{RS}[\mathbb{F}, D, K]$ and $\mathrm{Cen}(U;m)$ as in Definition 23.6, for every word $U : D \to \mathbb{F}$,*

$$\mathrm{Cen}(U;m) = \sum_{c\in C} \binom{s_c(U)}{m}.$$

Moreover, the support locators lying above a fixed codeword ray c are exactly

$$(W,N) = H\,(G_c, G_c c), \qquad G_c = \ell_{D\setminus S_c(U)}, \qquad H \mid \ell_{S_c(U)}, \qquad \deg H = s_c(U) - m.$$

Consequently the fiber size of the forgetful map from support locators to saturated codeword rays is $\binom{s_c(U)}{m}$ above c.

Proof. Let (W,N) be counted by $\mathrm{Cen}(U;m)$. Since W is monic, divides ℓ_D, and has degree $n-m$, there is a unique m-set $T \subseteq D$ such that $W = \ell_{D\setminus T}$. Since $W \mid N$ and $\deg(N/W) < K$, write $N = Wc$ for a unique codeword $c \in C$. On every $x \in T$, the value $W(x)$ is nonzero; the identity $WU = N = Wc$ on D therefore gives $U(x) = c(x)$. Thus $T \subseteq S_c(U)$.

Conversely, if $c \in C$ and $T \subseteq S_c(U)$ has $|T| = m$, then $W = \ell_{D\setminus T}$ and $N = Wc$ satisfy all defining conditions of $\mathrm{Cen}(U;m)$. For fixed c, the number of such supports is $\binom{s_c(U)}{m}$, and summing over c gives the first identity.

For the fiber statement, write $G_c = \ell_{D\setminus S_c(U)}$. If $T \subseteq S_c(U)$, then

$$\ell_{D\setminus T} = \ell_{D\setminus S_c(U)}\ell_{S_c(U)\setminus T} = G_c H,$$

where $H = \ell_{S_c(U)\setminus T}$, hence $H \mid \ell_{S_c(U)}$ and $\deg H = s_c(U) - m$. The converse is the same construction reversed, because every monic divisor H of $\ell_{S_c(U)}$ of degree $s_c(U) - m$ is the locator of $S_c(U) \setminus T$ for a unique m-set $T \subseteq S_c(U)$. □

Corollary 23.8 (raw support BC is not the MCA object)**.** *In the setting of Theorem 23.7, if U agrees with a single codeword c on $n-d$ positions and with no other codeword on m or more positions, then*

$$|\,\mathrm{Ray}(U;m)| = 1, \qquad \mathrm{Cen}(U;m) = \binom{n-d}{m}.$$

Thus an exponentially large raw support-locator family may represent one MCA ray. A split-pencil proof for MCA must therefore count saturated primitive line-rays after quotient, prefix, tangent, and common-support branches have been paid; a raw support census is a list/moment object, not the final MCA numerator.

Proof. The ray statement is the hypothesis. The support-census identity is Theorem 23.7 with one nonzero summand $\binom{s_c(U)}{m} = \binom{n-d}{m}$. The final sentence is just the same equality interpreted in MCA terms: MCA counts a bad slope once, whereas $\mathrm{Cen}(U;m)$ counts all size-m supports below the same saturated agreement ray. □

Definition 23.9 (line rays)**.** *In the setting of Definition 23.6, for an affine line $U_Z = u + Zv$ and a set of finite slopes $E \subseteq \mathbb{F}$, define*

$$\mathrm{LineRay}_E(u,v;m) = \{(z,c) \in E \times C : \ |\{x \in D : u(x) + zv(x) = c(x)\}| \geq m\}.$$

Write $s_{z,c} = |\{x \in D : u(x) + zv(x) = c(x)\}|$.

Proposition 23.10 (line-ray saturation identity)**.** *With notation as in Definition 23.9, for every $E \subseteq \mathbb{F}$,*

$$\sum_{z\in E} \mathrm{Cen}(U_z;m) = \sum_{(z,c)\in \mathrm{LineRay}_E(u,v;m)} \binom{s_{z,c}}{m}.$$

Consequently

$$|\{z \in E : \exists c\ (z,c) \in \mathrm{LineRay}_E(u,v;m)\}| \le |\,\mathrm{LineRay}_E(u,v;m)| \le \sum_{z\in E} \mathrm{Cen}(U_z;m).$$

The two inequalities are the exact deduplication losses between raw support locators, saturated codeword rays, and slope numerators.

Proof. Apply Theorem 23.7 separately to the word $U_z = u + zv$ for each $z \in E$, and then sum over z. The first inequality forgets the codeword coordinate of a line ray, and the second follows because each line ray has $s_{z,c} \ge m$, hence contributes at least one support to the right-hand side of the identity. Corollary 23.8 shows that the second inequality can be exponentially loose for a single slope and a single codeword ray, so an MCA upper ledger must either count slopes directly or control this saturation factor. □

Theorem 23.11 (Q discharges the shift-pair ledger)**.** *Let $N_w(z) = |\,\mathrm{Fib}_w(z)|$, let*

$$\overline{F} = \binom{n}{m}|\mathbb{B}|^{-w},$$

and let P_e denote the full off-diagonal shift-pair stratum in the imported decomposition (23.4), namely

$$P_e = \sum_{\substack{R\subseteq D\\ |R|=m-e}} \mathrm{sp}_w(e; D\setminus R).$$

If Q holds in max-fiber form with constant κ,

$$\max_z N_w(z) \le \kappa\overline{F},$$

then necessarily $\kappa\overline{F} \ge 1$ whenever $\binom{D}{m}$ is nonempty, and

$$\sum_{e=w+1}^{\min(m,n-m)} P_e \le \left(\kappa - \frac{1}{\overline{F}}\right)\binom{n}{m}\overline{F}.$$

Every quotient-removed or primitive shift-pair subledger satisfies the same upper bound. Hence SP is not an independent residual input in any closing theorem that already proves Q as a max-fiber theorem.

Proof. The imported exact second-moment identity (23.4) says

$$\sum_z N_w(z)^2 = \binom{n}{m} + \sum_{e=w+1}^{\min(m,n-m)} P_e.$$

On the other hand,

$$\sum_z N_w(z)^2 \le (\max_z N_w(z)) \sum_z N_w(z) \le \kappa\overline{F}\binom{n}{m},$$

because $\sum_z N_w(z) = \binom{n}{m}$. Since at least one fiber is nonempty when $\binom{D}{m} \ne \varnothing$, the hypothesis also implies $\kappa\overline{F} \ge 1$, so the displayed right-hand side is nonnegative in the nonvacuous case. Subtracting the diagonal term $\binom{n}{m}$ gives the displayed inequality. Any primitive or quotient-removed ledger is a sub-sum of the full off-diagonal ledger, so it is bounded a fortiori. □

Theorem 23.12 (SP: unconditional primitive shift-pair bound)**.** *Use the notation of the imported exact second-moment ledger. For $w + 1 \le e \le \min(m, n - m)$, put*

$$T_e(n,m) = \binom{n}{m-e}\binom{n-m+e}{e}\binom{n-m}{e}.$$

Then the total off-diagonal contribution at distance e satisfies

$$P_e^{\mathrm{quot}} + P_e^{\mathrm{prim}} \le T_e(n,m).$$

If $D \subseteq \mathbb{B}$, then the top stratum $e = w+1$ also satisfies the sharper constant-shift bound

$$P_{w+1}^{\mathrm{quot}} + P_{w+1}^{\mathrm{prim}} \le (|\mathbb{B}|-1)\binom{n}{m-w-1}\binom{n-m+w+1}{w+1}.$$

Consequently

$$\Gamma_2 \le \frac{1}{\overline{F}} + \frac{\sum_{e=w+1}^{\min(m,n-m)} T_e(n,m)}{\binom{n}{m}\overline{F}},$$

with the option of replacing the $e = w+1$ term by the sharper displayed top-stratum bound.

Proof. Fix e. In the imported exact second-moment decomposition (23.4), an off-diagonal pair is encoded by a common part $R \subseteq D$ of size $m-e$, then by two disjoint degree-e root sets inside $D \setminus R$. There are $\binom{n}{m-e}$ choices for R. Once R is fixed, the ambient set $D \setminus R$ has size $n-m+e$; the first root set has at most $\binom{n-m+e}{e}$ choices and the second, disjoint from it, has at most $\binom{n-m}{e}$ choices. This ignores the depth equation and the quotient/primitive first-match assignment, so it is an upper bound for $P_e^{\mathrm{quot}} + P_e^{\mathrm{prim}}$.

When $e = w+1$, the imported decomposition (23.4) shows that every top-stratum shift pair has the form $A - B = c \in \mathbb{F}^\times$. If both polynomials split over $D \subseteq \mathbb{B}$, then both have coefficients in $\mathbb{B}$, hence $c \in \mathbb{B}^\times$. For each fixed R, choose A by choosing its e-element root set in $D \setminus R$, and choose $c \in \mathbb{B}^\times$; the polynomial $B = A - c$ is then determined. Requiring B to split over $D \setminus R$, to be disjoint from A, and then assigning the pair to either the quotient or primitive ledger can only reduce the count. This gives the top-stratum bound. The displayed Γ_2 inequality is the imported exact second-moment ledger with the total quotient-plus-primitive off-diagonal contribution in every stratum replaced by the just-proved upper bound. ☐

Proposition 23.13 (size of the proper Q bound at the active adjacent rows)**.** *At the four active adjacent agreements, the imported Q packing estimate* [Cho26, Secs. 19–21] *gives the following one-term upper bounds for the remaining normalized prefix overhead:*

row	w	$\log_2 R_Q^{\mathrm{pack}}$ *allowed by the bound*	*spare margin*
KoalaBear MCA	67471	1660789.024	22.1969
KoalaBear list	67471	1660789.017	22.0109
Mersenne-31 MCA	67447	1660926.120	3.2589
Mersenne-31 list	67447	1660926.114	3.0730.

Thus the unconditional packing proof is a real theorem but is far too weak to supply the adjacent safe ledger.

Proof. For each row, $n = 2097152$, $K = k+1 = 1048577$ on the MCA route, $K = k = 1048576$ on the list route, and $w = a_+ - K$. The base-field orders are $2^{31} - 2^{24} + 1$ for KoalaBear and $2^{31} - 1$ for Mersenne-31. The imported Q packing inequality [Cho26, Secs. 19–21] gives

$$\log_2 R_Q^{\mathrm{pack}} \le w \log_2 |\mathbb{B}| - \log_2 \binom{a_+}{\lfloor w/2 \rfloor} - \log_2 \binom{n-a_+}{\lfloor w/2 \rfloor}.$$

Substituting the exact field orders and

$$a_+ \in \{1116048, 1116047, 1116024, 1116023\}$$

gives the displayed values. For these four rows the Johnson-ball summand

$$J_i = \binom{a_+}{i}\binom{n-a_+}{i}$$

is increasing on $0 \le i \le t$. Indeed

$$\frac{J_i}{J_{i-1}} = \frac{(a_+ - i + 1)(n - a_+ - i + 1)}{i^2},$$

and at $i = t$ this ratio is already > 900 in all four rows, while the ratio is larger for smaller i. Hence $V_t(n, a_+) \le (t+1)J_t$, so replacing the one-term denominator by the full ball can lower the displayed logarithmic upper bound by less than $\log_2(t+1) < 16$ bits. This is negligible compared with the gap between the displayed million-bit overhead and the row margins. $\square$

Definition 23.14 (row-sharp Q atom bound)**.** *Fix a deployed adjacent row, let a_+ be its adjacent agreement, let K be the list dimension used on that route, and put $w = a_+ - K$. For a first-match residual prefix-boundary family $\mathcal{P}_Q(z) \subseteq \mathrm{Fib}_w(z)$, define*

$$R_Q^{\max} = |\mathbb{B}|^w \max_z \frac{|\mathcal{P}_Q(z)|}{\binom{n}{a_+}}.$$

The row-sharp Q atom bound with bit margin Δ_Q is

$$R_Q^{\max} \le 2^{\Delta_Q}.$$

Equivalently, if the entire row budget is hypothetically assigned to Q, the sufficient integer form is

$$\max_z |\mathcal{P}_Q(z)| \le B^*.$$

Any non-Q cell paid in the same first-match ledger only decreases the available Δ_Q.

Proposition 23.15 (exact finite Q atom target at the four adjacent rows)**.** *At the active adjacent agreements, the full-budget Q atom target is as follows:*

row	w	$\left\lceil \binom{n}{a_+} \lvert\mathbb{B}\rvert^{-w} \right\rceil$	B^*	$B^*/\left\lceil \binom{n}{a_+} \lvert\mathbb{B}\rvert^{-w} \right\rceil$
KoalaBear MCA	67471	57198030366	274980728111395087	4807520.9295
KoalaBear list	67471	65065153468	274980728111395087	4226236.5253
Mersenne-31 MCA	67447	1752700	16777215	9.5722
Mersenne-31 list	67447	1993678	16777215	8.4152

In bits, these are exactly the spare margins already printed in the row table:

$$22.1969, \quad 22.0109, \quad 3.2589, \quad 3.0730.$$

Thus the binding Q problem is the Mersenne-31 list row: after all first-match payments, the remaining primitive prefix-boundary atom must be at most about 8.42 *times its average size, and less if any other cell consumes budget.*

Proof. For each row, substitute $n = 2^{21}$, $k = 2^{20}$, the displayed adjacent agreement a_+, and $K = k+1$ on the MCA route and $K = k$ on the list route. The average prefix fiber is $\binom{n}{a_+}|\mathbb{B}|^{-w}$, and the integer lower-route replay uses its ceiling. The budgets are

$$B^*_{\mathrm{KB}} = \left\lfloor \frac{(2^{31} - 2^{24} + 1)^6}{2^{128}} \right\rfloor, \qquad B^*_{\mathrm{M31}} = \left\lfloor \frac{(2^{31} - 1)^4}{2^{100}} \right\rfloor.$$

The displayed ratios are the exact integer quotients evaluated as decimals. Taking base-two logarithms of $B^*/(\binom{n}{a_+}|\mathbb{B}|^{-w})$ gives the four printed bit margins. $\square$

Proposition 23.16 (Exact fixed-union payments at the active rows)**.** *Let $t = n - a_+$.*

(a) Fixed-word Q and list charts. *Let $K = k+1$ on an MCA route and $K = k$ on a list route, put $R_Q = n - K$, and let $\mathcal{A}$ be a subfamily of one boundary-Q prefix fiber, or of one arbitrary-word list, whose error supports are all contained in a fixed union U of size $R_Q + \nu$. Then Theorem 4.7 gives a full-budget payment for every integer nullity in the ranges below:*

(FU-Q/list)

row	R_Q	$R_Q - t$	$0 \le \nu \le$	$\max_{\nu \text{ in range}} \lvert\mathcal{A}\rvert$
KoalaBear MCA	1048575	67471	4983	94008
KoalaBear list	1048576	67471	4983	94632
Mersenne-31 MCA	1048575	67447	4980	3555853
Mersenne-31 list	1048576	67447	4980	4735771

Every number in the last column is below the corresponding challenge budget B^* of Proposition 23.15. In particular, a global Q counterexample at the binding Mersenne-31 list row cannot be contained in one support union of nullity at most 4980.

(b) Fixed-line MCA charts. *For the actual MCA code, $R = n-k = 1048576$. The unconditional multiplicity compiler Theorem 4.5 pays the following fixed unions within the full row budget:*

(FU-MCA)

row	$0 < \nu \le$	$\max_{\nu \text{ in range}} \lvert Z\rvert$	B^*
KoalaBear MCA	2	2847909263951	274980728111395087
Mersenne-31 MCA	1	8150817	16777215

If the direction is rank-regular on its union, so that $d_U(y_1) = R$, the directional compiler Corollary 6.28 pays the much larger ranges

(FU-MCA-reg)

row	$0 \le \nu \le$	$\max_{\nu \text{ in range}} \lvert Z\rvert$
KoalaBear MCA	4982	94008
Mersenne-31 MCA	4979	3555853

The direction-defect threshold of Corollary 6.29 pays more than the rank-regular case. With the full row budget, selected exact thresholds are

(FU-KB-def)

KoalaBear ν	0	1000	4000	4500	4979	4980	4982
$J_{B^*}(\nu)$	4341	3466	853	418	3	2	0

and

(FU-M31-def)

Mersenne-31 ν	0	1000	4000	4500	4979
$J_{B^*}(\nu)$	4338	3463	849	415	0

Thus an unpaid fixed-union MCA chart is either above the displayed nullity range or has defect $j > J_{B^*}(\nu)$; it need not be declared exceptional merely because $j > 0$. At the Mersenne row, two disjoint one-circuit charts cost at most $2 \cdot 8150817 = 16301634 < 16777215$, leaving 475581 units of the full budget.

Proof. For a boundary-Q fiber, the imported exact witness-list dictionary [Cho26, Secs. 18–21] associates to every support a distinct codeword of $\mathrm{RS}[\mathbb{F}, D, K]$ agreeing with one received word on that support. Hence a subfamily whose error supports lie in U is a subset of the affine syndrome fiber $\mathcal{L}_U(y,t)$ in Theorem 4.7. The same statement is tautological for an arbitrary-word list. Substituting $N = R_Q + \nu$ and $h = N - t$ into (4.9), and evaluating every integer in the displayed ranges, gives (FU-Q/list). At the next nullity the Johnson denominator is nonpositive in every row: $\nu = 4984$ for KoalaBear and $\nu = 4981$ for Mersenne-31.

For MCA, substitute $R = 1048576$ and the two deployed values of t into (4.5). The largest full-budget nullities are 2 and 1, giving (FU-MCA). Under rank regularity, substitute the same row data into (6.28); its denominator is positive through $\nu = 4982$ and 4979, respectively, and the displayed maxima occur at the final integer in each range. Finally substitute the row budgets into (6.30) to obtain Eqs. (FU-KB-def) and (FU-M31-def). All comparisons are exact integer comparisons and are reproduced by the release verifier. □

Proposition 23.17 (Exact recursive affine-span payments at the active rows)**.** *In the boundary-Q witness list, or in an arbitrary-word list for the same route, suppose the associated codewords lie in one affine subspace of* $\mathrm{RS}[\mathbb{F}, D, K]$ *of dimension s. The recursive compiler Theorem 4.8 gives the following full-budget payments:*

(AF-Q/list)

row	$0 \le s \le$	$\max_{s \text{ in range}} \lvert\mathcal{L}_A\rvert$	*compiler value at the next s*
KoalaBear MCA	14	47876303026096432	743896698428332665
KoalaBear list	14	47876942243704279	743907339955962922
Mersenne-31 MCA	6	14115447	219426634
Mersenne-31 list	6	14115528	219428099

The middle column is below the corresponding full challenge budget in every row. The final column only records that this compiler no longer fits at dimension 15 or 7; it is not a lower bound on an actual list. Consequently a Q counterexample surviving the full-budget payments must have affine codeword-span dimension at least 15 on a KoalaBear row and at least 7 on a Mersenne-31 row.

Proof. Apply Theorem 4.8 with $m = a_+$, with $K = k + 1$ on an MCA boundary route and $K = k$ on a list route. Evaluate the single rational expression in (4.10) by exact integer products. The value is increasing over each displayed range, so the maximum occurs at its last dimension. Comparison with the exact budgets in Proposition 23.15 gives the claim; the next-dimension values mark the limit of the recursive compiler. □

Remark 23.18 (Retraction of the active affine-span MCA payments). The former KoalaBear and Mersenne-31 MCA affine-span payments used the refuted ordered-basis denominator of Proposition 6.9. Their falling- and rising-product evaluations were arithmetically correct evaluations of that expression, but they are not upper bounds for MCA slope families. No affine-rank lower wall for a surviving MCA chart is claimed here. The ordinary affine-span list payments immediately above are unaffected.

Proposition 23.19 (Exact rank-flat and common-zero reductions at the active rows)**.** *Consider the first affine dimensions not paid by Proposition 23.17: $s = 15$ on KoalaBear and $s = 7$ on Mersenne-31. If such an affine Q/list chart exceeds the full row budget, Corollary 4.10 forces a proper direction subcode with the following coarse support ceiling:*

(RF-active)

row	s	$\Theta^{\mathrm{coarse}}_{s,B^*}$	$d_j \le$	$d_j - (n - K) \le$
KoalaBear MCA	15	152252	1133356	84781
KoalaBear list	15	152252	1133357	84781
Mersenne-31 MCA	7	232423	1213551	164976
Mersenne-31 list	7	232423	1213552	164976

Thus an over-budget first-unpaid affine chart is not generalized-weight uniform: it contains a proper direction subcode supported on a union much smaller than the full domain.

There is also an exact common-zero refinement beyond the former fixed-union boundary. Write the common-zero set of the affine direction space as $z = K - \nu$, and let b be the number of those coordinates on which the common affine value disagrees with the received word. At the first nullity not paid by Proposition 23.16, the common-zero Johnson compiler gives

(CZ-active)

row	ν	$\min b$	*bound*	*residual* b
KoalaBear MCA	4984	2	537111	$b \le 1$
KoalaBear list	4984	2	558125	$b \le 1$
Mersenne-31 MCA	4981	7	590532	$b \le 6$
Mersenne-31 list	4981	7	616024	$b \le 6$

At nullity 5000*, the corresponding paid mismatch thresholds are $b \ge 102$ on the two KoalaBear rows and $b \ge 126$ on the two Mersenne-31 rows.*

Proof. For (RF-active), use the coarse form of Corollary 4.10. The integer roots are

$$\left\lfloor \left(\frac{n^{\underline{15}}}{(B^*_{\mathrm{KB}} + 1)(67471 + 15)} \right)^{1/14} \right\rfloor = 152252$$

and

$$\left\lfloor \left(\frac{n^{\underline{7}}}{(B^*_{\mathrm{M31}} + 1)(67447 + 7)} \right)^{1/6} \right\rfloor = 232423.$$

Adding the exact radii $t = n - a_+$ gives the support and nullity ceilings in the table.

For (CZ-active), substitute the row data into (4.20) and choose the least nonnegative integer b for which the denominator is positive and the integer floor is at most the row budget. All displayed comparisons, including the nullity- 5000 thresholds, are exact and are reproduced by the verifier. □

Proposition 23.20 (Exact codimension-one recursion at the active rows)**.** *Consider a first-unpaid affine Q/list chart: dimension* 15 *on a KoalaBear route and dimension* 7 *on a Mersenne-31 route. Suppose the low-support direction subcode forced by Proposition 23.19 has support closure of codimension one in the chart direction space. Write $j = 14$ or $j = 6$ for the dimension* *of that support-saturated hyperplane, d for its support size, e for the support increment to the whole chart, and b_0 for the common mismatch count outside the whole support. Then the exact full-budget payments are*

(CR-active)

row	j	$d-(n-K) \le$	*e paid from*	*b_0 paid from*	*largest residual nullity*
KoalaBear MCA	14	84781	8565	9813	93345
KoalaBear list	14	84781	8565	9813	93345
Mersenne-31 MCA	6	164976	73266	40074	238241
Mersenne-31 list	6	164976	73267	40074	238242

Here "e paid from" means that every chart with at least that support increment is below the full row budget, and similarly for b_0. The largest residual nullity is $d+e-(n-K)$ after both payments have failed. More explicitly, every surviving codimension-one branch lies in

(CR-KB) $e \le 8564$, $b_0 \le 9812$ *on either KoalaBear route,*

(CR-MCA) $e \le 73265$, $b_0 \le 40073$ *on the Mersenne-31 MCA route,*

(CR-list) $e \le 73266$, $b_0 \le 40073$ *on the Mersenne-31 list route.*

The corresponding certified maxima at the first paid integers are

(CR-budget)

row	C_e	C_{b_0}	B^*
KoalaBear MCA	274959863953460332	274954310643125130	274980728111395087
KoalaBear list	274963447501329540	274957707108580869	274980728111395087
Mersenne-31 MCA	16777189	16776155	16777215
Mersenne-31 list	16777174	16776251	16777215

where C_e is the closed bound at the first paid support increment and C_{b_0} is the all-extender bound at the first paid mismatch count. The immediately preceding integer fails for the displayed MDS-soft certificate in every row; this records the sharp integer transition of this compiler, not a lower bound on an actual list.

Proof. The support closure has the same support as the direction subcode supplied by Proposition 23.19; hence its support size is in the exact ranges

row	d range
KoalaBear MCA	$1048589 \le d \le 1133356$
KoalaBear list	$1048590 \le d \le 1133357$
Mersenne-31 MCA	$1048581 \le d \le 1213551$
Mersenne-31 list	$1048582 \le d \le 1213552.$

For the MDS-soft profile Π of Corollary 5.3, the largest initial factor is $A = d - t + b_0$ and $Q = w + 1 + b_0$. Uniformly over the displayed ranges,

$$(j-1)Q - A \ge \begin{cases} 13 \cdot 67472 - 152252 = 724884, & \text{KoalaBear}, \\ 5 \cdot 67448 - 232423 = 104817, & \text{Mersenne-31}. \end{cases}$$

The final term grows by $(j-2)b_0$, so Lemma 5.2 proves profile interpolation for every e and b_0 in the active ranges.

The closed bound (5.11) decreases with e and with b_0, while the all-extender bound (5.12) decreases with b_0. It therefore suffices to use $b_0 = 0$ for the support-increment payment and to test the displayed mismatch threshold for the all-extender payment. Exact rational comparison over every integer d in the four finite ranges gives the maxima in (CR-budget). At the preceding support increments the maxima are respectively

$$274997419229066860, \quad 275001003273987385, \quad 16777303, \quad 16777288,$$

and at the preceding mismatch counts they are

$$275005816112956831, \quad 275009213214651183, \quad 16777247, \quad 16777343.$$

Each exceeds its row budget. Finally combine the support ceiling with $e \le e_{\text{paid}} - 1$ to obtain the residual nullities in (CR-active). The release verifier performs all comparisons with exact integers. $\square$

Proposition 23.21 (Active higher-codimension flag payment)**.** *Consider a first-unpaid affine Q/list chart: dimension* 15 *on a KoalaBear route and dimension* 7 *on a Mersenne-31 route. If the chart is over the full row budget, choose a generalized-weight minimizer supplied by Corollary 4.10. By Lemmas 6.2 and 6.3, its support closure admits the exact coset-free flag certificate of Theorem 6.4 in every codimension.*

For support-closure codimension at least two, the coarse all-extender certificate already pays every chart with

(23.6)

$$b_0 \geq 70230 \quad \text{on either KoalaBear route,} \qquad b_0 \geq 108962 \quad \text{on either Mersenne-31 route.}$$

At the first paid integers, the worst exact bounds are

(HF-active)

row	b_0	*certified maximum*	B^*
KoalaBear MCA	70230	274974976450914526	274980728111395087
KoalaBear list	70230	274975238687487221	274980728111395087
Mersenne-31 MCA	108962	16776934	16777215
Mersenne-31 list	108962	16776950	16777215

The preceding mismatch integer fails the same uniform certificate, with values respectively

$$275004931457252515, \quad 275005193722392530, \quad 16777600, \quad 16777616.$$

These are transitions of the all-extender flag certificate, not lower bounds on actual lists. Below (23.6), the exact binary-pattern linear program (6.8) and the closed tensor certificate (6.12) remain available for the actual support-layer profile.

Proof. The rank-flat conclusion supplies a generalized-weight minimizer of some dimension $j < s$. It is support-saturated by Lemma 6.2, and therefore has a support-saturated flag to the whole chart by Lemma 6.3. For codimension at least two, use (6.9), replace the flag-support product by $(n-j)^{\underline{s-j}}$, and use the exact support ceilings in (RF-active). The resulting bound is

$$\frac{d^{\underline{j}}(n-j)^{\underline{s-j}}}{(d-t+b_0)\prod_{i=1}^{j-1}(w+i+b_0)(w+1+b_0)^{s-j}}.$$

For fixed j its maximum over the allowed support interval occurs at an endpoint: the ratio of the d-dependent factor at $d+1$ and d changes sign at most once, since the numerator of the comparison is

$$(j-1)d - j(t-b_0) + j - 1.$$

Checking the two endpoints for $1 \leq j \leq 13$ on KoalaBear and $1 \leq j \leq 5$ on Mersenne-31 gives the exact transitions displayed above. The independent verifier performs these integer comparisons without floating point. □

Proposition 23.22 (Exact paving-basis MCA payments at the active rows)**.** *Let $j = R - d_U(y_1)$ be the direction defect of a separated fixed-union MCA chart. With the full row budget, the paving threshold of Corollary 6.31 is*

(PAV-KB)

KoalaBear ν	0–9	10	11	12	13
$J^{\mathrm{pav}}_{B^*}(\nu)$	67471	67468	67423	66715	55724

and

(PAV-M31)

Mersenne-31 ν	0–1	2	3	4	5
$J^{\mathrm{pav}}_{B^*}(\nu)$	67447	67432	67213	63797	10700.

Consequently every separated KoalaBear direction defect is paid through nullity 9*. At Mersenne nullity* 2*, only the top fifteen possible defects* $67433 \leq j \leq 67447$ *remain outside this compiler. At every nullity the certified range is the maximum of the paving threshold and the directional-Johnson threshold in Eqs.* (FU-KB-def) *and* (FU-M31-def)*.*

Proof. Substitute $R = 1048576$, the deployed radii, and the exact row budgets into (6.32). At the last paid defects in the nontrivial columns, the paving bounds are respectively

ν	$J^{\mathrm{pav}}_{B^*}(\nu)$	bound at $J^{\mathrm{pav}}_{B^*}(\nu)$
10	67468	215264294199835875
11	67423	273052400452931367
12	66715	274631447525088415
13	55724	274966795861149003

and

ν	$J^{\mathrm{pav}}_{B^*}(\nu)$	bound at $J^{\mathrm{pav}}_{B^*}(\nu)$
2	67432	15838845
3	67213	16764415
4	63797	16774561
5	10700	16776956

The next defect exceeds the budget in every displayed nonterminal column. All calculations use exact binomial integers. $\square$

Proposition 23.23 (moment-order floor for a finite Q proof)**.** *Let*

$$\tau = \frac{\sum_z |\mathcal{P}_Q(z)|}{\binom{n}{m}} \in (0, 1]$$

be the mass of the pruned residual relative to the full support slice. A proof based only on the moment criterion of Theorem 23.1 must satisfy the necessary condition

$$r(\Delta_Q - \log_2 \tau) \geq w \log_2 |\mathbb{B}|.$$

In the full-mass case $\tau = 1$, this reduces to

$$r \geq \left\lceil \frac{w \log_2 |\mathbb{B}|}{\Delta_Q} \right\rceil.$$

For the active rows the corresponding full-mass floors are

row	w	Δ_Q *bits*	*minimum* r
KoalaBear MCA	67471	22.1969	94196
KoalaBear list	67471	22.0109	94991
Mersenne-31 MCA	67447	3.2589	641593
Mersenne-31 list	67447	3.0730	680397

Thus fixed moments cannot prove the finite adjacent rows in the full-mass model. After first-match pruning, the displayed row floors remain applicable only when $\tau = 1$; otherwise the residual mass must be carried explicitly, or one must use an off-diagonal or falling-factorial moment or isolate sparse-heavy fibers as a separate cell.

Proof. Normalize the residual masses by the full slice and put $\mu(z) = |\mathcal{P}_Q(z)|/\binom{n}{m}$, so $\sum_z \mu(z) = \tau$. Convexity over at most $|\mathbb{B}|^w$ prefix values gives

$$|\mathbb{B}|^{w(r-1)} \sum_z \mu(z)^r \geq \tau^r.$$

In the notation of Theorem 23.1, this is $\Gamma_r \geq \tau^r$. The finite criterion requires

$$\log_2 \Gamma_r + w \log_2 |\mathbb{B}| \leq r\Delta_Q.$$

Substituting the lower bound and rearranging gives the stated necessary condition. Setting $\tau = 1$ yields the table. The numerical entries use the exact row prime in $\log_2 |\mathbb{B}|$. $\square$

Proposition 23.24 (the BC moving-root theorem does not pay boundary Q)**.** *In the boundary-Q profile of the imported boundary-Q dictionary* [Cho26, Secs. 18–21]*, the unknown divisor is*

$$Q_z + A, \qquad \deg A \leq \omega - w - 1, \qquad \omega = n - a_+.$$

Thus the coefficient parameter space has dimension $\omega - w$. At the active rows this dimension is

row	$\omega = n - a_+$	w	$\omega - w$
KoalaBear MCA	981104	67471	913633
KoalaBear list	981105	67471	913634
Mersenne-31 MCA	981128	67447	913681
Mersenne-31 list	981129	67447	913682

Therefore Theorem 23.4 and Corollary 23.5 cannot be applied directly to boundary Q: those theorems count split parameters on a projective one-parameter locator pencil, while Q is a high-dimensional affine divisor slice. To route Q through BC one would need an additional theorem decomposing this affine slice into paid one-parameter pencils with a first-match bound on the number of relevant pencils. No such theorem is present in this manuscript.

Proof. The imported boundary-Q dictionary [Cho26, Secs. 18–21] states exactly that a prefix fiber is counted by the divisibility condition $Q_z + A \mid X^n - \beta$ with $\deg A \le \omega - w - 1$. A polynomial of degree at most $\omega - w - 1$ has $\omega - w$ free coefficients. Substituting the four values of a_+, K, and $w = a_+ - K$ gives the table. Since $\omega - w$ is larger than nine hundred thousand in all four cases, the family is not a one-parameter pencil. The moving-root theorem is still valid for every line inside this affine space, but a line-by-line decomposition without a bound on the number of lines gives no row budget. $\square$

Part 5. Order-32 line rank and rational-atom classification

The rest of the paper changes the MCA completion mechanism. Instead of adding many local cell budgets, it seeks a maximum-type classification: every large bad-slope set is either one global affine codeword line, one coherent rational atom, or one primitive spread family. This part proves the exact rank dictionary, the dyadic quotient branch, low-degree and low-complexity atom extraction, and atom coherence.

24. Rank-theoretic setup

Let $D \subseteq \mathbb{F}$ consist of n distinct evaluation points and let

$$C = \mathrm{RS}[\mathbb{F}, D, k] = \{(h(x))_{x \in D} : h \in \mathbb{F}[X],\ \deg h < k\}.$$

Fix distinct finite slopes $\gamma_1, \dots, \gamma_L$ and a received line $r_\gamma = r_0 + \gamma r_1$. Suppose $h_i \in \mathbb{F}[X]_{<k}$ agrees with r_{γ_i} on a support S_i of size m; shrinking supports to exact size m does not affect the rank calculation. Put

$$I_x = \{i : x \in S_i\}.$$

The tuple is globally affine if $h_i = A + \gamma_i B$ for some $A, B \in \mathbb{F}[X]_{<k}$.

25. The rank-two reduced intersection matrix

25.1. Local relation spaces. For $I \subseteq [L]$, define

$$\mathcal{K}_\gamma(I) := \left\{ c \in \mathbb{F}^L : \mathrm{supp}(c) \subseteq I, \quad \sum_{i=1}^{L} c_i = 0, \quad \sum_{i=1}^{L} c_i \gamma_i = 0 \right\}.$$

Since the slopes are distinct,

$$\dim \mathcal{K}_\gamma(I) = (|I| - 2)_+.$$

Let $v_k(x) = (1, x, \dots, x^{k-1}) \in \mathbb{F}^k$.

Definition 25.1 (rank-two reduced intersection matrix)**.** *Choose any basis $\mathcal{B}_x$ of $\mathcal{K}_\gamma(I_x)$ for every $x \in D$. The matrix*

$$\mathcal{M}^{(2)}(D, \mathcal{S}, \gamma)$$

has Lk columns, grouped into L blocks of length k, and one row

$$c \otimes v_k(x)$$

for every $x \in D$ and $c \in \mathcal{B}_x$.

Changing the local bases left-multiplies the corresponding row blocks by invertible matrices, so the kernel and rank are intrinsic.

Theorem 25.2 (exact kernel dictionary)**.** *For $h_i \in \mathbb{F}[X]_{<k}$, let $\mathbf{h}$ be the concatenated coefficient vector. Then*

$$\mathbf{h} \in \ker \mathcal{M}^{(2)}(D, \mathcal{S}, \gamma)$$

if and only if, for every $x \in D$, there exist scalars $a_x, b_x \in \mathbb{F}$ such that

$$h_i(x) = a_x + \gamma_i b_x \qquad (i \in I_x).$$

In particular, every family of codeword explanations of one received line belongs to the kernel.

Proof. For fixed x, the local equations are

$$\sum_{i \in I_x} c_i h_i(x) = 0 \qquad (c \in \mathcal{K}_\gamma(I_x)).$$

The orthogonal complement of $\mathcal{K}_\gamma(I_x)$ in $\mathbb{F}^{I_x}$ is exactly

$$\operatorname{Span}\{\mathbf{1}_{I_x}, (\gamma_i)_{i \in I_x}\}.$$

Thus the local value vector is affine in the slope exactly when all local rows annihilate it. Applying this independently at each coordinate proves the claim. For a received line, take $a_x = r_0(x)$ and $b_x = r_1(x)$. □

Proposition 25.3 (row count and trivial kernel)**.** *The matrix has*

$$\sum_{x \in D} (r_x - 2)_+ \geq Lm - 2n$$

rows. It always contains the $2k$-dimensional kernel

$$h_i = A + \gamma_i B, \qquad A, B \in \mathbb{F}[X]_{<k}.$$

Consequently its maximum useful rank is $(L-2)k$, and the equation surplus over the nontrivial unknown dimension is

$$\boxed{L(m-k) - 2(n-k).}$$

Proof. Since $(r-2)_+ \geq r - 2$,

$$\sum_x (r_x - 2)_+ \geq \sum_x r_x - 2n = \sum_i |S_i| - 2n = Lm - 2n.$$

A globally affine tuple satisfies every local relation. Distinct slopes make the map $(A, B) \mapsto (A + \gamma_i B)_i$ injective, so this kernel has dimension $2k$. Subtracting $(L-2)k$ from the row lower bound gives the displayed surplus. □

26. Why the deployed critical order is exactly 32

The actual Reed–Solomon dimension on both active MCA rows is

$$n = 2^{21} = 2097152, \qquad k = 2^{20} = 1048576.$$

The adjacent safe agreements are

$$m_{\mathrm{KB}} = 1116048, \qquad m_{\mathrm{M31}} = 1116024.$$

Proposition 26.1 (critical order)**.** *For either deployed row,*

$$L_* := \left\lfloor \frac{2(n-k)}{m-k} \right\rfloor + 1 = 32.$$

More precisely,

row	$m-k$	$31(m-k) - 2(n-k)$	$32(m-k) - 2(n-k)$
KoalaBear MCA	67472	-5520	61952
Mersenne-31 MCA	67448	-6264	61184

Thus order 31 *can remain underdetermined for dimension reasons, whereas order* 32 *has a positive row surplus.*

Proof. Direct substitution into Proposition 25.3. □

27. What full relative rank would buy

Theorem 27.1 (one global code-line block). *Let Z be a set of bad slopes whose explaining codewords lie on one global codeword line,*

$$h_\gamma = A + \gamma B.$$

Assume the received line is support-wise MCA-bad at agreement m. Then

$$\boxed{|Z| \le n - m + 1.}$$

Proof. Let

$$C_0 := \{x \in D : r_0(x) = A(x),\ r_1(x) = B(x)\}, \qquad c := |C_0|.$$

MCA-badness gives $c < m$. Outside C_0, the equation

$$r_0(x) - A(x) + \gamma(r_1(x) - B(x)) = 0$$

has at most one finite solution γ. Counting agreements over all $\gamma \in Z$ therefore yields

$$|Z|m \le |Z|c + (n - c).$$

Hence

$$|Z| \le \frac{n-c}{m-c} \le n - m + 1,$$

where the last expression is the maximum over integers $0 \le c \le m - 1$. □

Corollary 27.2 (conditional generic closure). *Suppose every* 32*-subfamily of a set of bad explanations has full relative rank, meaning that the kernel of its rank-two matrix is exactly the global affine kernel. Then all explanations lie on one global codeword line and*

$$|Z| \le n - m + 1.$$

At the deployed rows the bounds are

$$981105 \qquad \textit{and} \qquad 981129,$$

respectively.

Proof. Any 32 explanations are globally affine. If there are more than 32, fix two slopes and place any third slope in a 32-subset containing the fixed pair; the unique codeword line through the fixed pair contains the third. Apply Theorem 27.1. □

The Mersenne-31 challenge numerator is 16777215, so the generic order-32 branch would fit with more than fifteen million units of margin. The difficult issue is therefore not the generic branch. It is classification and exclusive payment of specialized rank defects.

28. A necessary partition inequality

The following obstruction identifies the correct rank-two analogue of the partition conditions in higher-order MDS theory.

For a partition $\mathcal{P}$ of $[L]$, put

$$c(\mathcal{P}) := \sum_{P \in \mathcal{P}} \min\{2, |P|\},$$

and for $I \subseteq [L]$ define

$$w_{\mathcal{P}}(I) := \sum_{P \in \mathcal{P}} \min\{2, |I \cap P|\} - \min\{2, |I|\}.$$

Proposition 28.1 (partition obstruction). *If*

$$\sum_{x\in D} w_{\mathcal{P}}(I_x) < k\big(c(\mathcal{P})-2\big)$$

for some partition $\mathcal{P}$, then the rank-two matrix has kernel dimension strictly larger than $2k$. Consequently full relative rank requires

$$\boxed{\sum_{x\in D} w_{\mathcal{P}}(I_x) \ge k\big(c(\mathcal{P})-2\big) \quad \textit{for every partition } \mathcal{P}.}$$

Proof. Consider tuples that are independently affine on each block of $\mathcal{P}$: a singleton block contributes one free degree-$<k$ polynomial, and a block of size at least two contributes two. This space has dimension $kc(\mathcal{P})$. At coordinate x, forcing the blockwise-affine local values on I_x to lie on one affine slope line imposes at most $w_{\mathcal{P}}(I_x)$ scalar conditions. Hence the intersection with the rank-two kernel has dimension at least

$$kc(\mathcal{P}) - \sum_x w_{\mathcal{P}}(I_x).$$

Under the strict inequality this exceeds $2k$. □

For generic evaluation points, higher-order MDS and row-span constrained local-property machinery make closely related partition inequalities sufficient for the expected rank and for curve-decoding phenomena [BGM23, GGSW26]. The deployed domains are not generic. The point of the next sections is to identify what specializations must be removed before a sufficiency theorem can be true. Recent work explicitly notes that identifying deterministic Reed–Solomon evaluation sets with the random-evaluation curve-decoding behavior remains open [GGSW26].

29. Exact dyadic quotient descent

Theorem 29.1 (pair descent on a multiplicative dyadic domain). *Assume* $\operatorname{char}\mathbb{F}\neq 2$ *and*

$$D = D_+ \sqcup (-D_+).$$

Suppose the incidence pattern is pair-invariant:

$$I_x = I_{-x} \qquad (x\in D_+).$$

Write uniquely

$$h_i(X) = u_i(X^2) + Xv_i(X^2),$$

where

$$\deg u_i < \left\lceil \frac{k}{2}\right\rceil, \qquad \deg v_i < \left\lfloor \frac{k}{2}\right\rfloor.$$

Then the rank-two kernel on D is naturally the direct sum of two rank-two kernels on the squared domain

$$D^{(2)} := \{x^2 : x\in D_+\},$$

with the common incidence sets $I_x = I_{-x}$ and degree bounds $\lceil k/2\rceil$ and $\lfloor k/2\rfloor$. In particular, every nontrivial pair-invariant defect descends to a nontrivial defect on the half-size quotient in at least one parity component.

Proof. Fix $x\in D_+$ and $c\in \mathcal{K}_\gamma(I_x)=\mathcal{K}_\gamma(I_{-x})$. The two local equations are

$$\sum_i c_i\big(u_i(x^2)+xv_i(x^2)\big)=0, \qquad \sum_i c_i\big(u_i(x^2)-xv_i(x^2)\big)=0.$$

Because $2x\neq 0$, adding and subtracting gives

$$\sum_i c_iu_i(x^2)=0, \qquad \sum_i c_iv_i(x^2)=0.$$

The converse follows by reversing the calculation. The even/odd polynomial decomposition is a vector-space isomorphism, so the full kernel is the direct sum claimed. The global affine kernels descend componentwise; hence any excess kernel dimension appears in at least one component. □

Iterating Theorem 29.1 proves exact descent for every fully periodic dyadic incidence atom. This is the rigorous quotient branch of the proposed specialization theorem. What fails is the assertion that every non-descending atom must be a simple pole.

30. Rational atoms and profile depth

Let

$$\Lambda_i(X) := \prod_{x \in S_i} (X - x)$$

be the monic support locator of degree m.

Definition 30.1 (scalar-locator rational atom). *A family is a scalar-locator rational atom if there exist*

$$Q, A, B \in \mathbb{F}[X], \qquad c_i = c_0 + c_1\gamma_i,$$

with $Q(x) \neq 0$ for all $x \in D$, such that

$$\boxed{Qh_i + c_i\Lambda_i = A + \gamma_i B \qquad (i \in [L]).}$$

Proposition 30.2 (rational atoms are MCA explanations). *Every atom of Definition 30.1 is explained by the rational received line*

$$r_\gamma(x) = \frac{A(x) + \gamma B(x)}{Q(x)}.$$

The codeword h_i agrees with r_{γ_i} on all roots of Λ_i.

Proof. At $x \in S_i$, the locator term vanishes and division by $Q(x)$ is valid. □

Proposition 30.3 (converse under the scalar degree profile). *Put $m = k + d$. Suppose*

$$r_\gamma(x) = \frac{A(x) + \gamma B(x)}{Q(x)}$$

on D, where Q has no root in D, $\deg Q = s \leq d$, and $\deg A, \deg B \leq m$. If $h_i \in \mathbb{F}[X]_{<k}$ agrees with r_{γ_i} on the m roots of Λ_i, then

$$A + \gamma_i B - Qh_i = c_i\Lambda_i$$

for a scalar c_i, and c_i is affine in γ_i.

Proof. The left side has degree at most m and vanishes at the m distinct roots of Λ_i, so it is a scalar multiple of Λ_i. Since $\deg(Qh_i) \leq k + s - 1 \leq m - 1$, its degree-$m$ coefficient is zero. Comparing degree-m coefficients gives

$$c_i = [X^m]A + \gamma_i[X^m]B.$$

□

Corollary 30.4 (scaled-locator prefix depth). *Under the hypotheses of Proposition 30.3, the coefficients of*

$$c_i\Lambda_i$$

in degrees

$$k + s, k + s + 1, \ldots, m$$

are affine functions of γ_i. Apart from the leading scaling coefficient, this is a depth-$(d - s)$ affine prefix constraint.

Proof. The product Qh_i has degree at most $k + s - 1$, so the displayed high coefficients of $c_i\Lambda_i$ equal those of $A + \gamma_i B$. □

Thus a simple pole, $s = 1$, is only one endpoint of a profile continuum. At the opposite endpoint $s = d$, only the leading scaling is forced, while the locator residues modulo Q must lie on a projective affine line. Any correct rank-defect classification must account for the entire continuum or prove that the already-declared first-match cells remove all but selected endpoints.

31. An exact degree-two rank defect over $\mathbb{F}_{17}$

We now give a completely explicit obstruction to an unqualified "quotient or simple pole" theorem.

Theorem 31.1 (degree-two support-wise MCA atom). *Let*

$$\mathbb{F} = \mathbb{F}_{17}, \qquad D = \mu_8 = \{1, 9, 13, 15, 16, 8, 4, 2\}, \qquad k = 4, \quad m = 6.$$

Take slopes

$$(\gamma_0, \gamma_1, \gamma_2, \gamma_3, \gamma_4) = (0, 1, 4, 7, 2)$$

and supports

i	S_i
0	$\{1, 15, 16, 8, 4, 2\}$
1	$\{1, 9, 13, 15, 16, 8\}$
2	$\{1, 13, 15, 16, 4, 2\}$
3	$\{1, 9, 15, 16, 4, 2\}$
4	$\{1, 9, 13, 15, 8, 4\}$.

Let

$$h_0 = h_1 = 0,$$
$$h_2 = 9 + 13X + 8X^2 + 4X^3,$$
$$h_3 = 3 + 10X + 14X^2 + 7X^3,$$
$$h_4 = 11 + 8X + 9X^2 + 6X^3.$$

Define a received line by the following values, in the displayed order of D:

x	1	9	13	15	16	8	4	2
$r_0(x)$	0	16	6	0	0	0	0	0
$r_1(x)$	0	1	11	0	0	0	5	12.

Then:

(i) *h_i agrees with $r_0 + \gamma_i r_1$ on S_i for every i;*
(ii) *the received line is support-wise MCA-bad at agreement $m = 6$;*
(iii) *the rank-two matrix has rank* 11, *whereas full relative rank is* 12*;*
(iv) *the defect is not first-level quotient-periodic;*
(v) *it has no scalar-locator rational representation with* $\deg Q \le 1$, *but it has one with* $\deg Q = 2$.

Proof. Direct substitution proves (i). For (ii), observe that r_0 is zero at exactly the six points S_0. On any other six-point set, a degree-<4 polynomial agreeing with r_0 would have at least four zero roots and also take a prescribed nonzero value, which is impossible; with five zero roots the same conclusion is immediate. Hence the only possible six-point common explanation set for the first coefficient is S_0. On S_0, the word r_1 is zero at $1, 15, 16, 8$ and nonzero at $4, 2$, so no degree-<4 polynomial can agree with it there. Thus no pair of codeword polynomials simultaneously explains the received line on six coordinates.

Part (i) and Theorem 25.2 give a nontrivial kernel vector. It is not globally affine because $h_0 = h_1 = 0$ at slopes 0 and 1, while $h_2 \neq 0$. Exact row reduction gives rank 11; one 11×11 pivot minor has determinant $9 \in \mathbb{F}_{17}^{\times}$. This proves (iii).

For (iv), the incidence sets at 1 and $-1 = 16$ are different: the former contains all five slopes, while the latter omits slope γ_4. Thus the hypothesis of Theorem 29.1 fails already at the first dyadic level.

For (v), let

$$Q = 12 + 3X + 5X^2,$$
$$A = 13 + 10X + 7X^2 + 13X^3 + 5X^4 + 11X^5 + 9X^6,$$
$$B = 7 + 5X + 6X^2 + 14X^3 + 3X^4 + 15X^5 + X^6,$$

and $c_i = 9+\gamma_i$. The polynomial Q is nonzero on every point of D, and exact multiplication gives

$$Qh_i + c_i\Lambda_i = A + \gamma_i B \qquad (i = 0, \ldots, 4).$$

For a denominator bound s, the identities of Definition 30.1 form a homogeneous coefficient system in Q, A, B, c_0, c_1. At $s = 0$ and $s = 1$ the matrices have full column ranks 17 and 18, respectively. At $s = 2$ the 35×19 system has rank 18 and its one-dimensional kernel is generated by the displayed certificate. Hence degree two is minimal within this atom model. $\square$

Remark 31.2 (the common-support qualifier is essential). The example has the common support factor

$$H = (X-1)(X-15) = X^2 + X + 15.$$

It divides every locator and also $A, B, h_0, \ldots, h_4$. Therefore Theorem 31.1 refutes an *unqualified* quotient-or-pole theorem, but it does not refute a theorem stated only after the common-support first-match branch has been removed. Instead, it proves that the qualifier cannot be omitted and that rational denominator profiles can be hidden inside common-support reductions.

The incidence pattern also satisfies every partition inequality of Proposition 28.1, with minimum slack 1. Thus the defect is a genuine specialization phenomenon rather than a failure of the obvious partition count. This finite claim is included in the verifier.

32. Maximal supports and the slope-degree barrier

For each explanation, take its *maximal* agreement support

$$\widehat{S}_i := \{x \in D : h_i(x) = r_0(x) + \gamma_i r_1(x)\}, \qquad |\widehat{S}_i| \geq m.$$

Put

$$\widehat{I}_x := \{i : x \in \widehat{S}_i\}, \qquad C_\cap := \bigcap_{i=1}^{L} \widehat{S}_i, \qquad c := |C_\cap|.$$

Let $R_0, R_1 \in \mathbb{F}[X]_{<n}$ be the interpolation polynomials of r_0, r_1 on D. Interpolating the codeword polynomials coefficientwise in the slope gives the unique polynomial

$$H(X,Z) \in \mathbb{F}[X,Z], \qquad \deg_X H < k, \qquad \deg_Z H < L,$$

with

$$H(X,\gamma_i) = h_i(X) \qquad (i \in [L]).$$

Define the slope-error polynomial

$$E(X,Z) := H(X,Z) - R_0(X) - ZR_1(X).$$

Lemma 32.1 (zero slope fibers are exactly common agreements)**.** *For $x \in D$,*

$$E(x,Z) \equiv 0 \quad \Longleftrightarrow \quad x \in C_\cap.$$

Proof. If $x \in C_\cap$, then $E(x,\gamma_i) = 0$ for all L distinct slopes. Since $\deg_Z E(x,Z) < L$, the polynomial is zero. Conversely, if $E(x,Z)$ is zero, then evaluating at every γ_i gives

$$h_i(x) = R_0(x) + \gamma_i R_1(x) = r_0(x) + \gamma_i r_1(x),$$

so x belongs to every maximal support. $\square$

Theorem 32.2 (maximal-support slope-degree barrier)**.** *Assume the explanations are not globally affine: there do not exist $A, B \in \mathbb{F}[X]_{<k}$ with*

$$h_i = A + \gamma_i B \qquad (i \in [L]).$$

Then $E \neq 0$. Writing

$$r := \deg_Z E,$$

one has $c < k$ and

$$\boxed{Lm \leq Lc + r(n-c).}$$

Equivalently,

$$\boxed{r \geq \left\lceil \frac{L(m-c)}{n-c} \right\rceil.}$$

In particular, after the common-support branch is removed $(c = 0)$*,*

$$r \geq \left\lceil \frac{Lm}{n} \right\rceil.$$

Proof. If $c \geq k$, choose two slopes and let $A, B \in \mathbb{F}[X]_{<k}$ be the unique codeword line through their polynomials. For every i, the polynomial $h_i - A - \gamma_i B$ has degree $< k$ and vanishes on $C_\cap$, hence is zero. This contradicts non-affinity, so $c < k$.

For $x \notin C_\cap$, Lemma 32.1 says $E(x, Z)$ is nonzero. It has the $|\widehat{I}_x|$ distinct roots γ_i indexed by agreements at x, whence

$$|\widehat{I}_x| \leq r.$$

Counting maximal-support incidences gives

$$Lm \leq \sum_i |\widehat{S}_i| = \sum_{x \in C_\cap} L + \sum_{x \notin C_\cap} |\widehat{I}_x| \leq Lc + r(n-c).$$

It remains only to justify that $E \neq 0$. If it were zero, then H would be affine in Z and all $h_i = H(X, \gamma_i)$ would lie on one global codeword line. □

Corollary 32.3 (the active primitive degree is at least eighteen)**.** *For* $L = 32$ *and either deployed adjacent row, a non-affine explanation with no common agreement coordinate satisfies*

$$\boxed{18 \leq r \leq 31.}$$

Indeed,

$$\left\lceil \frac{32 m_{\mathrm{KB}}}{n} \right\rceil = \left\lceil \frac{32 m_{\mathrm{M31}}}{n} \right\rceil = 18.$$

Thus every slope-degree-≤ 17 explanation is already global-affine or common-support. This removes the entire low-degree branch before any determinant specialization argument begins.

33. Local incidence complexity

For the rational extraction theorem, choose exact m-subsupports

$$S_i \subseteq \widehat{S}_i, \qquad |S_i| = m,$$

and set

$$I_x := \{i : x \in S_i\}, \qquad r_x := |I_x|.$$

Let N_a count coordinates of incidence multiplicity a, let

$$N_{\geq 2} := |\{x : r_x \geq 2\}|, \qquad U := \bigcup_i S_i, \qquad u := |U|.$$

Definition 33.1 (two-cover incidence complexity)**.** *Define*

$$\chi(\mathcal{S}) := \sum_{x \in D} \min\{2, r_x\} = N_1 + 2N_{\geq 2}.$$

Equivalently,

$$\boxed{\chi(\mathcal{S}) = 2u - N_1 = 2n - 2N_0 - N_1.}$$

Proposition 33.2 (exact relation to the rank-two row count)**.** *For exact* m*-supports,*

$$\sum_{x \in D} (r_x - 2)_+ = Lm - \chi(\mathcal{S}).$$

Proof. At multiplicity zero or one, both sides contribute zero after subtracting $\min\{2, r_x\}$. At multiplicity at least two, the difference is $r_x - 2$. Sum over x and use $\sum_x r_x = Lm$. □

The quantity χ is the number of scalar conditions needed to force one common rational received line on the occupied coordinates: two conditions at a multiply occupied coordinate and one at a singleton.

34. Support-collapsed rational interpolation

Let

$$\Lambda_i(X) := \prod_{x\in S_i}(X-x)$$

be the monic locator of S_i.

Definition 34.1 (scalar-locator certificate)**.** *A scalar-locator certificate of denominator degree at most s is a tuple*

$$(Q, A, B, c_0, c_1)$$

with

$$\deg Q \le s, \qquad \deg A, \deg B \le m,$$

not all zero, such that, with $c_i = c_0 + c_1\gamma_i$,

$$\boxed{Qh_i + c_i\Lambda_i = A + \gamma_i B \qquad (i \in [L]).}$$

It is a rational atom when $Q \neq 0$ and $(c_0, c_1) \neq (0, 0)$, a pure locator pencil when $Q = 0$, and globally affine when $(c_0, c_1) = (0, 0)$.

When Q has no root on D, a rational atom is explained by the received line $(A + \gamma B)/Q$. A root of Q on D is retained as a separate denominator-root first-match branch rather than being silently divided away.

Theorem 34.2 (low-complexity atom extraction)**.** *Assume $L \ge 2$. Let*

$$d = m - k$$

and fix an integer $0 \le s \le d$. If

$$\boxed{\chi(\mathcal{S}) < 2m + s + 3,}$$

then the explanations admit a scalar-locator certificate of denominator degree at most s.

More precisely, one of the following holds:

(i) *the codeword explanations are globally affine;*
(ii) *they lie in a pure locator pencil;*
(iii) *they form a rational atom, possibly with a denominator-root degeneracy on D.*

Proof. Use as unknowns the coefficients of

$$Q \in \mathbb{F}[X]_{\le s}, \qquad A, B \in \mathbb{F}[X]_{\le m}.$$

There are

$$(s + 1) + 2(m + 1) = 2m + s + 3$$

unknowns.

At a coordinate x with $r_x \ge 2$, impose the two homogeneous linear equations

$$A(x) = Q(x)r_0(x), \qquad B(x) = Q(x)r_1(x).$$

At a singleton coordinate, with $I_x = \{i\}$, impose

$$A(x) + \gamma_i B(x) = Q(x)h_i(x).$$

No equation is needed at an unoccupied coordinate. The total number of equations is at most $\chi(\mathcal{S})$. Under the strict inequality, the homogeneous system has a nonzero solution (Q, A, B).

For every i and every $x \in S_i$, the imposed equations give

$$A(x) + \gamma_i B(x) = Q(x)h_i(x).$$

The polynomial

$$P_i := A + \gamma_i B - Qh_i$$

has degree at most m because

$$\deg(Qh_i) \le s + k - 1 \le m - 1.$$

It vanishes on the m distinct roots of Λ_i, so

$$P_i = c_i\Lambda_i$$

for a scalar c_i. Comparing coefficients of X^m gives

$$c_i = [X^m]A + \gamma_i[X^m]B,$$

which is affine in γ_i. This is the required certificate.

If $Q = 0$, nontriviality and two distinct slopes rule out $c_0 = c_1 = 0$, so the certificate is a pure locator pencil. If $Q \neq 0$ but $c_0 = c_1 = 0$, then

$$Qh_i = A + \gamma_i B.$$

Using two slopes shows Q divides both A and B, and cancellation gives a global affine codeword line. The remaining case is a rational atom. □

Remark 34.3 (relation to rational-function interpolation). The proof is a support-collapsed Berlekamp–Welch argument: it interpolates only the two received-line coefficients on multiply occupied coordinates, rather than imposing a separate condition for every support incidence. This is compatible with the rational-function and curve-decoding viewpoint in recent proximity-gap work [BCHKS25, GG25], but the theorem above is elementary and finite; no random-evaluation or asymptotic result is imported.

Corollary 34.4 (full denominator profile). *Taking $s = d = m - k$, every explanation with*

$$\boxed{\chi(\mathcal{S}) < 3m - k + 3}$$

is globally affine, a pure locator pencil, or a scalar-locator rational atom of denominator degree at most $m - k$.

Corollary 34.5 (support-union form). *If*

$$2u - N_1 < 3m - k + 3,$$

then the same alternatives hold. In particular, it suffices that

$$u \leq \left\lfloor \frac{3m - k + 2}{2} \right\rfloor.$$

Corollary 34.6 (exact deployed extraction thresholds). *For the active* 32*-tuples, put*

$$\Xi := 3m - k + 3.$$

Then

row	Ξ	$2n - \Xi$	*automatic union ceiling*	*union nullity ceiling*
KoalaBear MCA	2299571	1894733	1149785	101209
Mersenne-31 MCA	2299499	1894805	1149749	101173

where the last column subtracts $n - k = 1048576$ from the union ceiling.

Equivalently, extraction is automatic if

$$2N_0 + N_1 \geq 1894734$$

on KoalaBear, or if

$$2N_0 + N_1 \geq 1894806$$

on Mersenne-31.

These are classification thresholds, not challenge-numerator payments. Their use is to show that a large low-complexity support family belongs to one atom instead of being charged as many unrelated cells.

35. The degree-two counterexample lies at the extraction threshold

The degree-two $\mathbb{F}_{17}$ example of Theorem 31.1 has

$$n = 8, \qquad k = 4, \qquad m = 6, \qquad L = 5.$$

Its five listed supports are maximal. Their incidence multiplicities are

$$(5, 3, 3, 5, 4, 3, 4, 3),$$

so

$$\chi = 16 \quad \text{while} \quad 3m - k + 3 = 17.$$

Thus Theorem 34.2 forces a denominator of degree at most

$$d = m - k = 2,$$

which is exactly the minimal degree established by Theorem 31.1.

The common support has size $c = 2$. The coefficientwise slope interpolation of the five codewords has degree $r = 4$, and

$$\left\lceil \frac{5(6-2)}{8-2} \right\rceil = 4.$$

Hence the example also attains equality in Theorem 32.2. The counterexample is therefore not an unexplained accident: it sits simultaneously on the common-support slope-degree boundary and the low-complexity rational-extraction boundary.

36. Collision pencils for two atom certificates

Suppose the same indexed explanations

$$(\gamma_i, h_i, \Lambda_i), \qquad i \in I,$$

satisfy two nontrivial scalar-locator certificates

$$Qh_i + (c_0 + c_1\gamma_i)\Lambda_i = A + \gamma_i B,$$

$$Q'h_i + (c_0' + c_1'\gamma_i)\Lambda_i = A' + \gamma_i B'.$$

Assume $|I| \geq 2$, $Q, Q' \neq 0$, and both scalar pairs are nonzero. Define

$$P_0 := Q'c_0 - Qc_0', \quad P_1 := Q'c_1 - Qc_1',$$
$$R_0 := Q'A - QA', \quad R_1 := Q'B - QB'.$$

Eliminating h_i gives the collision identity

$$\boxed{(P_0 + \gamma_i P_1)\Lambda_i = R_0 + \gamma_i R_1.}$$

Let

$$G := \{x \in D : R_0(x) = R_1(x) = 0\}, \qquad g := |G|,$$

and

$$H := \{x \in G : P_0(x) = P_1(x) = 0\}, \qquad h := |H|.$$

Theorem 36.1 (atom collision rigidity). *If $R_0 = R_1 = 0$ and the certificates share at least two distinct slopes, then the certificates are projectively identical.*

Otherwise, if $g < m$, then

$$\boxed{|I| \leq \left\lfloor \frac{n-g}{m-g} \right\rfloor}.$$

Moreover, every coordinate in $G \setminus H$ belongs to at least $|I| - 1$ of the supports S_i.

If both denominators have degree at most d, then two distinct certificates satisfy

$$h \leq d.$$

Proof. First suppose $R_0 = R_1 = 0$. Then

$$(P_0 + \gamma_i P_1)\Lambda_i = 0$$

as a polynomial. Since $\Lambda_i \neq 0$, two distinct slopes imply $P_0 = P_1 = 0$. Thus

$$Q'(c_0, c_1) = Q(c_0', c_1').$$

Choose a nonzero component of (c_0, c_1). Because $Q, Q' \neq 0$, this identity gives a scalar $\lambda \in \mathbb{F}^\times$ with

$$Q' = \lambda Q, \qquad (c_0', c_1') = \lambda(c_0, c_1).$$

The equations $R_0 = R_1 = 0$ then give

$$A' = \lambda A, \qquad B' = \lambda B.$$

So the certificates are projectively identical.

Now assume $(R_0, R_1) \neq (0, 0)$. At every root of Λ_i,

$$R_0(x) + \gamma_i R_1(x) = 0.$$

After removing the at most g roots in G, each S_i contributes at least $m - g$ incidences. At a coordinate outside G, the affine equation

$$R_0(x) + \gamma R_1(x) = 0$$

has at most one slope solution. Hence

$$|I|(m - g) \leq n - g,$$

which proves the bound.

For $x \in G \setminus H$, the polynomial

$$P_0(x) + \gamma P_1(x)$$

is a nonzero affine function of γ and therefore vanishes at at most one of the shared slopes. The collision identity then forces $\Lambda_i(x) = 0$ for all other i, proving the near-sunflower assertion.

Finally, if the certificates are distinct, then $(P_0, P_1) \neq (0, 0)$ by the first part. Both have degree at most d, so their common root set in D has size at most d. □

Corollary 36.2 (primitive atom uniqueness)**.** *If $G = \varnothing$, two distinct atom certificates share at most*

$$\left\lfloor \frac{n}{m} \right\rfloor = 1$$

explanation at either deployed row. Thus two primitive certificates sharing two slopes are projectively identical.

37. Affine lines versus rational atoms

A similar argument prevents a primitive rational atom from masquerading as a global affine codeword line.

Theorem 37.1 (affine–atom collision)**.** *Suppose $h_i = A_0 + \gamma_i B_0$ for $i \in I$, and the same explanations satisfy a nontrivial atom certificate*

$$Qh_i + (c_0 + c_1\gamma_i)\Lambda_i = A + \gamma_i B, \qquad (c_0, c_1) \neq (0, 0).$$

Put

$$E_0 := A - QA_0, \qquad E_1 := B - QB_0,$$

and let

$$G_{\mathrm{aff}} := \{x \in D : E_0(x) = E_1(x) = 0\}, \qquad g_{\mathrm{aff}} := |G_{\mathrm{aff}}|.$$

If $|I| \geq 2$, then $(E_0, E_1) \neq (0, 0)$, and if $g_{\mathrm{aff}} < m$,

$$\boxed{|I| \leq \left\lfloor \frac{n - g_{\mathrm{aff}}}{m - g_{\mathrm{aff}}} \right\rfloor.}$$

Every point of G_{aff} belongs to at least $|I| - 1$ supports. In particular, a primitive atom and a global affine line share at most one explanation.

Proof. Substitution gives

$$(c_0 + c_1\gamma_i)\Lambda_i = E_0 + \gamma_i E_1.$$

If $E_0 = E_1 = 0$, then $c_0 + c_1\gamma_i = 0$ for every i, since $\Lambda_i \neq 0$. A nonzero affine function of the slope has at most one root, contradicting $|I| \geq 2$.

The root-incidence count for the pencil $E_0 + \gamma E_1$ is identical to the proof of Theorem 36.1. At a common fixed root, all supports except possibly the unique slope with $c_0 + c_1\gamma = 0$ contain the coordinate. □

38. Coherence or a quantified near-sunflower

The graph whose vertices are the 32-subsets of a fixed slope set and whose edges join subsets with 31 common slopes is connected. Collision rigidity can therefore propagate a local classification globally.

Theorem 38.1 (order-32 coherence)**.** *Let Z be a set of at least 32 bad slopes. Suppose every 32-subset of the corresponding explanations is classified as either*

(a) *one global affine codeword line, or*
(b) *one nontrivial scalar-locator rational atom of denominator degree at most $d = m - k$.*

Assume all cross-fixed sets arising between distinct local classifications have already been removed by the first-match rule. Then the entire set Z lies on one global affine codeword line or in one projective rational atom certificate.

Proof. Adjacent 32-subsets share 31 explanations. Two affine codeword lines sharing two distinct slopes are identical. By Corollary 36.2, two primitive atom certificates sharing two explanations are identical. By Theorem 37.1, a primitive atom and an affine line cannot share two explanations. Therefore adjacent vertices of the 32-subset graph receive the same type and the same structure. Connectedness propagates that structure to every 32-subset and hence to every explanation in Z. □

Without the primitive hypothesis, failure of coherence is not unstructured.

Corollary 38.2 (exact near-sunflower alternative on a 31-overlap)**.** *Suppose two distinct local classifications share $q = 31$ explanations. Then at least one classification is an atom, and their cross-fixed set has size at least*

$$g_{31} := \left\lceil \frac{31m - n}{30} \right\rceil.$$

For two rational atoms, at least $g_{31} - d$ of those coordinates belong to at least 30 of the 31 supports. For an affine-line/atom collision, all g_{31} fixed coordinates have that property.

At the deployed rows,

row	g_{31}	$d = m - k$	$g_{31} - d$
KoalaBear MCA	1083345	67472	1015873
Mersenne-31 MCA	1083320	67448	1015872

Thus every failure of local atom coherence produces a 30-of-31 near-sunflower on more than one million domain coordinates.

Proof. Rearrange the collision inequality

$$31(m - g) \leq n - g$$

to obtain

$$g \geq \left\lceil \frac{31m - n}{30} \right\rceil.$$

For two atoms, Theorem 36.1 gives a collision core H of size at most d; every point outside it lies in at least thirty supports. The affine–atom statement follows from Theorem 37.1. □

For reference, if two classifications share all 32 explanations, the corresponding fixed-root thresholds are

row	$g_{32} = \lceil (32m - n)/31 \rceil$	$g_{32} - d$
KoalaBear MCA	1084400	1016928
Mersenne-31 MCA	1084375	1016927

39. The proved part of the relative order-32 theorem

The results can now be stated as one relative classification theorem for the actual explanation vector, which is weaker than classifying every vector in the rank-two kernel but is sufficient for MCA.

Theorem 39.1 (partial relative order-32 explanation theorem)**.** *Fix either deployed adjacent MCA row and* 32 *distinct bad slopes. After maximalizing supports, removing their common-support branch, and selecting exact m-subsupports, any non-global-affine explanation outside the pure-locator, denominator-root, and scalar-locator rational-profile cells must satisfy both*

$$\boxed{18 \le \deg_Z E \le 31}$$

and

$$\boxed{\chi(\mathcal{S}) \ge 3m - k + 3.}$$

Numerically, the second inequality is

$$\chi(\mathcal{S}) \ge 2299571 \quad \textit{on KoalaBear},$$

and

$$\chi(\mathcal{S}) \ge 2299499 \quad \textit{on Mersenne-31}.$$

Moreover, if every residual 32*-subset is classified as affine or rational, then either all bad slopes belong to one coherent structure or a* 31*-overlap enters the quantified near-sunflower branch of Corollary 38.2.*

Proof. The slope-degree statement is Corollary 32.3. The incidence-complexity statement is the contrapositive of Corollary 34.4. The coherence alternative follows from Theorem 38.1 and Corollary 38.2. □

This proves the relative theorem in two substantial regimes:

- low slope degree ($r \le 17$) is impossible outside global-affine/common-support structure;
- low two-cover complexity ($\chi < 3m - k + 3$) automatically produces a rational or pure-locator atom.

It also discharges the separate atom-coherence obligation once the near-sunflower first-match cell is applied.

Part 6. Primitive spread abundance and the exact MCA endgame

The rank and atom theorems reduce a large bad-slope set to one coherent global structure, except for a quantified exceptional set. This part proves that universal spread exclusion is false, identifies the correct spread-abundance target, localizes rational atoms to a canonical owner, and bounds broad classes of spread-core corrections.

40. A critical spread core exists

A universal theorem excluding primitive spread critical tuples on dyadic domains would be stronger than the deployed abundance statement. The following exact example shows that such a universal prohibition is false.

Proposition 40.1 (exact dyadic critical spread core)**.** *Let*

$$\mathbb{F} = \mathbb{F}_{17}, \qquad D = \mathbb{F}_{17}^{\times} = \langle 3 \rangle, \qquad n = 16, \quad k = 8, \quad m = 10.$$

The critical order is

$$L_* = \left\lfloor \frac{2(n-k)}{m-k} \right\rfloor + 1 = 9.$$

*There exist nine distinct slopes, nine codeword explanations, and exact size-*10 *witness supports with all of the following properties.*

(i) *The received pair is support-wise MCA-bad at agreement* 10*.*
(ii) *The maximal agreement supports have empty common intersection.*
(iii) *The rank-two reduced intersection matrix has rank* 55*, whereas full relative rank is* $(9-2)8 = 56$*.*
(iv) *The two-cover complexity is* 32*, above the extraction threshold of Corollary 34.4* $3m - k + 3 = 25$*.*
(v) *The incidence pattern is not periodic at any nontrivial level of the dyadic tower.*
(vi) *There is no scalar-locator certificate with denominator degree* 0*,* 1*, or* $2 = m - k$*.*
(vii) *The coefficientwise slope interpolation has degree* 8*, above the common-support-free lower barrier* $\lceil 9m/n \rceil = 6$*.*
(viii) *The direction code contains a word of weight* $9 = n - k + 1$*, so the minimum-support excess is exactly one.*

Moreover, for this same received line every slope in $\mathbb{F}_{17}$ *has a degree-*<8 *codeword agreeing on at least ten coordinates.*

Exact certificate. Order the domain as

$$(1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6).$$

Use slopes $\gamma_i = i$ for $0 \le i \le 8$ and the following exact size-10 supports, written by domain indices:

i	S_i
0	$0, 3, 4, 5, 7, 9, 10, 11, 12, 13$
1	$1, 3, 5, 6, 7, 10, 11, 12, 13, 14$
2	$1, 4, 6, 7, 9, 10, 11, 12, 13, 15$
3	$0, 1, 2, 4, 5, 6, 7, 9, 14, 15$
4	$0, 2, 3, 4, 5, 6, 7, 9, 12, 13$
5	$0, 3, 4, 5, 6, 7, 8, 9, 10, 14$
6	$1, 2, 3, 5, 6, 8, 9, 11, 13, 15$
7	$0, 3, 4, 5, 6, 7, 11, 12, 14, 15$
8	$2, 3, 4, 5, 6, 7, 8, 9, 10, 14$

The codeword coefficient vectors, constant coefficient first, are

i	1	X	X^2	X^3	X^4	X^5	X^6	X^7
0	1	13	16	16	9	16	7	12
1	13	9	14	12	6	4	2	13
2	2	2	2	4	2	12	0	1
3	13	3	3	4	0	13	16	5
4	1	4	11	4	2	16	9	16
5	16	13	15	13	2	15	5	7
6	3	3	0	13	0	8	12	4
7	3	2	6	8	10	13	10	12
8	1	12	2	14	9	10	9	4

and the received coefficient words on the ordered domain are

$$r_0 = (5, 16, 0, 0, 0, 0, 0, 0, 5, 7, 8, 0, 8, 10, 0, 6),$$

$$r_1 = (6, 14, 15, 0, 0, 0, 2, 13, 14, 12, 6, 3, 15, 13, 6, 15).$$

Direct substitution verifies every claimed support. Maximalizing the nine supports gives sizes 10 or 11 and empty total intersection.

For the rank statement, form the rank-two matrix of Definition 25.1. It has 58 rows and 72 columns. The displayed explanation vector supplies one kernel direction independent of the 16-dimensional global-affine kernel, so its rank is at most 55; exact Gaussian elimination over $\mathbb{F}_{17}$ gives rank 55. The scalar-locator systems at denominator degrees $0, 1, 2$ have column counts and ranks

$$(25, 25), \qquad (26, 26), \qquad (27, 27),$$

respectively. Exact interpolation gives slope degree 8, and direct evaluation of all pairwise codeword differences gives minimum weight 9.

Finally, exhaustive interpolation over the $\binom{16}{10}$ candidate common supports finds no ten-coordinate support on which both r_0 and r_1 are degree-< 8 evaluations. Enumerating the $\binom{16}{8}$ interpolation bases at each of the seventeen slopes finds at least one agreement-10 codeword for every slope. The accompanying verifier repeats all calculations in exact arithmetic. □

Remark 40.2 (what the example does and does not refute). The example refutes the *universal* prohibition of primitive spread critical tuples. It does not refute a row-sharp theorem for the two deployed length-2^{21} domains. It also shows why the minimum-distance refinement is relevant but not by itself decisive: the hard spread core lies exactly in the support-excess-1 branch, the same branch in which the boundary top seam survives by Corollary 7.9.

41. The deployed spread-abundance target

Definition 41.1 (first-match primitive spread core)**.** *A set of* 32 *explanations is a* first-match primitive spread core *if it satisfies the residual conditions of Theorem 39.1: no global affine codeword line, common-support factor, exact dyadic quotient descent, scalar-locator certificate of denominator degree at most $m-k$, denominator-root, extension, tangent, field-drop, or quantified near-sunflower atom; slope degree lies between* 18 *and* 31*; and two-cover complexity is at least* $3m - k + 3$.

A universal nonexistence theorem is false by Proposition 40.1. The finite MCA theorem requires the weaker and more natural abundance statement.

Problem 41.2 (deployed spread-abundance theorem)**.** Intermediate status. *This is the line-level abundance consequence required from the final spread-component routing problem Problem 53.1; it is not an additional terminal input to the completion certificate.*

For either deployed domain, prove that if a support-wise MCA-bad received line contains one first-match primitive spread core, then its total number of bad challenge slopes is at most the row budget B^*.

A stronger useful form would bound the number of slopes extending a fixed core by a quantity below $B^* - 32$, with every failure routed to one canonical quotient, extension, rational-owner, clone, or near-sunflower certificate.

Remark 41.3. The distinction is essential. The finite theorem does not need local higher-order-MDS rank at every critical tuple. It needs a global statement that a line carrying one genuine spread defect cannot carry too many further slopes. The exact example in Proposition 40.1 shows that any proof must use the deployed domain and field geometry, not only the abstract critical-order dimension count.

42. Rational atoms have a canonical owner

Consider one coherent rational atom certificate

$$Qh_i + (c_0 + c_1\gamma_i)\Lambda_i = A + \gamma_i B, \qquad i \in I, \tag{42.1}$$

where Λ_i is the locator of a witness support S_i, $|S_i| \geq m$, and Q has no root on D. Put

$$\widetilde{r}_0(x) = \frac{A(x)}{Q(x)}, \qquad \widetilde{r}_1(x) = \frac{B(x)}{Q(x)}.$$

The line

$$\widetilde{r}_\gamma = \widetilde{r}_0 + \gamma \widetilde{r}_1$$

is the *owner line* of the atom.

Definition 42.1 (fixed owner set). *Define*

$$G = \{x \in D : r_0(x) = \widetilde{r}_0(x),\ r_1(x) = \widetilde{r}_1(x)\}, \qquad g = |G|.$$

Theorem 42.2 (exclusive rational-owner localization). *Retain* (42.1) *and assume every scalar $c_0 + c_1\gamma_i$ is nonzero; zero-scalar slopes are a separate global-affine degeneracy and contribute at most one slope. Then:*

(a) *for every $x \in D \setminus G$, at most one support S_i contains x;*
(b) *if $g < m$, then*

$$\boxed{|I| \le \left\lfloor \frac{n-g}{m-g} \right\rfloor \le n - m + 1;}$$

(c) *if $g \ge m$, at most $n - g$ supports meet $D \setminus G$, while the supports contained in G form a support-wise MCA-bad instance for the punctured code*

$$C|_G \cong \mathrm{RS}[\mathbb{F}, G, k].$$

Consequently

$$\boxed{|I| \le n - g + B_{\mathrm{MCA}}(\mathrm{RS}[\mathbb{F}, G, k], m).}$$

Proof. On S_i, the locator vanishes, and (42.1) gives

$$h_i(x) = \widetilde{r}_0(x) + \gamma_i \widetilde{r}_1(x).$$

Since S_i is also an agreement support for the original received line,

$$(r_0 - \widetilde{r}_0)(x) + \gamma_i(r_1 - \widetilde{r}_1)(x) = 0.$$

For $x \notin G$ this is a nonzero affine equation in γ_i, hence has at most one solution. This proves (a).

If $g < m$, every support uses at least $m - g$ coordinates outside G. By (a), those outside incidences are disjoint across slopes, so

$$|I|(m - g) \le n - g.$$

The quotient $(n - g)/(m - g)$ is increasing for $g < m$, and at $g = m - 1$ equals $n - m + 1$. This proves (b).

Now assume $g \ge m$. Distinct supports meeting $D \setminus G$ use disjoint nonempty outside-coordinate sets, so there are at most $n - g$ of them. Every remaining support lies in G and witnesses the punctured owner line there. Since $g \ge m > k$, evaluation on G is injective and the punctured code is Reed–Solomon of dimension k. If the punctured pair admitted a common size-m explaining support, the same support would explain the original pair on D, contradicting support-wise MCA-badness. The contained slopes are therefore bounded by the punctured MCA numerator. □

Corollary 42.3 (all small-owner atoms are paid). *For either deployed row, every coherent rational atom with*

$$g \le 2m - k$$

has at most n bad slopes. Explicitly, the paid owner ranges are

row	$2m - k$	*atom bound*
KoalaBear MCA	1183520	2097152
Mersenne-31 MCA	1183472	2097152

Both fit their challenge budgets; on Mersenne-31 the exact residual margin is

$$16777215 - 2097152 = 14680063.$$

Proof. The case $g < m$ is Theorem 42.2(b). For $m \le g \le 2m-k$, the punctured error count is $g-m$ and

$$2(g-m) \le g-k.$$

CAP25 v13.2 proves, by its half-distance pincer and unique-decoding correlated-agreement input, that the punctured MCA numerator is at most g in this range [Cho26]. Apply Theorem 42.2(c):

$$|I| \le (n-g) + g = n.$$

□

43. A self-contained minimum-distance owner payment

The corollary uses the half-distance proximity theorem of CAP25 v13.2. The next statement is elementary and retains the actual minimum support of the punctured direction code.

Theorem 43.1 (triple-collapse from direction minimum support)**.** *Assume $g \ge m$ and restrict to the atom slopes whose supports lie in G. Let*

$$C'_G = \operatorname{Span}\{h_i - h_j : i, j \in I\} \subseteq \mathrm{RS}[\mathbb{F}, G, k]$$

be their direction code. If it is nonzero, write

$$d_1(C'_G) = g - k + a, \qquad a \ge 1.$$

If

$$\boxed{3m - 2g \ge k - a + 1,}$$

then all contained codeword explanations lie on one global codeword line. Hence the full atom, including supports meeting $D \setminus G$, satisfies

$$\boxed{|I| \le n - m + 1.}$$

Equivalently, this payment applies whenever

$$g \le \left\lfloor \frac{3m - k + a - 1}{2} \right\rfloor.$$

Proof. For three distinct slopes define the affine deviation

$$p_{ijk} = (\gamma_j - \gamma_k)h_i + (\gamma_k - \gamma_i)h_j + (\gamma_i - \gamma_j)h_k.$$

It belongs to C'_G. On $S_i \cap S_j \cap S_k$, all three codewords equal the owner line at their slopes, so p_{ijk} vanishes. Inclusion–exclusion gives

$$|S_i \cap S_j \cap S_k| \ge 3m - 2g.$$

A nonzero word of C'_G has at most

$$g - d_1(C'_G) = k - a$$

zeros. Under the displayed hypothesis, p_{ijk} must therefore be zero for every triple.

Fix two slopes. The vanishing of every triple deviation shows that every h_i lies on the same global codeword line $H_0 + \gamma_i H_1$. Let $C_0 \subseteq G$ be the coordinate set on which this codeword line equals the owner line coefficientwise. Support-wise MCA-badness gives $|C_0| < m$. Outside C_0, equality of the two affine lines determines at most one slope. Thus the contained slopes satisfy

$$|I_{\text{in}}|(m - |C_0|) \le g - |C_0|, \qquad |I_{\text{in}}| \le g - m + 1.$$

There are at most $n-g$ crossing supports by Theorem 42.2, and hence

$$|I| \le (n-g) + (g-m+1) = n - m + 1.$$

□

Corollary 43.2 (exact support-excess-one thresholds). *At $a = 1$, Theorem 43.1 pays owners through*

row	$\lfloor (3m-k)/2 \rfloor$
KoalaBear MCA	1149784
Mersenne-31 MCA	1149748

Larger direction support excess increases the threshold by one owner coordinate for every two units of excess.

Remark 43.3 (connection with the minimum-distance sharpening of Section 7). For the boundary prefix dictionary, Proposition 7.8 proves that a shift pair of exchange defect r can occur only when $r \geq a$, where a is the direction-code support excess; the top seam $r = 1$ is therefore confined to $a = 1$. The counterexample in Proposition 40.1 also has $a = 1$. Thus the difficult seam is now consistently localized: large a is paid by the sharpened flag and owner bounds, while the exact-minimum-distance branch remains the genuine image-counting problem.

44. Owner pencils and a seven-point rich-incidence bound

The exclusive atom theorem also requires control of competing owner certificates. Theorem 36.1 gives uniqueness once two slopes are shared. The following theorem gives a complementary geometric bound when owners vary in a projective pencil.

Let

$$(Q_\tau, A_\tau, B_\tau) = (Q_0, A_0, B_0) + \tau(Q_1, A_1, B_1), \qquad \tau \in \mathbb{P}^1(\mathbb{F}),$$

be a projective pencil of owner triples. At a coordinate $x \in D$, set

$$F_x(\gamma, \tau) = A_\tau(x) + \gamma B_\tau(x) - Q_\tau(x)(r_0(x) + \gamma r_1(x)).$$

This has bidegree at most $(1,1)$ on $\mathbb{P}^1_\gamma \times \mathbb{P}^1_\tau$.

Theorem 44.1 (component-free owner-pencil rich points). *Assume no F_x is identically zero and no two coordinate curves $F_x = 0$ share an irreducible component. Then the number of parameter pairs (γ, τ) for which at least m coordinate equations vanish is at most*

$$\boxed{\left\lfloor \frac{2\binom{n}{2}}{\binom{m}{2}} \right\rfloor = \left\lfloor \frac{2n(n-1)}{m(m-1)} \right\rfloor.}$$

For both deployed MCA rows this number is exactly 7.

Proof. Two bidegree-$(1,1)$ curves with no common component have intersection number at most

$$(1,1) \cdot (1,1) = 2$$

on $\mathbb{P}^1 \times \mathbb{P}^1$. A parameter pair lying on at least m coordinate curves contributes at least $\binom{m}{2}$ incidences with unordered coordinate-curve pairs. Summing first over parameter pairs and then over the $\binom{n}{2}$ curve pairs gives

$$N_{\text{rich}} \binom{m}{2} \leq 2\binom{n}{2}.$$

The deployed integer evaluations are direct. □

Remark 44.2 (the shared-component branches are explicit). A nonzero bidegree-$(1,1)$ polynomial is either irreducible or factors into one vertical and one horizontal linear factor. Hence a shared component is one of:

(i) a fixed-slope vertical component;
(ii) a fixed-owner horizontal component;
(iii) a common irreducible $(1,1)$ component, in which case the two coordinate equations are proportional and form a coordinate-clone cell.

Identically zero F_x are fixed owner coordinates. Thus every failure of Theorem 44.1 is already a named structural branch rather than an unpriced multiplicity.

45. The exact large-owner atom target

After Corollary 42.3, only

$$g \geq 2m - k + 1$$

remains for a coherent rational atom. If at most 31 exceptional slopes are reserved for the local-to-global first-match transition, Theorem 42.2 shows that the required bound for the owner-contained slopes is

$$\boxed{B_{\mathrm{owner}}(g) \leq B^* - 31 - (n - g).} \tag{45.1}$$

At the first unproved owner size and at $g = n$, the targets are

row	$g_{\min} = 2m - k + 1$	target at $g_{\min}$	target at $g = n$
KoalaBear MCA	1183521	274980728110481425	274980728111395056
Mersenne-31 MCA	1183473	15863505	16777184

The Mersenne-31 row is binding.

For the full-owner boundary-prefix atom, use the CAP25 convention $K = k+1$ and $w = m-K$. The average prefix-fiber sizes are

$$\overline{F} = \binom{n}{m} |\mathbb{B}|^{-w}.$$

Their high-precision values and the full-owner multiplicative targets are

row	w	$\log_2 \overline{F}$	$(B^* - 31)/\overline{F}$
KoalaBear MCA	67471	35.7352464170	4807520.929566
Mersenne-31 MCA	67447	20.7411470346	9.57219783037

Thus the exact remaining Mersenne theorem is a factor-9.5722 maximum-image or maximum-fiber statement before any additional cell consumes budget. Owner localization has removed every smaller owner and all multi-owner component-free pencils, but it does not prove this full-owner prefix bound.

Problem 45.1 (exclusive large-owner atom image bound). *For every first-match coherent rational atom on either deployed domain with $g > 2m - k$, prove (45.1). In the full-owner simple-pole branch it suffices to prove the row-sharp challenge-slope image bound*

$$|\{U_z(\alpha) - \Lambda_M(\alpha) : M \in \mathrm{Fib}_w(z)\} \cap \Gamma| \leq B^* - 31.$$

For Mersenne-31, the stronger support-fiber estimate need only lose a factor below 9.57219783037 *from the exact average.*

46. A conditional exact adjacent-row theorem

We can now state the final certificate without reverting to a sum of many unrelated local budgets.

Definition 46.1 (exception routing input). *Say that exception routing holds if, for every bad received line with more than* 31 *slopes, there is a set E of at most* 31 *slopes such that every remaining local order-32 classification is primitive and Theorem 38.1 applies; every denominator-root, extension, coordinate-clone, imperfect-quotient, and quantified near-sunflower failure is either included in E or assigned to the spread-abundance or large-owner input with the same received-line owner retained.*

Theorem 46.2 (exact conditional adjacent-row closure). *Fix one of the two deployed MCA rows. Assume the following three inputs.*

(S): *The deployed spread-abundance theorem, Problem 41.2.*
(A): *The exclusive large-owner atom image bound, Problem 45.1.*
(E): *Exception routing in the sense of Definition 46.1.*

Then

$$\boxed{B_{\mathrm{MCA}}(C,m) \le B^*.}$$

Together with the unconditional CAP25 v13.2 witness at agreement $m-1$, this proves that the row's exact adjacent MCA threshold is m.

Proof. Let Z be the set of bad challenge slopes. If $|Z| \le 31$, the claim is immediate. If Z contains a primitive spread core, assumption (S) pays the whole line.

Otherwise remove the exceptional set E supplied by exception routing. Every 32-subset of $Z \setminus E$ is classified by Theorem 39.1 as a global affine codeword line or one primitive rational atom. The overlap graph of 32-subsets is connected, and Theorem 36.1 forces one coherent global structure.

If that structure is a global affine codeword line, the standard coordinate-injection argument gives at most $n-m+1$ slopes. Adding $|E| \le 31$ remains far below either budget.

Suppose instead that it is one rational atom with owner-set size g. For $g < m$, Theorem 42.2 gives at most $n-m+1$ atom slopes. For $m \le g \le 2m-k$, Corollary 42.3 gives at most n. Again the additional 31 slopes fit both budgets. Finally, for $g > 2m-k$, there are at most $n-g$ crossing slopes and assumption (A), namely (45.1), pays the contained slopes; adding the reserved 31 gives exactly at most B^*.

CAP25 v13.2 already proves an unsafe witness at $m-1$, so safety at m identifies the adjacent threshold. □

47. Core interpolant and correction setup

Let

$$C = \mathrm{RS}[\mathbb{F}, D, k], \qquad |D| = n,$$

and let

$$r_\gamma = r_0 + \gamma r_1 \in \mathbb{F}^D$$

be support-wise MCA-bad at agreement $m > k$. Put

$$d = m-k, \qquad R = n-k, \qquad t = n-m.$$

Fix one first-match primitive spread core in the sense of Problem 41.2

$$(\gamma_i, h_i, S_i)_{i=1}^{32},$$

where the slopes are distinct, $h_i \in \mathbb{F}[X]_{<k}$, and S_i is the *maximal* agreement support of h_i with r_{γ_i}. The common-support branch has already been removed, so

$$\bigcap_{i=1}^{32} S_i = \varnothing. \tag{47.1}$$

There is a unique coefficientwise interpolant

$$H(X,Z) = \sum_{j=0}^{31} H_j(X) Z^j, \qquad H(X,\gamma_i) = h_i(X), \qquad \deg H_j < k. \tag{47.2}$$

For $x \in D$ define

$$E_x(Z) = H(x,Z) - r_0(x) - Z r_1(x) \tag{47.3}$$

and

$$q_x = |\{i : x \in S_i\}|.$$

Every core agreement contributes a root of E_x, so

$$\sum_{x \in D} q_x = \sum_{i=1}^{32} |S_i| \ge 32m. \tag{47.4}$$

Moreover, (47.1) and maximality imply

$$E_x \ne 0 \quad \text{for every } x \in D. \tag{47.5}$$

Indeed, $E_x \equiv 0$ would put x in every maximal core support.

The deployed values are

row	m	d	$n-m$	B^*
KoalaBear MCA	1116048	67472	981104	274980728111395087
Mersenne-31 MCA	1116024	67448	981128	16777215

with $n = 2^{21}$ and $k = 2^{20}$.

48. Only fifty-eight slopes can use the core interpolant

Theorem 48.1 (core-compatible abundance bound)**.** *Let Z_0 be the set of slopes $\gamma \in \mathbb{F}$ for which the codeword $H(X, \gamma)$ agrees with r_γ on at least m coordinates. Then*

$$\boxed{|Z_0| \le \left\lfloor \frac{31n}{m} \right\rfloor = 58.} \tag{48.1}$$

More precisely, outside the 32 *core slopes there are at most*

$$\boxed{\left\lfloor \frac{31n - 32m}{m} \right\rfloor = 26} \tag{48.2}$$

additional core-compatible slopes on either deployed row.

Proof. By (47.5), every E_x has at most 31 roots. A slope in Z_0 contributes at least m coordinate roots, hence

$$|Z_0| m \le \sum_{x \in D} \deg E_x \le 31n,$$

which proves (48.1).

For a non-core slope, every agreement coordinate consumes a root distinct from the q_x core roots already present at x. The remaining root capacity is at most

$$\sum_x (31 - q_x) \le 31n - 32m$$

by (47.4). Dividing by m gives (48.2). □

Remark 48.2. The theorem uses maximal supports only to exclude identically zero coordinate error polynomials. It is independent of quotient, rational-owner, or minimum-distance estimates. Thus any over-budget spread line must obtain essentially all of its extra slopes from nonzero codeword corrections to H.

49. An unconditional theorem for one complete correction ray

Fix a nonzero $P \in \mathbb{F}[X]_{<k}$. Consider all parameter pairs (γ, c) for which

$$h_{\gamma,c}(X) = H(X, \gamma) + cP(X) \tag{49.1}$$

agrees with r_γ on at least m coordinates. At $x \in D$ the agreement equation is

$$E_x(\gamma) + cP(x) = 0. \tag{49.2}$$

No component-freeness assumption will be imposed.

For $P(x) \neq 0$, put

$$f_x(Z) = \frac{E_x(Z)}{P(x)} \in \mathbb{F}[Z]_{\le 31}.$$

Coordinates with the same f_x form a *clone class*. Coordinates with $P(x) = 0$ form the vertical class.

Lemma 49.1 (clone-class dichotomy)**.** *Let $C_f = \{x : P(x) \neq 0,\ f_x = f\}$. Then exactly one of the following holds.*

(i) $|C_f| \le k - 1$*;*

(ii) *the codeword curve*

$$G_f(X,Z) = H(X,Z) - f(Z)P(X)$$

has slope degree at most one. In this case the graph $c = -f(\gamma)$ is a global affine codeword line.

At most two clone classes of type (ii) *can have size at least k.*

Proof. For every $x \in C_f$,

$$G_f(x,Z) = r_0(x) + Zr_1(x)$$

as a polynomial identity in Z. If G_f has a nonzero coefficient of Z^j for some $j \geq 2$, that coefficient is a nonzero degree-$< k$ polynomial vanishing on C_f, so $|C_f| \leq k-1$. Otherwise $G_f = A_f + ZB_f$ is a global affine codeword line.

Clone classes are disjoint subsets of D. A class of type (ii) not covered by (i) has at least k coordinates. Since $n = 2k$, there can be at most two. □

Lemma 49.2 (global affine-line block)**.** *For any fixed global codeword line $A + \gamma B$, the number of slopes for which it agrees with r_γ on at least m coordinates is at most*

$$n - m + 1. \tag{49.3}$$

Proof. Let C_0 be the coordinate set on which $A = r_0$ and $B = r_1$. Support-wise MCA-badness gives $|C_0| < m$. Outside C_0, agreement of the two affine lines determines at most one slope. Hence

$$N(m - |C_0|) \leq n - |C_0|,$$

and the right quotient is maximized at $|C_0| = m - 1$. □

Theorem 49.3 (complete correction-ray bound)**.** *For either deployed row, the number of rich parameter pairs (γ, c) satisfying* (49.1) *is at most*

$$\boxed{2(n-m+1) + \left\lfloor \frac{31\binom{n}{2}}{(k-1)(m-k+1)} \right\rfloor.} \tag{49.4}$$

The final fraction equals 963 *on both rows. Thus one full correction ray contributes at most*

row	*ray bound*	B^* *– ray bound*
KoalaBear MCA	1963173	274980728109431914
Mersenne-31 MCA	1963221	14813994

In particular, a single correction ray can never be the residual spread-abundance obstruction.

Proof. Remove all rich points lying on a type-(ii) clone graph whose class has at least k coordinates. By Lemmas 49.1 and 49.2, there are at most two such graphs and each contributes at most $n - m + 1$ slopes.

Consider a remaining rich point and choose exactly m of its agreement coordinates. Partition them into the vertical class and the clone classes meeting the point. The vertical class has size at most $k-1$, because a nonzero degree-$< k$ polynomial has at most $k-1$ zeros. Every clone class present at the point has size at most $k-1$: type-(i) classes have this bound, while every larger type-(ii) graph was removed.

Since $m < 2(k-1)$ on both rows, a partition of m elements into parts of size at most $k-1$ has at least

$$(k-1)(m-k+1) \tag{49.5}$$

heterogeneous unordered pairs. This is the minimum obtained from the two-part partition $(k-1, m-k+1)$.

A heterogeneous pair contributes to at most 31 rich parameter points. For two graph coordinates from different clone classes, the condition is

$$f_x(\gamma) = f_y(\gamma),$$

a nonzero polynomial equation of degree at most 31, after which c is determined. For one vertical and one graph coordinate, $E_x(\gamma) = 0$ has at most 31 roots and again the graph equation determines c. We never count vertical–vertical pairs or pairs inside one clone class.

Double counting rich-point/heterogeneous-pair incidences gives

$$N_{\mathrm{res}}(k-1)(m-k+1) \le 31\binom{n}{2}.$$

Adding the at most two affine-line blocks proves (49.4). Counting parameter pairs only overestimates the number of distinct slopes. □

Remark 49.4 (minimum-support refinement). Suppose every nonzero codeword in the span of P and the high slope coefficients $H_2, \ldots, H_{31}$ has support at least $R + a$. Then $k - 1$ in the proof can be replaced by $k - a$. Writing $q = k - a$, the exact cross-pair denominator is

$$\Xi(m, q) = \binom{m}{2} - u\binom{q}{2} - \binom{r}{2}, \qquad m = uq + r, \quad 0 \le r < q,$$

and there are at most $\lfloor n/(q + 1)\rfloor$ large affine clone classes. In the active residual wedge $m \le 2q$, so $\Xi(m, q) = q(m - q)$. The exact-minimum-distance case $a = 1$ is the worst case in the range left by the minimum-distance sharpening of Section 7; the same section also proves that the boundary top seam occurs only at $a = 1$.

50. Higher-dimensional correction spaces

Let $W \le C$ have dimension s, and consider all rich points

$$(\gamma, P) \in \mathbb{F} \times W$$

for which $H(X, \gamma) + P(X)$ agrees with r_γ on at least m coordinates. Choose a basis $P_1, \ldots, P_s$ of W and write $P = \sum c_j P_j$. The coordinate equation

$$F_x(Z, \mathbf{c}) = E_x(Z) + \sum_{j=1}^{s} c_j P_j(x) = 0 \tag{50.1}$$

has bidegree at most $(31, 1)$ on $\mathbb{P}^1_Z \times \mathbb{P}^s_{\mathbf{c}}$.

50.1. **The proper-intersection compiler.**

Definition 50.1 (proper correction space)**.** *The correction space W is* proper for the core *if every $s + 1$ coordinate hypersurfaces from (50.1) have empty or zero-dimensional common intersection in $\mathbb{P}^1 \times \mathbb{P}^s$.*

Theorem 50.2 (proper-intersection correction compiler)**.** *If W is proper for the core, then the number N_W of rich parameter points satisfies*

$$\boxed{N_W \le \left\lfloor 31(s+1)\frac{\binom{n}{s+1}}{\binom{m}{s+1}} \right\rfloor}. \tag{50.2}$$

At the deployed rows this pays every proper correction space through

row	$s \le$	*bound at final s*	*bound at $s+1$*
KoalaBear MCA	50	148068539552473273	283695391248936790
Mersenne-31 MCA	15	11989622	23938324

The next values only mark the limit of this compiler.

Proof. The divisor class of each projective hypersurface is at most

$$31H_Z + H_W.$$

In the Chow ring of $\mathbb{P}^1 \times \mathbb{P}^s$,

$$(31H_Z + H_W)^{s+1} = 31(s+1)H_Z H_W^s,$$

because $H_Z^2 = 0$ and $H_W^{s+1} = 0$. Hence every proper intersection of $s+1$ coordinate hypersurfaces has degree at most $31(s+1)$.

A rich point lies on at least m hypersurfaces and therefore contributes at least $\binom{m}{s+1}$ incidences with coordinate $(s+1)$-subsets. Summing first over rich points and then over the $\binom{n}{s+1}$ coordinate subsets proves (50.2). $\square$

Lemma 50.3 (what a nonproper tuple means)**.** *Let T be a set of $s+1$ coordinates.*

(i) *If the evaluation functionals of T on W have rank below s, then a nonzero word of W vanishes on a subset spanning the same evaluation flat.*

(ii) *If their rank is s and the common intersection is positive-dimensional, choose any evaluation basis $B \subset T$. There is a unique polynomial curve $P_B(Z) \in W \otimes \mathbb{F}[Z]_{\le 31}$ solving the equations on B, and every equation in T vanishes identically on that curve.*

Thus every failure of properness is either an evaluation rank-flat or an exact polynomial clone component.

Proof. Part (i) is the annihilator statement for the column matroid of the evaluation map. For (ii), the constant evaluation matrix on B is invertible, so solving its equations gives $P_B(Z)$ coefficientwise and with degree at most 31. Substitution into the remaining equation gives a univariate polynomial. A positive-dimensional intersection forces that polynomial to vanish identically. $\square$

50.2. **A clone-tolerant basis-curve compiler.** Let

$$V = \mathrm{Span}(W, H_2, \dots, H_{31}) \le C$$

and write

$$d_1(V) = R + a, \qquad a \ge 1. \tag{50.3}$$

We say that W *absorbs the high core* if $H_j \in W$ for every $j \ge 2$.

Theorem 50.4 (clone-tolerant basis-curve bound)**.** *Assume $s \ge 1$ and that W does not absorb the high core. Put*

$$M_a = \left\lfloor \frac{31(R+a)}{d+a} \right\rfloor. \tag{50.4}$$

Then

$$\boxed{N_W \le \left\lfloor M_a \frac{n^{\underline{s}}}{(d+a)^{\overline{s}}} \right\rfloor}. \tag{50.5}$$

No component-freeness assumption is required.

Proof. Fix a rich point and an exact size-m agreement support S. For every j-dimensional subcode $U \le W$, strict increase of generalized Hamming weights and (50.3) give

$$|\,\mathrm{supp}_D(U)| \ge R + a + j - 1.$$

At most $t = n - m$ of these support coordinates lie outside S, so the restricted code $W|_S$ has

$$d_j(W|_S) \ge d + a + j - 1.$$

Greedily choosing independent evaluation columns therefore gives at least

$$\frac{(d+a)^{\overline{s}}}{s!} \tag{50.6}$$

unordered evaluation bases $B \subset S$.

For each such B, solve the basis equations to obtain a unique polynomial correction curve

$$P_B(Z) \in W \otimes \mathbb{F}[Z]_{\le 31}.$$

Let C_B be the coordinates whose equations vanish identically after this substitution, and put $c_B = |C_B|$. Since W does not absorb all high core coefficients, the codeword curve $H + P_B$ has a nonzero coefficient of Z^j for some $j \geq 2$. That coefficient lies in V, so it has at most $k - a$ zeros. Hence

$$c_B \leq k - a. \tag{50.7}$$

Every rich point on the curve $P = P_B(\gamma)$ agrees automatically on C_B and must obtain at least $m - c_B$ further agreements outside C_B. Each outside coordinate supplies a nonzero polynomial equation of degree at most 31. Root-capacity counting on this one curve gives

$$N(B)(m - c_B) \leq 31(n - c_B).$$

The quotient $(n - c)/(m - c)$ is increasing for $c < m$, and (50.7) yields

$$N(B) \leq \left\lfloor \frac{31(R + a)}{d + a} \right\rfloor = M_a.$$

Each rich point is incident to at least (50.6) basis curves. Globally there are at most $\binom{n}{s}$ evaluation bases, and each basis curve contains at most M_a rich points. Therefore

$$N_W \frac{(d + a)^{\underline{s}}}{s!} \leq M_a \binom{n}{s} = M_a \frac{n^{\underline{s}}}{s!},$$

which is (50.5). □

Corollary 50.5 (exact deployed transitions)**.** *At exact minimum support $a = 1$, Theorem 50.4 pays through*

row	$s \leq$	*bound at final s*	*bound at $s+1$*
KoalaBear MCA	9	13013823503882165	404431535289439486
Mersenne-31 MCA	3	14457337	449492838

The actual minimum-support excess enlarges the range. The first exact paid transitions beyond those dimensions are

row	s	a_{paid}	*bound at a_{paid}*	*bound at $a_{\text{paid}} - 1$*
KoalaBear MCA	10	2434	274948660791683460	274987992550612247
	11	25508	274953206296382843	274985734988552690
	12	50875	274977972290567960	275005854396152321
Mersenne-31 MCA	4	64258	16777099	16777608

All comparisons are exact integer comparisons with the row budgets.

Remark 50.6 (relation to the minimum-distance wedge). Proposition 7.3 proves that the first-unpaid charts already satisfy $a \leq 70831$ on KoalaBear and $a \leq 111528$ on Mersenne-31 at zero outside mismatch, with positive mismatch shrinking the residual further. It also confines the boundary top seam to $a = 1$. Thus Corollary 50.5 is not merely an abstract monotonicity statement: it removes correction dimensions 10–12 on substantial parts of the KoalaBear wedge and dimension 4 on a substantial part of the Mersenne wedge, while the genuine top-seam endpoint remains the exact-minimum-distance case.

51. The new spread residual

Theorem 51.1 (spread residual after core-relative incidence)**.** *Assume a deployed support-wise MCA-bad line contains a first-match primitive spread core and has more bad slopes than its row budget. Then all of the following must hold.*

(R1) *At most* 58 *bad slopes use the core interpolant itself.*
(R2) *The remaining corrections are not contained in one projective correction ray.*
(R3) *If their linear correction span W has dimension at most* 50 *on KoalaBear or* 15 *on Mersenne-31, then some $s + 1$ coordinate equations have a positive-dimensional common component.*

(R4) *If W does not absorb the high core, then its dimension and support excess lie beyond every transition in Corollary 50.5.*

(R5) *Every low-dimensional surviving component is therefore an evaluation rank-flat or an exact polynomial clone component in the sense of Lemma 50.3; it is not an unstructured failure of Bezout counting.*

Proof. Items (R1)–(R4) are the contrapositives of Theorems 48.1, 49.3, 50.2 and 50.4. Item (R5) is Lemma 50.3. ☐

The theorem changes the spread-abundance target. It is no longer necessary to control arbitrary extensions of a core. The only remaining spread mechanism is

$$\boxed{\begin{array}{c}\text{many projectively independent correction directions, together with}\\ \text{a high-dimensional correction span or a positive-dimensional}\\ \text{rank-flat/clone atlas surviving all first-match routes.}\end{array}}$$

Problem 51.2 (next spread theorem)**.** Intermediate status. *This is the local correction-component form of the final Problem 53.1. It is retained to state the geometric alternatives exposed by Theorem 51.1, but is not a separate terminal input.*

Prove that every positive-dimensional correction component from Theorem 51.1 either

(a) *yields a rational-owner, quotient, extension, field-drop, or quantified near-sunflower atom declared in Theorems 39.1 and 42.2 and Corollary 38.2;*

(b) *has correction directions contained in a bounded union of projective rays, so Theorem 49.3 pays it; or*

(c) *admits a proper quotient correction space to which Theorem 50.2 or Theorem 50.4 applies.*

For Mersenne-31 it is enough that the total surviving spread contribution be at most 16,777,215*; for KoalaBear the available budget is much larger.*

Part 7. Completion certificates and remaining row-sharp inputs

52. Integrated theorem status

Theorem 52.1 (Audited theorem status)**.** *All statements explicitly presented as theorems, propositions, lemmas, or corollaries in Parts I–VI are unconditional under their displayed hypotheses. In particular, the paper proves the complete support and profile compiler, the minimum-distance flag sharpening, the primitive-Q/Sidon implication and its shallow unconditional range, saturation, one-pencil moving-root BC, $Q \Rightarrow SP$, the order-*32 *rank dictionary, dyadic descent, rational-atom extraction and coherence, owner localization, the seven-point component-free owner-pencil bound, the* 58*-slope core-compatible bound, the complete correction-ray bound, and the displayed proper and clone-tolerant correction-space payments.*

No theorem in this paper proves either deployed adjacent MCA-safe row or either deployed adjacent list-safe row. The terminal open inputs used by the completion certificates are exactly Problems 45.1, 53.1, 53.2 and 54.1. The earlier Problems 41.2 and 51.2 are intermediate formulations subsumed by Problem 53.1, not additional terminal cells.

Remark 52.2 (Status of the v3 saturated-BC problem). The v3 problem labelled `prob:saturated-bc` is superseded in v4, not silently discharged. Its primitive one-parameter MCA clause is proved by Corollary 23.5. Its higher-dimensional MCA clause remains open and is represented jointly by Problems 45.1, 53.1 and 53.2; in particular, v4 still requires exhaustive same-owner routing from every surviving balanced-core component into that endgame. Its separate arbitrary-word list-interior clause is contained in Problem 54.1. Thus no archived problem is being banked as a live theorem, and no higher-dimensional balanced-core payment is claimed merely from the one-pencil moving-root bound.

53. The final MCA certificate

The order-32 method replaces the old sum over many independently budgeted MCA cells by a maximum-type endgame.

Problem 53.1 (Complete spread-component routing)**.** *This is the final v4 successor to the higher-dimensional MCA clause of the v3* `prob:saturated-bc`*. Together with Problems 45.1 and 53.2, it must provide the exhaustive same-owner routing that the old balanced-core cell left open. Its row-sharp conclusion subsumes the intermediate Problems 41.2 and 51.2 for purposes of the final certificate.*

Prove that every positive-dimensional correction component surviving Theorem 51.1 is routed, with the same received-line owner retained, into one of the following disjoint outcomes:

(a) *a rational-owner, quotient, extension, field-drop, or quantified near-sunflower atom;*
(b) *a bounded union of complete projective correction rays paid by Theorem 49.3;*
(c) *a proper quotient correction space paid by Theorem 50.2 or Theorem 50.4;*
(d) *an explicitly named residual component with a literal row-sharp slope bound.*

The total spread contribution must fit the row budget.

Problem 53.2 (Complete MCA exception routing)**.** *Verify the exception-routing input of Definition 46.1: denominator-root, extension, coordinate-clone, imperfect-quotient, and quantified near-sunflower failures must either be absorbed into the spread or large-owner bounds with a common owner retained, or occupy a set of at most* 31 *slopes.*

Theorem 53.3 (Exact MCA completion certificate)**.** *Fix either deployed MCA row. If Problems 45.1, 53.1 and 53.2 are proved with the literal row constants in Parts V–VI, then*

$$B_{\mathrm{MCA}}(m) \le B^*.$$

Together with the unsafe witness at agreement $m-1$ from CAP25 v13.2, the exact adjacent MCA threshold is m.

Proof. The hypotheses supply the spread-abundance and exception-routing inputs of Theorem 46.2; Problem 45.1 supplies its atom input. Apply that theorem and then the imported unsafe witness. □

54. The final list certificate

For a list row retain the disjoint summed ledger

$$U_{\mathrm{list}} = U_{\mathrm{paid}} + U_Q + U_{\mathrm{list-int}} + U_{\mathrm{ext}} + U_{\mathrm{new}}. \tag{54.1}$$

Here U_{paid} is the exact first-match sum of all tangent, common-support, quotient, planted, field-drop, fixed-union, affine-span, common-zero, rank-flat, codimension-one, support-flag, minimum-distance, puncturing, and other certified cells; U_Q is the row-sharp pruned boundary-prefix maximum; $U_{\mathrm{list-int}}$ is the high-nullity arbitrary-word interior codeword-ray term; U_{ext} is the exact extension projection payment; and U_{new} is zero unless an explicitly named residual cell survives.

Problem 54.1 (Row-sharp list completion)**.** *This is the final v4 successor to the arbitrary-word list-interior clause of the v3* `prob:saturated-bc`*; it also includes the remaining prefix and extension terms required by the v4 list ledger.*

For each deployed list row, prove exact integer bounds for the remaining high-nullity prefix fiber, arbitrary-word interior, and extension projection terms so that

$$U_{\mathrm{list}} \le B^*.$$

The prefix estimate must use the actual residual mass and the exact budget left after the certified compiler cells are charged.

Theorem 54.2 (Exact summed completion certificate)**.** *Fix one deployed list row. If an exhaustive first-match compiler proves, with no unresolved cell,*

$$U_{\mathrm{list}} \le B^*,$$

then the first safe integer agreement is the adjacent row $a_0 + 1$.

Proof. CAP25 v13.2 proves the unsafe inequality at a_0. Exhaustive first-match coverage and the displayed upper bound give safety at $a_0 + 1$. Monotonicity identifies the first safe integer agreement. $\square$

55. Conclusion

This paper gives one unified Grande Finale manuscript. It contains the exact threshold refinements, profile and Fourier compiler, primitive Q, support and minimum-distance recursions, saturation and moving-root geometry, the order-32 rank pivot, rational-atom extraction and coherence, owner localization, and the current spread-core incidence theory.

The remaining MCA problem has been compressed to three row-sharp inputs: route the positive-dimensional correction components isolated by Theorem 51.1, prove the large-owner image bound of Problem 45.1, and complete the finite exception routing of Problem 53.2. The binding numerical atom target is the Mersenne-31 full-owner factor 9.57219783037. The remaining list problem is the exact high-nullity prefix/interior/extension sum of Problem 54.1. Until those inputs are proved, no adjacent safe row is claimed.

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